Learn Onshape for FTC

Alan G. Smith

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Learn Onshape for FTC
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The most recent PDF is free at https://github.com/alan412/LearnOnshapeForFTC

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This book is dedicated to:

My wife who after suffering through my first two books encouraged me to write this one.

My father who spent many hours with me on the Vic 20, Commodore 64, and the robotic arm science project. Without his investment, I wouldn't be the engineer I am today.

FTC coaches everywhere that teach their students to think well and work hard

My sons Philip and Joshua who have shown me what is possible with Onshape and FTC

Whatever you do, work at it with all your heart, as working for the Lord, not for men.
Colossians 3:23 (NIV 1984)

Contents

I. Getting Started 1

1. Introduction 3
- 1.1. Why use CAD (and what is it)? . . . 3
- 1.2. Why Onshape? . . . 4
- 1.3. Roadmap for our time together . . . 4
- 1.4. Get your educational account . . . 5
- 1.5. Setup your account . . . 5

2. For coaches... 7
- 2.1. "Hudson Bay" Start . . . 8
- 2.2. Create a "team tiles" document . . . 9
- 2.3. Create your individual tile . . . 11
- 2.4. Put it in the assembly . . . 12

3. For students 15
- 3.1. Making your team tile . . . 15
- 3.2. Connecting your team tile to the rest of your team . . . 23
  - 3.2.1. Fastened Mates . . . 24

II. Modeling a simple Part 27

4. Spacers 29
- 4.1. A simple spacer . . . 29
  - 4.1.1. Make in CAD . . . 29
  - 4.1.2. 3D printing our part . . . 32
- 4.2. Using variables . . . 32

5. Simple Robot sign 35

6. Magnet Robot sign 41

7. Cover end of U-Channel 55

8. Ball holder 61

9. Funnel 69

III. Modeling your robot 73

10. Design your robot - Space Planning 75
10.1. Space planning for Drive train 76
10.2. Space planning for intake 80
10.3. Space planning for shooter 82
10.4. Space planning for electronics and battery 83
10.5. Some comments 85

11. Mecanum Drivetrain 87

12. Intake 101

13. Shooter 113
13.1. Making the shooter ramp 113
13.2. A mixed assembly (custom and COTS) 117

14. Bringing it all together 125
14.1. Assembling the pieces 125
14.2. Creating BOM (Bill of Materials) 137
14.3. Next steps 139

IV. Appendices 141

A. List of Keyboard Shortcuts mentioned 143
A.1. In 3D viewing 143
A.2. In Part Studio 143
A.3. In a Sketch 143
A.4. In an Assembly 144

B. Setting Workspace Units 145

C. Configurations 147

Index 151

Part I.

Getting Started

1. Introduction

This is a little dangerous, because unlike my other books on software where I worked on software professionally for many years, I have never been a professional mechanical engineer.

However, I still feel like many rookie teams could be helped with an introduction to Onshape. (an online CAD system) and specifically with how to use it well for FTC.

There are many excellent resources for Onshape, but a lot of them focus on many details instead of getting you to make useful things quickly. After you have gone through this book, I suspect many students will then go on to take several of the excellent courses at http://learn.onshape.com. But many students need quick "wins" and ways they see this is useful before they are ready for all of the detail there.

1.1. Why use CAD (and what is it)?

CAD stands for computer aided design. It allows us to put together designs and show them to our team mates. It falls in between making a drawing on a whiteboard or a piece of paper and actually building the robot.

It provides several significant benefits over a simple drawing.

1. It forces you to make sure that things can actually fit together. (One of the most important principles of robotics is that no two objects can occupy the same space at the same time. - The fancy name for this is impenetrability.[1])

2. You can use part of your design to create 3d printed parts or to send places for fabrication (or use yourself with a laser or a CNC machine) or to even print parts to bend sheet metal for.

3. It is faster than building for figuring out if an idea will work.

[1] yes, you can argue why this is not "technically" true. But for our purposes it is true enough. As my best friend says, "All education is a series of slightly less egregious lies."

4. You can add motion in to make sure things are possible to move. (If it works in CAD, it doesn't mean it will work in the real world. But if it doesn't work in CAD, it is highly unlikely to work in the real world.)

5. You can save versions of the design and compare them.

1.2. Why Onshape?

There are many CAD systems available and many of them are quite good. Why should you choose Onshape over Creo or Solidworks or Fusion 360 or many many others?

1. It runs completely in your browser which means that all installation headaches are gone. I can't overemphasize how important this is.

2. It is like Google Docs for CAD where it was designed from the ground up for collaboration and working on designs with other people. (which makes it great for FTC teams)

3. Many (most?) of the parts commonly used for FTC are already available to find easily thanks to the great work by Purple Gears (FTC #2901)

4. It is a solid choice that can accomplish most of what you need in a CAD system

5. They have excellent training materials and documentation available. (With this point, you might be asking why I am writing this book. I am writing it to help people with the small subset that is necessary and some good patterns. I have also found that many students give up on the training materials because of the amount of depth they go into.)

6. PTC generously offers a free educational version to all students on FTC teams. (Please do not take advantage of this to design commercial products.) I have a note from PTC's legal team confirming that this covers students that are homeschooled as well as coaches and mentors working with the team.

1.3. Roadmap for our time together

This book is split into 3 main parts.
Part I: Getting Started

This is the part you are in now, it explains briefly what we are doing and why as well as how to set things up for success. If you are a coach make sure you read chapter 2.

Part II: Modeling a simple Part

This part focuses on modeling individual parts that you might 3d print.

Part III: Modeling your robot

This part focuses on assembling parts together (both your own and standard ones)

I recommend you start at the beginning and follow along. I think that doing the exercises will be the best way to learn. Also, I get less detailed as I get to later chapters so if you have skipped earlier ones you may find it difficult to follow along.

1.4. Get your educational account

You should get an educational account. Some people will use the hobbyist version, and that can be fine but it doesn't have all the power of the educational account. To get an educational account go to https://onshape.com/edu

1.5. Setup your account

Go to the upper right where it will have your name. Press the down arrow and select "My account". Under Profile you can change the image that represents you (helpful to your team mates), select a Nickname, etc.

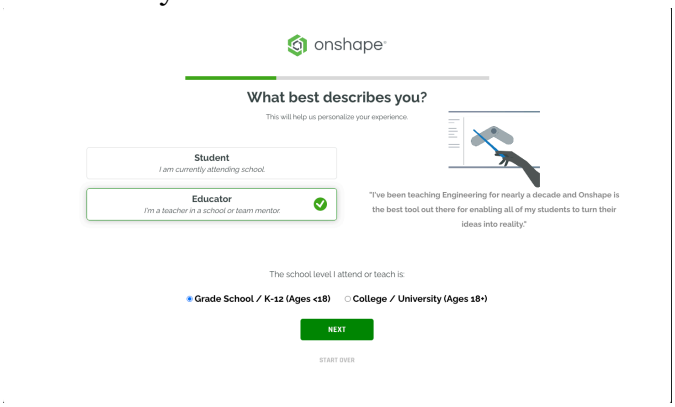
On the left hand side, select "Preferences", scroll down to Units and change "Length Default unit" to millimeter. (Both REV and GoBilda are based on metric so this will make your life easier.) . Be aware that this only changes new documents. (I describe how to change existing documents in Appendix B.)

2. For coaches...

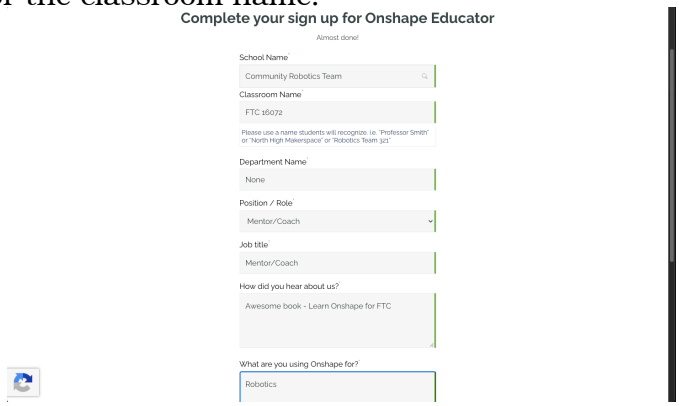
Get an educator account, and create a classroom for your team. That way there is a common place that all of your students on the team can put the design documents. I recommend making a folder for each year (for example BioBuzz 2026-2027).

Personally while I use the classroom, I don't setup different classes per year. To make a classroom, you need to get an Educator Plan. (They are free for FIRST teams). You need to create an account at https://onshape.com/edu

Make sure you select educator:

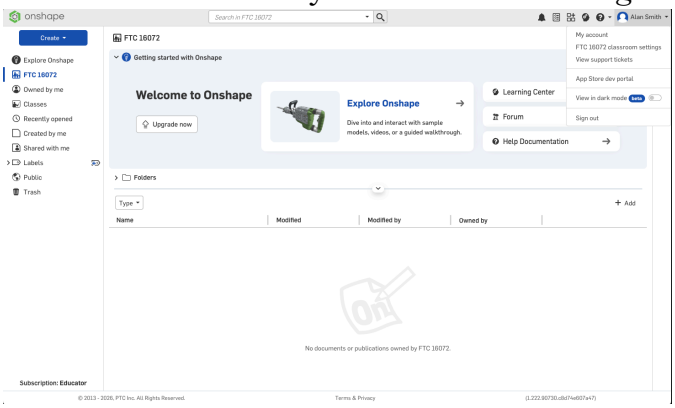


Then fill out all of the settings. Make sure you setup something recognizable for the classroom name.

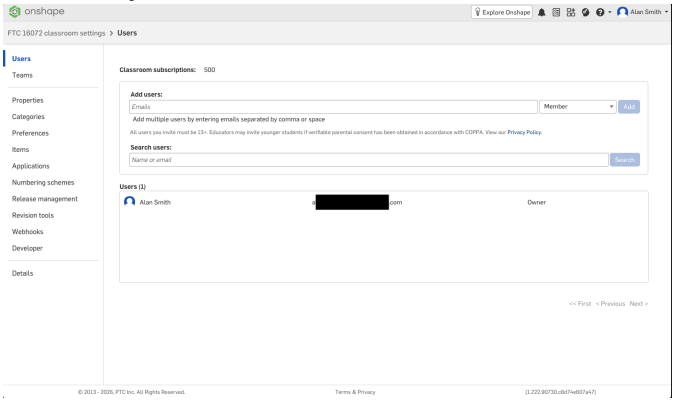


Then it'll send you an e-mail with a link to click to finish setting up. After you

have done that, you now have an educator account!! Now go to the upper right hand corner and select your classroom settings.



It will take you to the screen where you can add your students. I recommend adding them as "Member" and adding your other co-coaches as "Owners". If you add email addresses that already have Onshape accounts, it will add them immediately. Otherwise it will send them a link to setup an account.



The huge advantage of using a classroom is that all of the documents are owned by the classroom and you don't have to worry about a student remembering to share it withe others on the team.

2.1. "Hudson Bay" Start

Many years ago there was a fur trading company in Canada called the Hudson Bay Company. When they were going on an expedition, they would start by going 1 mile the first day and setting up camp. Why do this when there was so much they needed to do??

Simply, it let them find all of the things they had forgotten when it wasn't too

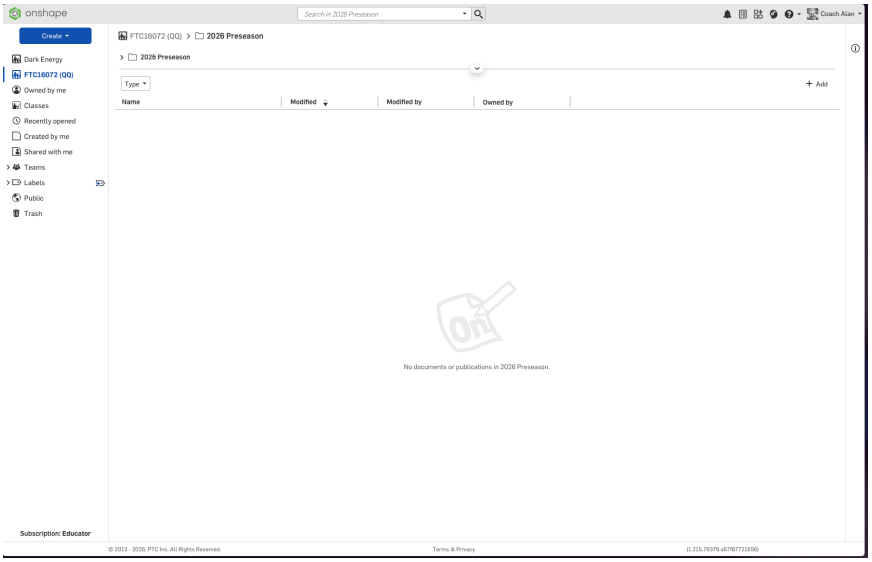
far to go back. And since it was a real start and a real camp setup, it found things that would not have been found with any number of checklists.

For my teams, our “Hudson Bay” start is that we have every student make a tile with their name on it in a document and then attach it to all of the other tiles. This doesn’t show that students have all of the CAD skills necessary, but it shows that their login works and they are able to do a few VERY common things.

So let’s begin with how to do this.

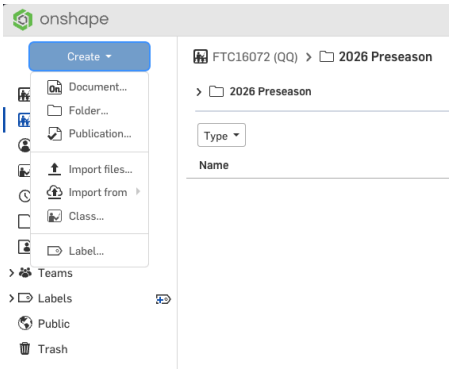
2.2. Create a “team tiles” document

I start each year by creating a new folder that is the year and preseason. (For example, you can see here I did 2026 Preseason)

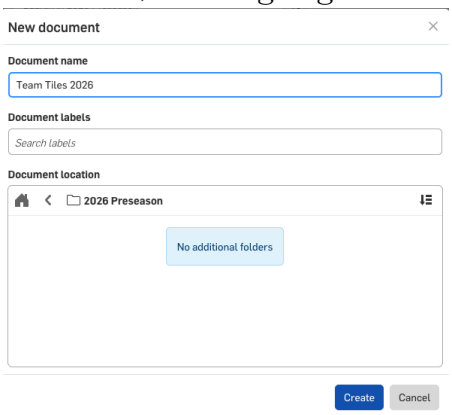


In order to do that, you click on the blue “Create” button in the upper left and then select folder and type in “2026 Preseason”. When you click on the “Create” button, you get a menu that looks like this:

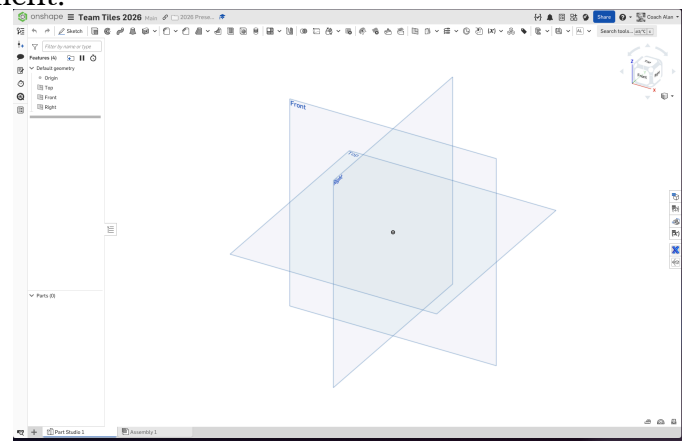
2. For coaches...



After we have done the folder, we'll come back to this and Create a document. In this case, we are going to call it team tiles:



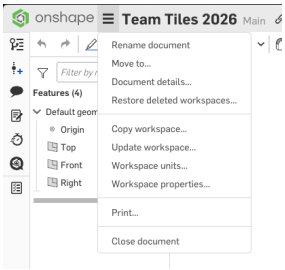
After we press create (blue button in lower right), it will take us to that document:



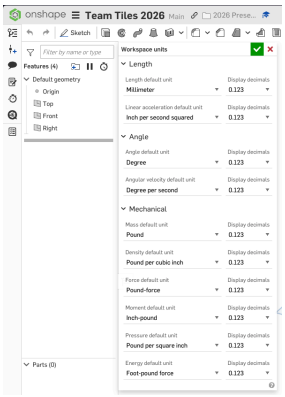
You'll see that this shows the name of the document in the upper right followed by what folder it is in.

The next thing we will do is make sure the workspace units are set to millimeters for length. (If you are using GoBilda or REV parts, you will be much happier if you work by default in metric.)

To do this, click on the "hamburger" menu (≡) in the upper left after the Onshape logo. You'll get something like this:



Click on Workspace units and you'll get something like this:



You could argue (and I would agree with you) that you should change all of the units to be metric. But I typically only change the other defaults when I need to use them since I don't use them in many documents. (Personally, I change my account to have millimeter as the default length unit under preference)

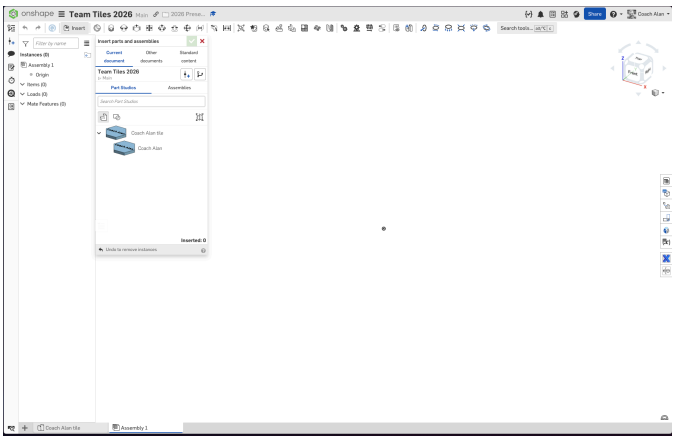
You'll notice that at the bottom of the screen you see two tabs - Part Studio 1 and Assembly 1. These are the two you get by default. First, click on the "Part Studio 1" and rename it "Coach Tile".

2.3. Create your individual tile

Since this is the same thing we'll have students do, go follow the steps for students in section 3.1 on page 15 and then come back here for the next step.

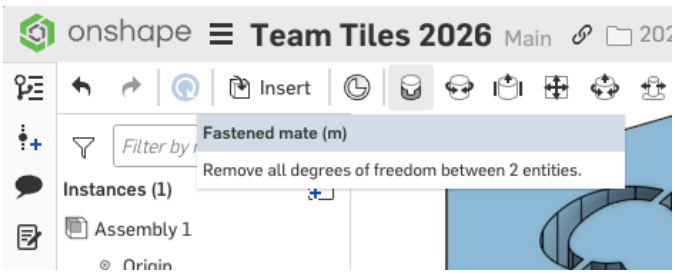
2.4. Put it in the assembly

Switch to the tab called “Assembly 1”. Now you want to select the word “Insert” on the toolbar (or press I) and you’ll select the part you made in the part studio.

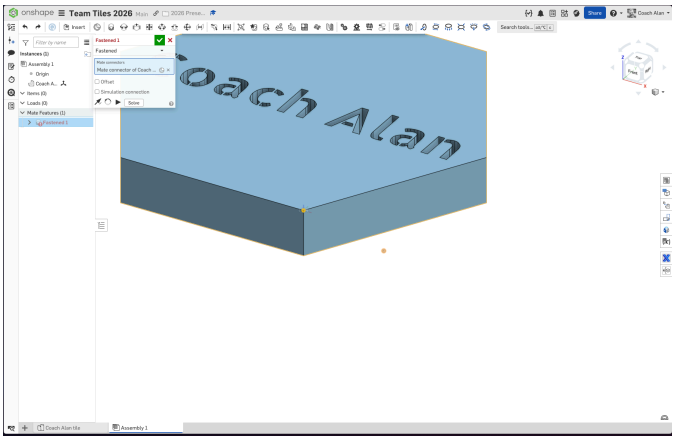


Yes, you could put multiple copies in but for now just put in one.

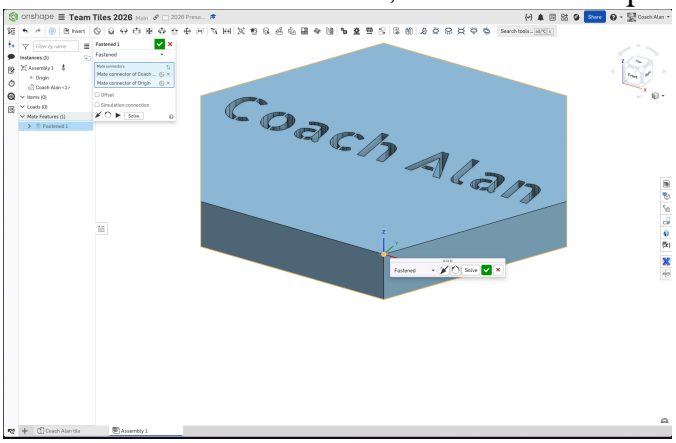
We’ll select a fastened mate (this means that the two things you are connecting can’t move at all with respect to each other.) . This is immediately to the right of Insert. (or you can press m)



Then we’ll select the lower right hand corner of our part (the top part) and the origin.



After we have selected both, we'll notice the part snaps to make that true.



Now we can press the green check. you'll see that there is no longer a symbol next to Coach Alan part saying that it has degrees of freedom.

Now we can close this (you don't have to save, like Google Docs it is always saved) and move on to our students making their own name tiles and attaching them.

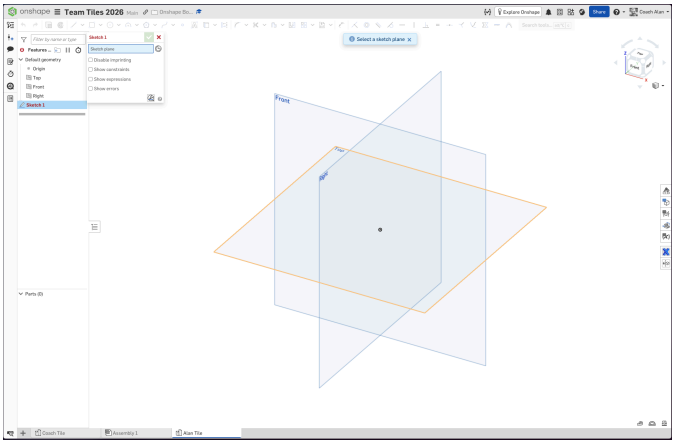
3. For students

If your coach hasn't done chapter 2, stop and get them to do that. If you don't think that it is likely they will ever do it, then play paper rock scissors on your team and get the winner to go through chapter 2.

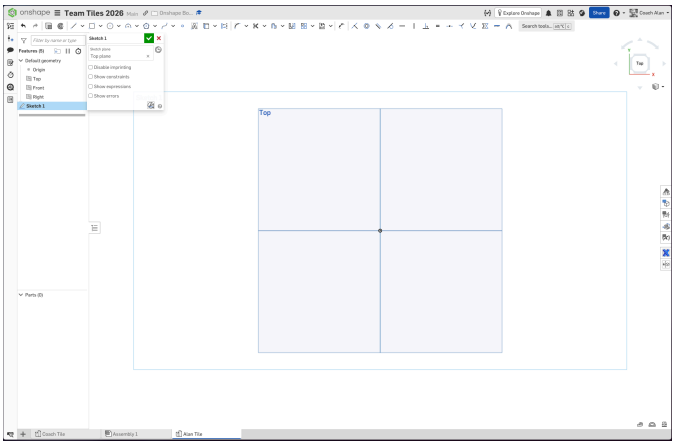
Open the Team Tiles document for this season that your coach created. In the lower right, select the + and then Create Part Studio. It will be highlighted and will be "Part Studio 1" (if all of the other Part Studios have been renamed). Right click on it select rename, and name it "<your name> Tile" (where you replace <your name> with your name...) . While you don't have to rename the part studios, doing so will make it easier to find things later so it is a good habit to get into.

3.1. Making your team tile

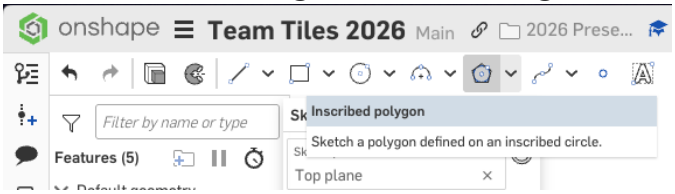
Select "Sketch" up on the top (or press Shift + S). It will then ask you to select a sketch plane (this is the plane you are drawing on). In this case, let's select the "Top" plane so that we are sketching it as if we were looking down on it.



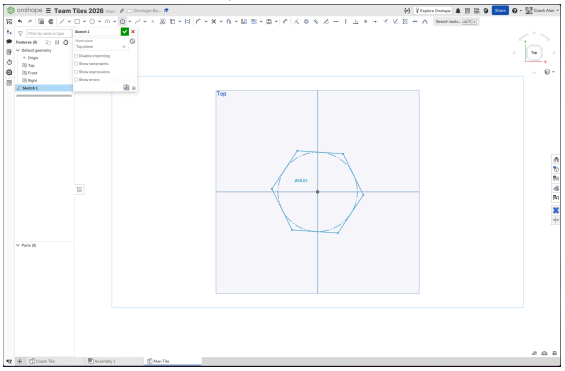
After this is done, press N on your keyboard. This will make it "normal" (perpendicular). I find that this makes it easier to work with as it now looks like I am working on a piece of paper instead of it being rotated in 3d space.



The next thing we are going to is to select an inscribed polygon. (Or a polygon that is defined with a circle on the inside.) . We do that by clicking on the picture of the polygon in the toolbar. (This is only visible when you are in a sketch. If you don't see the "Sketch" with a green check, you may have to double click on it in the feature list to get back to editing it.)

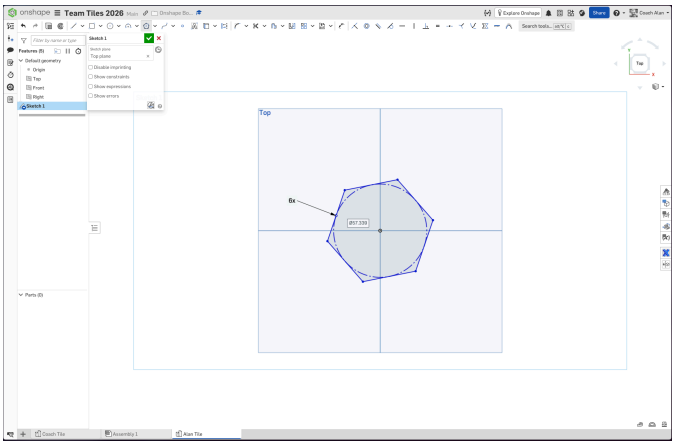


After we have clicked on this we will click on the origin (the circle with a dot in it in the middle)

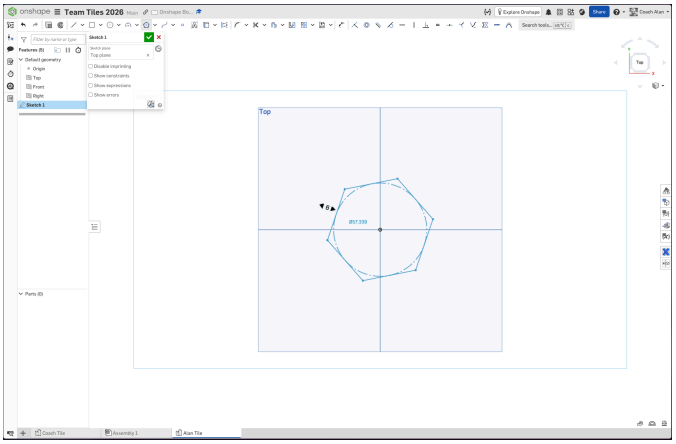


The next movement will change the size. It doesn't matter the size you make it because we will fix that in a minute.

After you click, the next thing it will ask is how many sides you want the polygon to be. The default is six and that is what we want, so we can just click to accept.

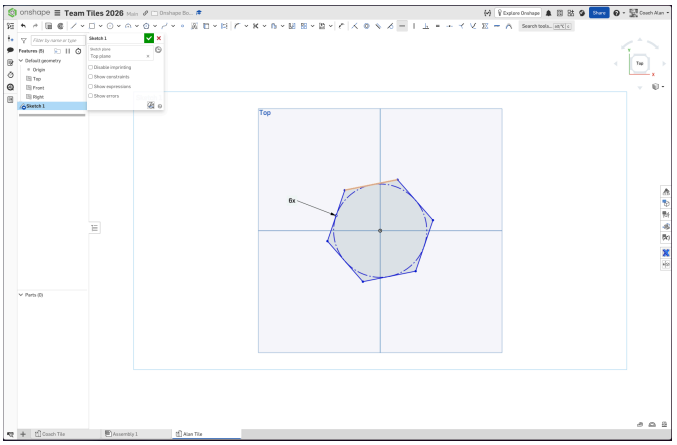


Then it will ask us to set the diameter of the circle. (I often mess up entering this here so I am going to show how to do it another way in a minute, but you can go ahead and enter in 70. This means that the circle (drawn with dashes and dots to indicate that it is for construction only) has a diameter of 70mm (or 7 cm).

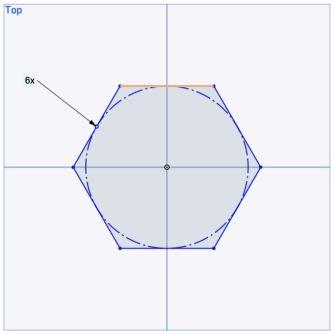


After you finish it, it will start making a second one. Since we only want one, press ESC.

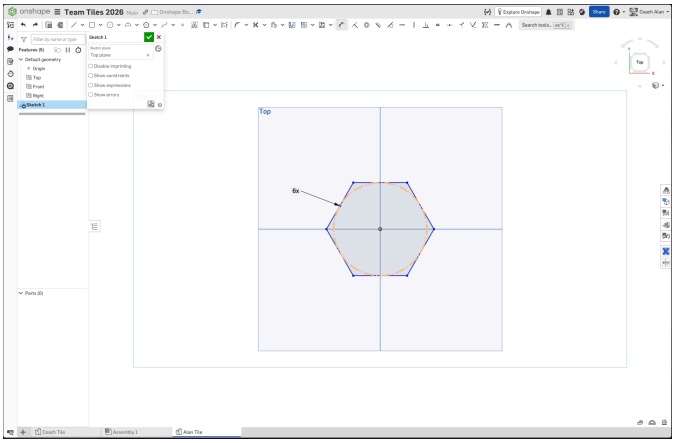
The next thing we are going to do is apply a "constraint". This means that we are saying something that must be true. For this, we are going to say that the top line must be horizontal. So we click on the horizontal line in the toolbar (or press H) and then on the top line of our hexagon.



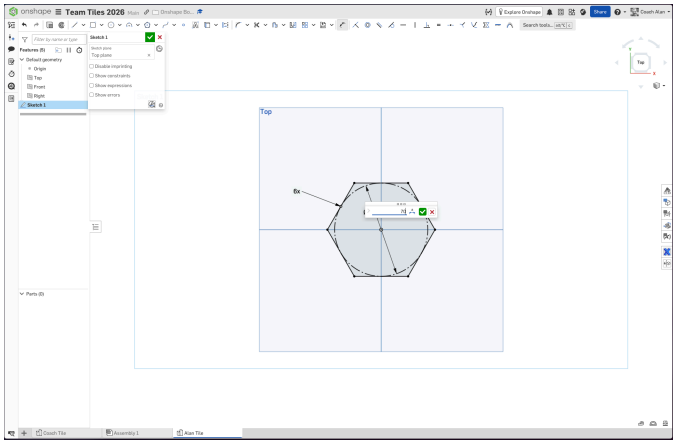
You should see that your hexagon now looks like:



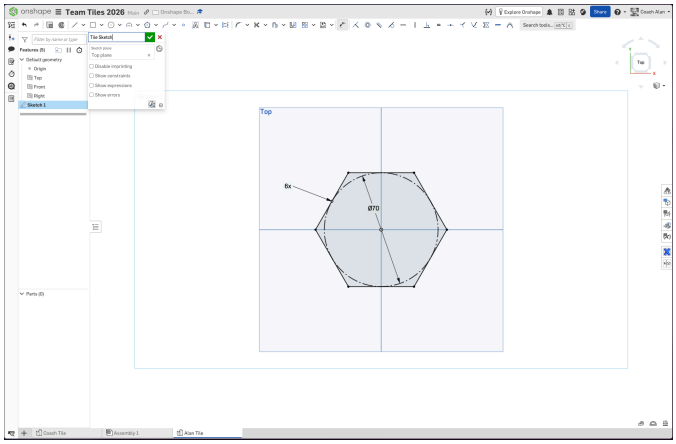
Now we are going to set the size of the circle. To do this, you can either select the dimension item or press D on your keyboard (for dimension) and then click on the circle.



You can then enter in the dimension as 70.

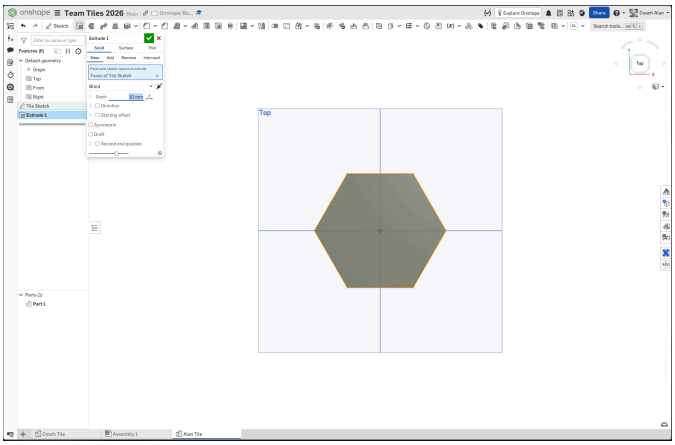


Then we can rename the sketch so we have some idea of what it is referring to and mark that the sketch is complete. I recommend something like "Tile Sketch".

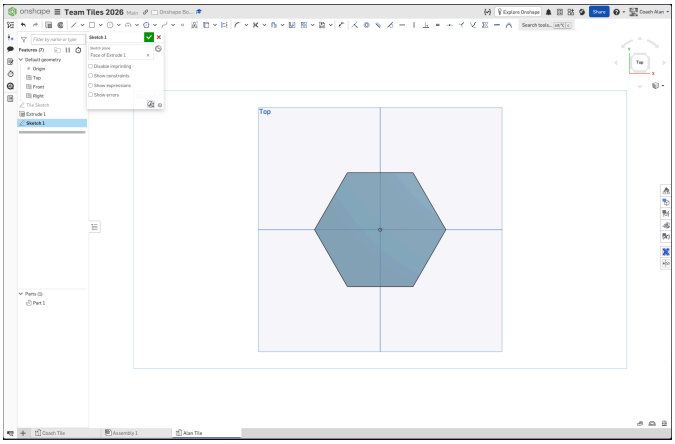


You can see here that the sketch lines are completely black. This is good because it means everything is defined and it can't change on you (more on that later). I renamed it to Tile Sketch and press the check button.

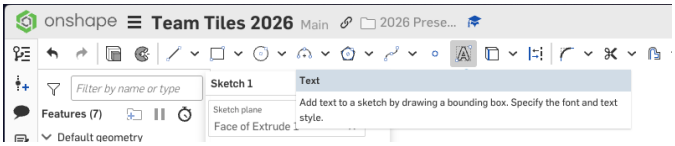
Now we are out of the sketch, and we will select extrude (the icon to the right of the word Sketch) in the toolbar (or press Shift+E). Extrude is like pushing a material through the sketch you just made and making it a certain size. Click on the sketch to say that is what you are extruding and make it a blind sketch of 10mm (1 cm). You'll see that now you have your first part down in the Parts list.



Now we need to make another sketch to have our name on it. (Yes, you could have combined this with the first sketch, but it is often helpful to have multiple small sketches.) So select sketch, and click on the top of your part.

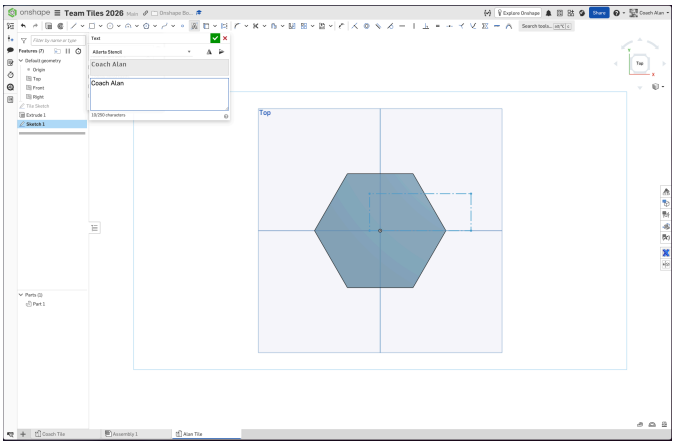


We now want to put text in our sketch. It is the icon that looks like the letter A in the toolbar.

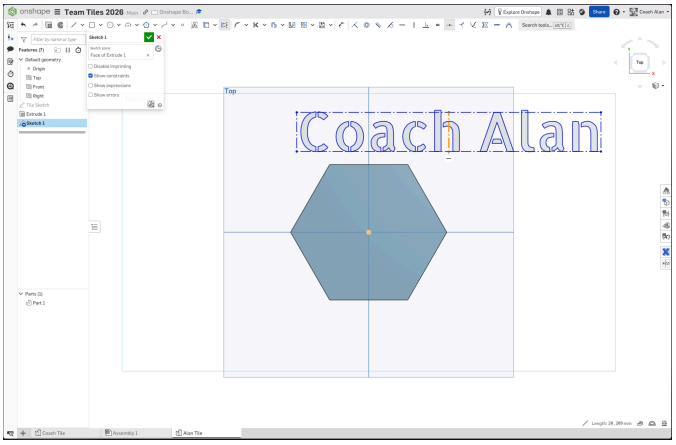


You then draw the bounding box. you can put it wherever and at whatever size because we are going to use constraints and dimensions to locate and size it.

After you have drawn the bounding box, type in the text and press the green check. (Obviously use your name instead of mine)

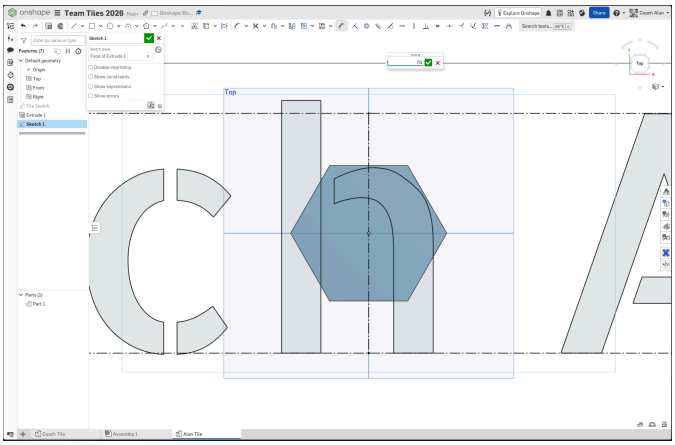


Now, click the midpoint constraint. (It is the one that looks like a line with a dot in the middle.) or you can press Shift+M. Click on the vertical construction line (construction lines are dashed lines) in the middle of the text and the origin.

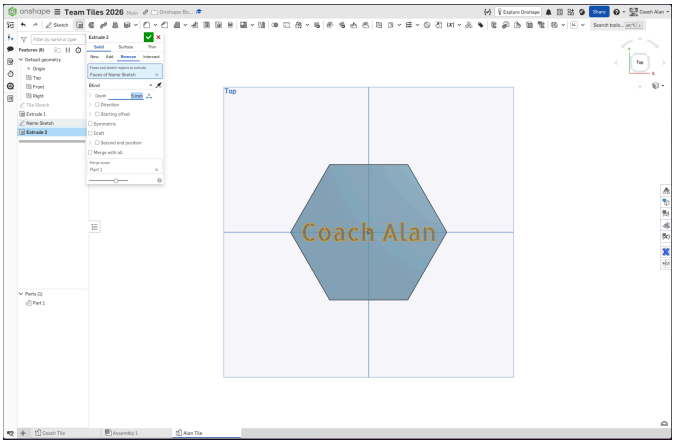


It will snap it to the origin.

Next, press d on your keyboard (for dimension) and the top line of the bounding box. Enter 70. (That is the radius of our circle, so we know it will fit.)



Now you can rename the sketch to be Name Sketch so you can find it. We extrude (Shift+E) just like we did before, but instead of "New", we will select "Remove". This will remove our sketch from the part we select. Notice we make the depth 5 so that it will just cut halfway through the part. Since we have a stencil font, it doesn't matter. But if we had any other font it would be important not to go all the way through or it would change our singular part into lots of little parts.

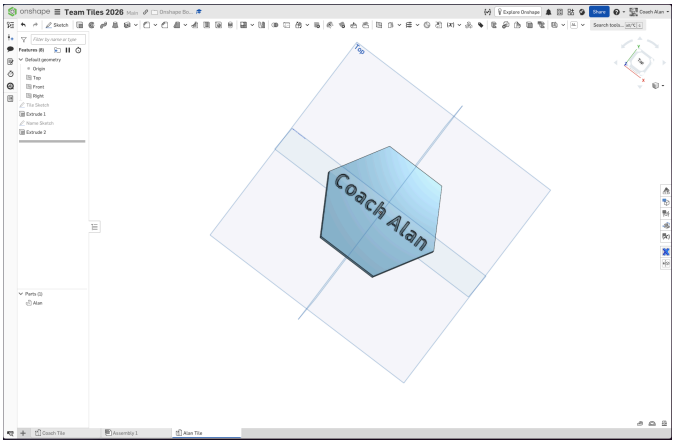


We can click on Part 1 and rename it to be our name. The reason why I do so much renaming is it makes things simple later. Instead of wondering what Part 1 is, it is obvious.

After you have renamed it, you can rotate by pressing the right mouse button and moving things around to see it from different angles. [1] (you can use the cube in the upper right hand to go to specific perspectives.) . You can also

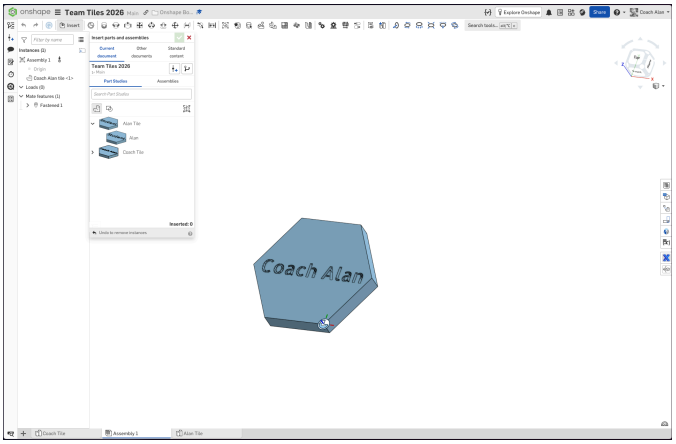
[1] I am convinced that more time has been spent in CAD rotating items as a fidget than actual work being done....

press P to show/hide the planes.

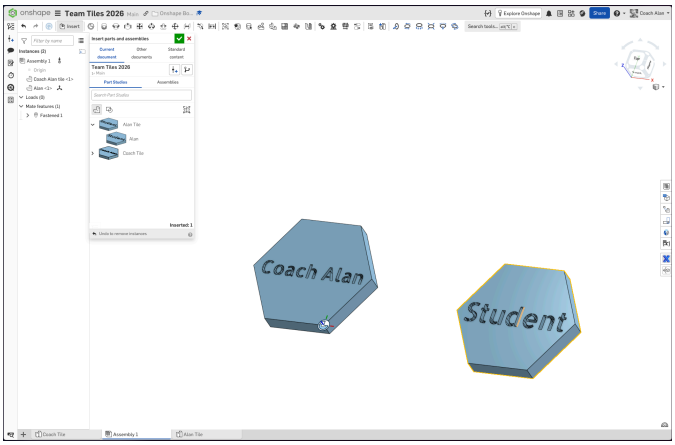


3.2. Connecting your team tile to the rest of your team

Go to the "Assembly 1" tab. Select Insert (or press I) . Expand under your Tile Part Studio and you'll see your part.



Select it and place it somewhere in the document that isn't overlapping other things (just to make it easier to connect)

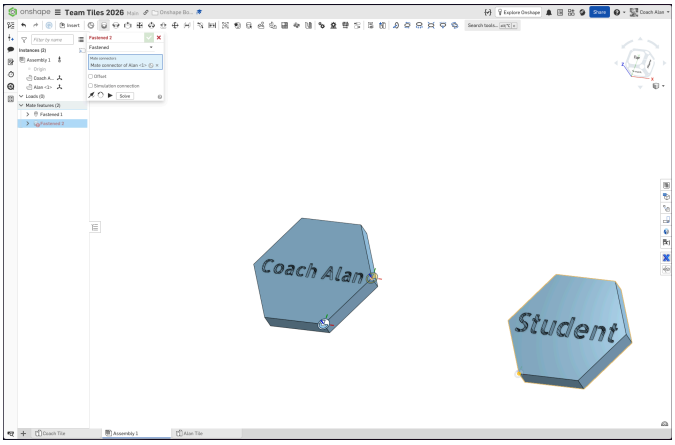


We'll accept this (by clicking on the green check mark) and then create a fastened mate (Press M)

3.2.1. Fastened Mates

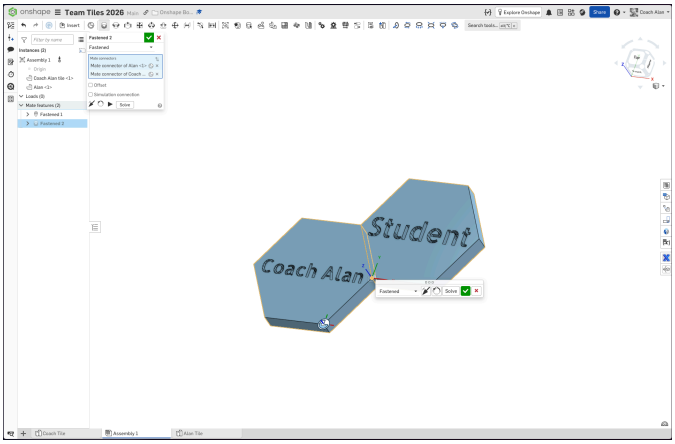
A fastened mate says that two parts are connected with no degrees of freedom (ie, they can't move or rotate in any way) . This is almost like saying they are glued together. It is the most common mate that we will use.

We need to select the point on each part where we want it fastened. Think about what two points you want touching. In the team tiles, you could pick the lower left corner of the "Student" tile and the right midpoint of the "Coach Alan" tile. For both of them, make sure the blue axis is pointing up. (This means you have selected the top part of the surface and they should both go together.)

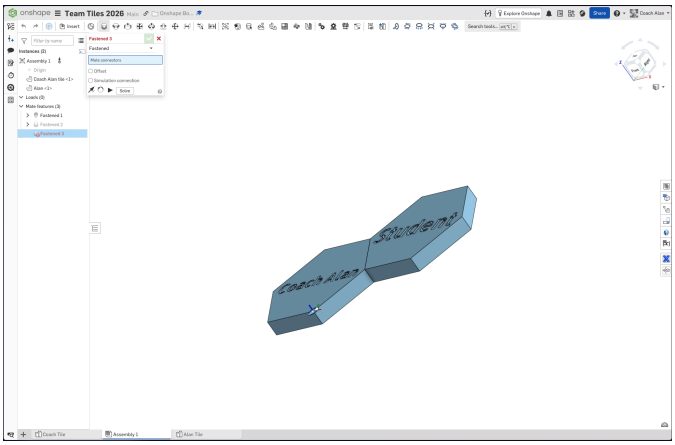


When you have the two points selected, you'll see the parts "snap" together. (This only shows the current mate. If you want to see what it looks like with

this mate and all others, then select “Solve”) . If it is what you want, press the green check mark.

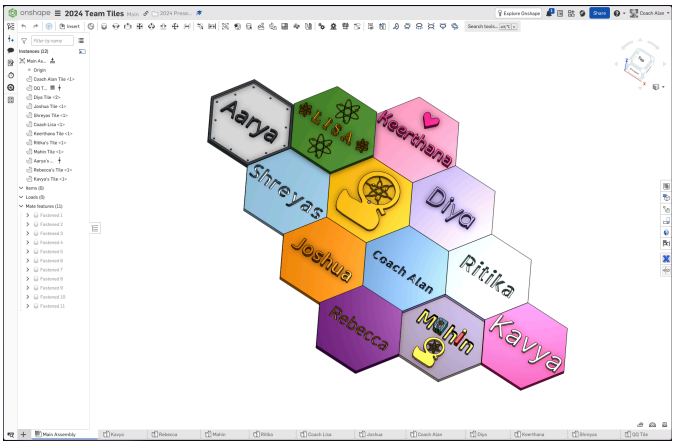


After it is fastened, you can rotate it around and it will stay connected.



As an example, here is a completed Team Tiles from one of the years that I coached. You can tell that many students decided to update their tiles to be a little fancy. One of the cool things is that if you update it on your part studio with just your tile that it will automatically update the assembly.

3. For students



While this was simple, I like starting each year with this because it makes sure that everyone has a working account and they can modify a team document.

Part II.

Modeling a simple Part

4. Spacers

We are going to start with the simplest useful thing that you can make in CAD and 3d print. This will help us go through the process with a simple thing that prints quickly so we can figure out where any problems are before we try more difficult things.

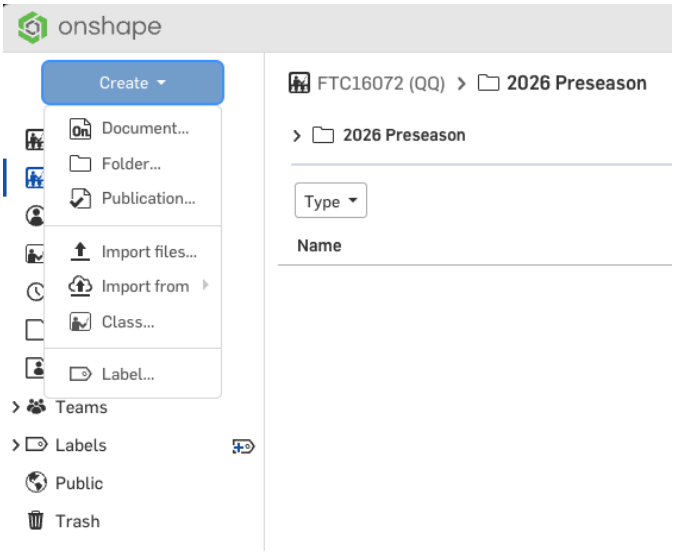
While you can buy spacers from GoBilda in a variety of sizes and shapes, you'll likely never have all of them and they are easy to design and print. A spacer is simply something that goes over a screw (or a shaft) that does not keep the screw (or shaft) from turning.

4.1. A simple spacer

Let's say that our screw is a little too long and we want to make sure that it doesn't stick out too far on the other side. GoBilda screws are M4 (which means they are 4mm in diameter). Let's say that we are trying to have the head 5mm from where it was.

4.1.1. Make in CAD

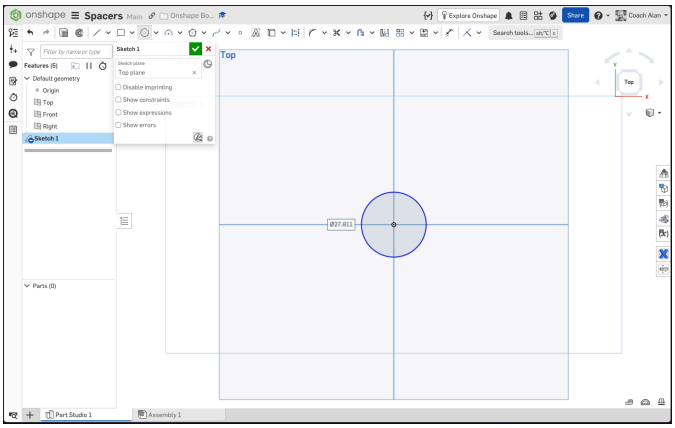
Let's go make a new document by selecting Create and Document. Call the document Spacers



Select “Sketch” up on the top (or press Shift + S). It will then ask you to select a sketch plane (this is the plane you are drawing on). In this case, let’s select the “Top” plane so that we are sketching it as if we were looking down on it.

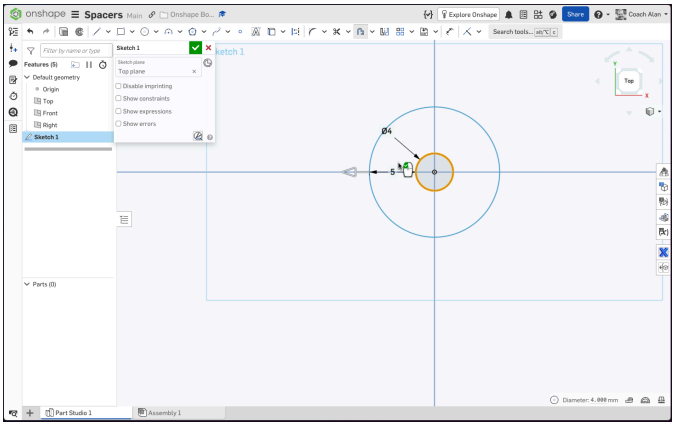
After this is done, press N on your keyboard. This will make it “normal” (perpendicular). I find that this makes it easier to work with as it now looks like I am working on a piece of paper instead of it being rotated in 3d space.

Select center point circle (or press C), then click on the origin, move it some distance and click (it doesn’t matter how far because we are about to set it.) . You’ll get something that looks like:

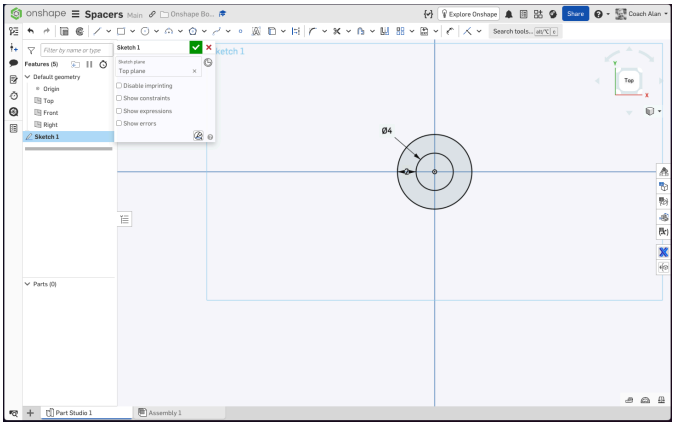


Type in “4” because that is what we want our inner diameter (often called ID) to be. If this is too small to see, you can press f to scale your window to fit just what you have (including the planes.)

While we could just make a second circle of the size of our outer diameter (often called OD), we are going to use offset instead because it is a very useful thing to know. Either select Offset (or press O) and then select our circle.



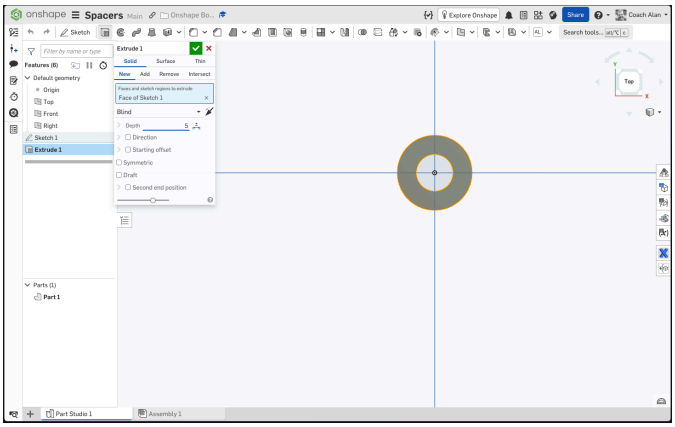
Accept by pressing the left mouse button. It will then bring up a box for you to type in the offset. While normally I like a thickness of 3mm, that feels overkill for a spacer whose ID is only 4mm, so I put in 2mm here. Your sketch should now look like:



Make sure yours has black circles here and not blue ones. The black lines means it is fully defined. The inner circle is fully defined because we said it started at the origin and have given it a diameter of 4mm. (The symbol ∅ means diameter). The outer circle is fully defined because it is an offset of 2mm from the inner shape (which is fully defined)

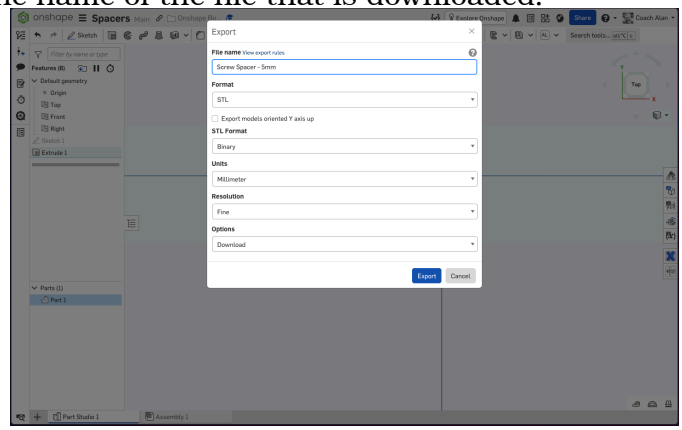
One cool thing about using offset is that if you decide to change the inner diameter, the outer diameter will automatically change to have the same offset.

Now click on the section between the two circles and select Extrude (or press Shift+E) and make the depth 5mm and press the green checkmark.



4.1.2. 3D printing our part

Right Click on “Part 1” and select Export... (This will be in the lower left where the set of Parts are.) I change the File name here so that my computer isn’t littered with Part Studio 1 - Part 1 that I can’t tell what it is. In this case, I’ll rename to “Screw Spacer - 5mm”. It is important to know that this only affects the name of the file that is downloaded.

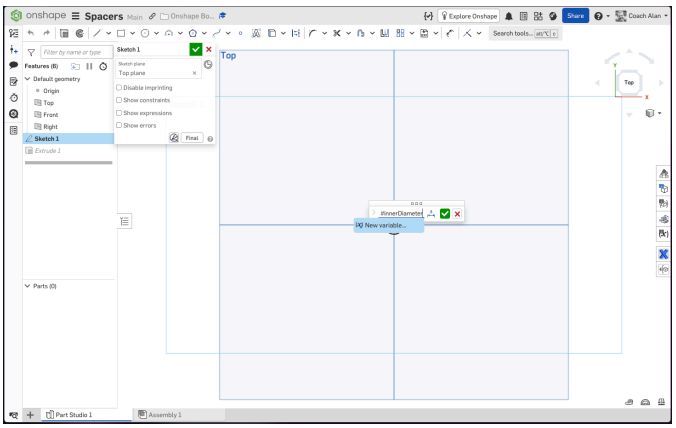


Press Export and it will download to the downloads directory on your computer. You can then bring it into your slicer. (This is 3d printer dependent). Make sure the bottom surface is flat on the bed so that it will print vertically.

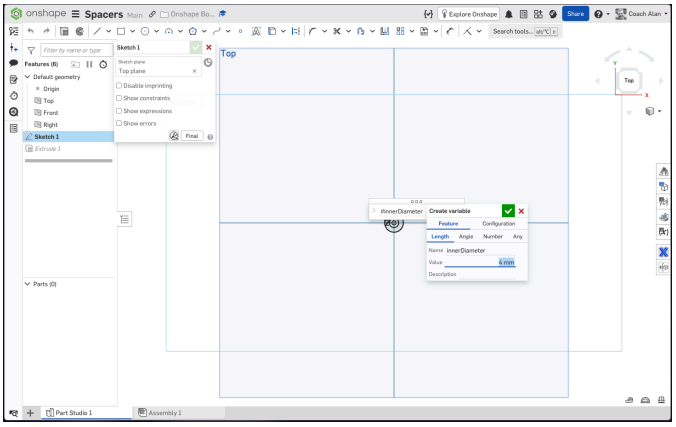
4.2. Using variables

While this is pretty cool, it is going to get old doing this over and over again for every spacer that we need. It turns out that we can do better. Let’s right click

on “Sketch 1” (we can rename sketches and if we have multiple ones we should) and select “Edit”. Click on where it says ∅4 and type in “#innerDiameter” . The # character means use a variable. (We can either choose from existing variables or we can create new ones here. But since we don’t have any yet, we will have to create a new one.) . Either select the green checkmark or Press Enter.

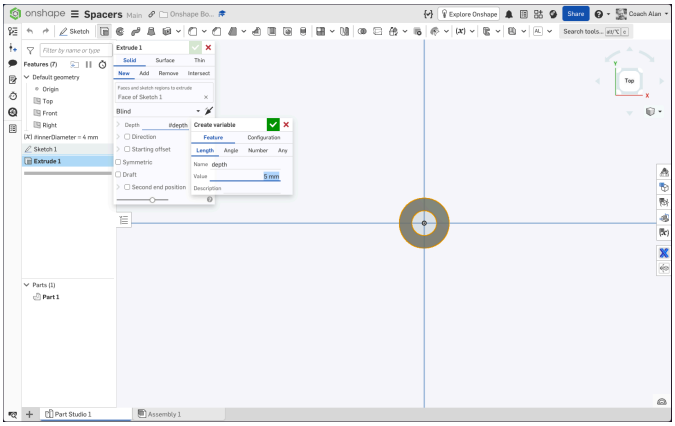


It will then ask us to create a variable. (We will have it as a length and a feature. You can see about Configurations in Appendix C.) . Variables in Onshape have units attached to them and you don’t have to use the default units of the document.



When we press all of the green check marks, we’ll see that there is a gray fx next to the diameter showing that it is calculated. Click the green check on the sketch.

You can double click on “Extrude 1” and change to #depth. That will have us create the variable.



Now you'll see two variables in the "feature list". They are evaluated from top to bottom, so your variable definition has to be before it is used. To change the value of a variable, you can double click it. So if you wanted to make a spacer that went over a 5mm D shaft for 10mm you would just change the innerDiameter to be 5mm and the depth to be 10mm and you can see the whole thing regenerate.

5. Simple Robot sign

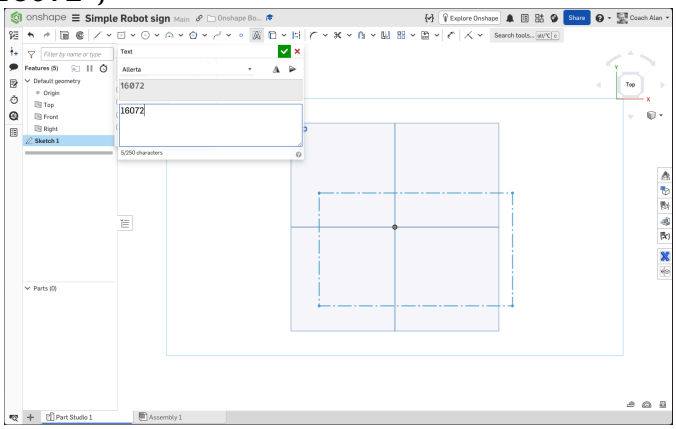
This may be obvious, but before you begin in CAD figure out what the problem is you are trying to solve. This is helpful to write down because it can be clarifying for what is important.

For our next part, we are going to make robot signs. From the competition manual (R401) we know that it has to be minimally 6.5 inches wide and 2.5 inches tall and the numeric part must be at 2.25 inches tall (+/- 0.5 inches). We need 2 red signs and 2 blue signs. So let's assume we have 3 different color filaments and we are going to make this with our 3D printer. To make this simple, we are going to attach it with velcro on the back. That way we can easily swap out the parts.

So let's create a new document called "Simple Robot sign" (if you don't remember how, reread chapter 4.)

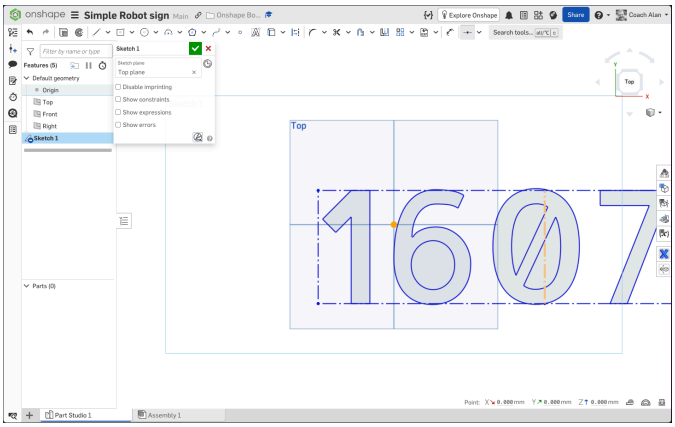
We will create a sketch on the top plane (again from chapter 4) and press N to make it so we are looking at it top down.

First, select Text and make the blue rectangle in the middle. (don't worry about size, we'll adjust later). Put in your team number. (The team I coach is #16072[1])

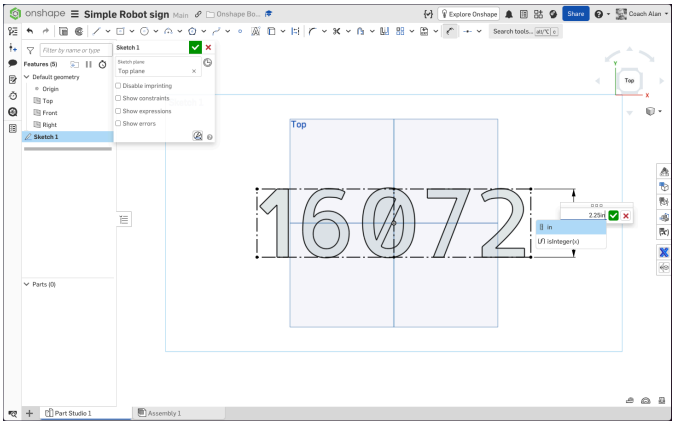


Now select Midpoint (or press Shift+M). Select the origin and then select the vertical construction line (construction lines are there just to help with construction and they are dashed) in the middle of the text.

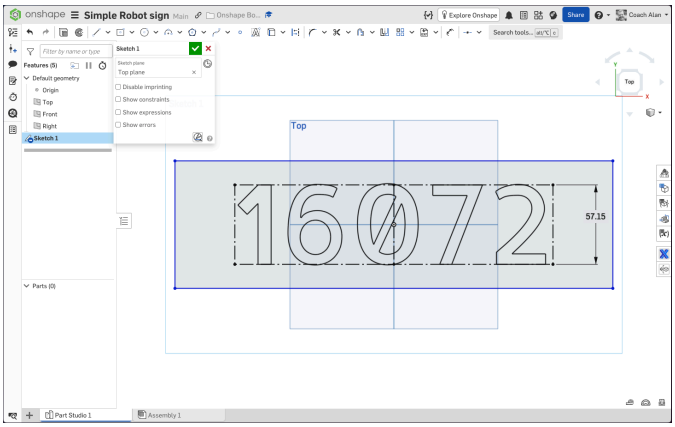
[1] Go Quantum Quacks!!



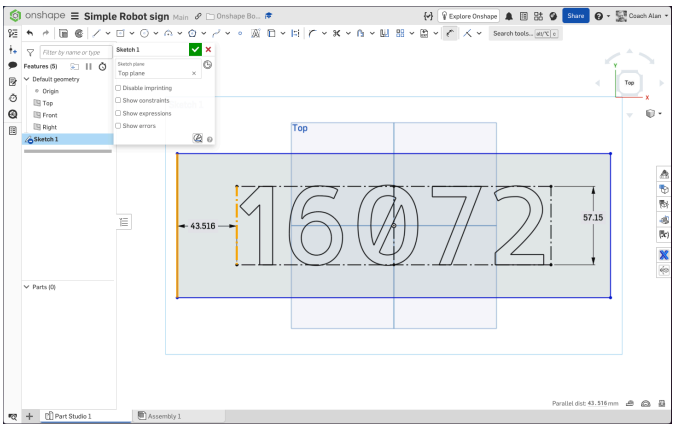
Let's give a dimension to the number since we know the height has to be at least 2.25 inches. Select Dimension (or press D) and click the right side and type in 2.25in.



Now add another rectangle (select center point rectangle or press R). Don't worry about its size. Jut make it bigger than the other rectangle. When it is drawn after clicking you can press escape to not need to enter in dimensions.

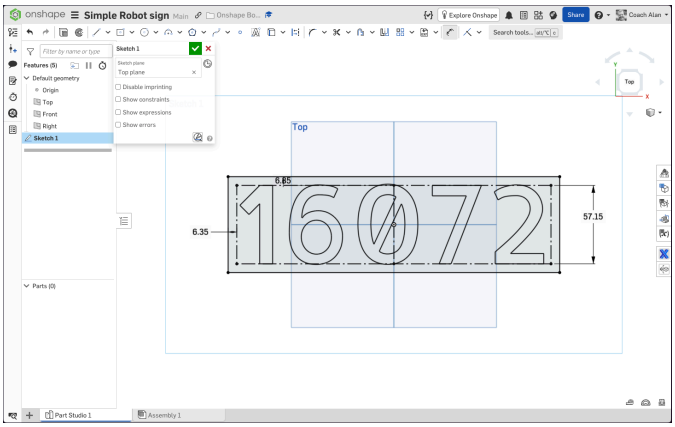


Now let's give a dimension between the left side and the text of .25 inches. (minimum spacing from the competition manual) . First we select Dimension (or press D) then we click on the left construction line of the text and the left line of the outer rectangle.

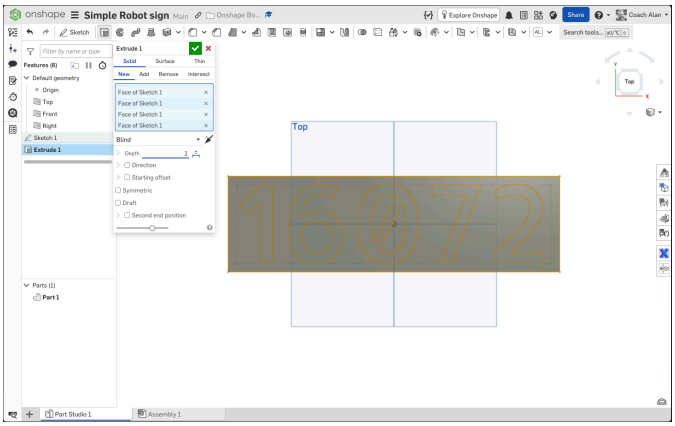


Now set this to be 0.25 inches. (You'll notice that it shows it as 6.35mm)

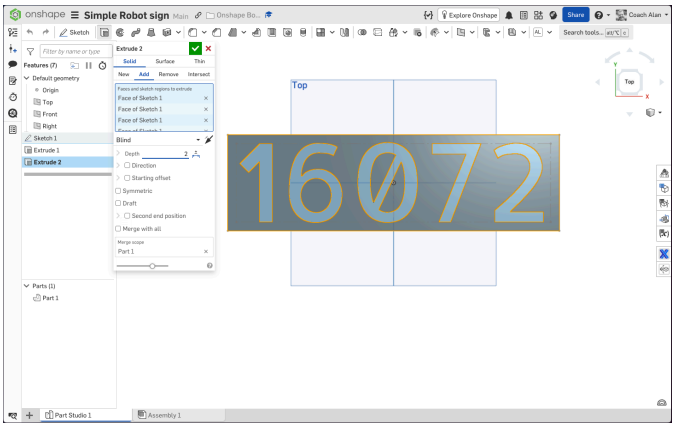
Now do the same thing between the top line of the text and the top line of the rectangle. Your sketch should now look like:



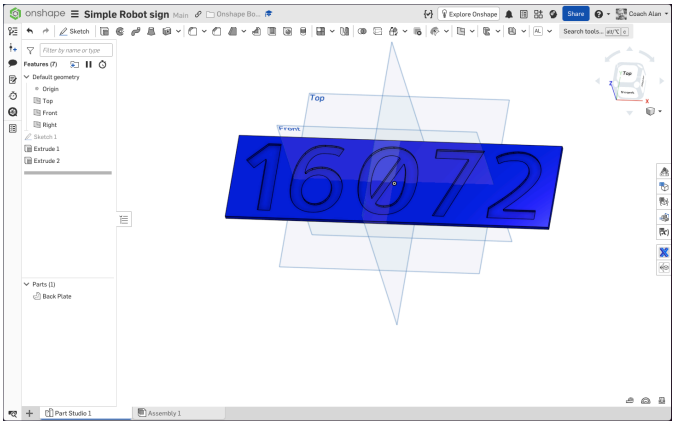
Now that we have our sketch, we'll go to making our parts. Select Extrude (or press Shift+E) and select everything so you have just a rectangle. Extrude it with a depth of 1mm.



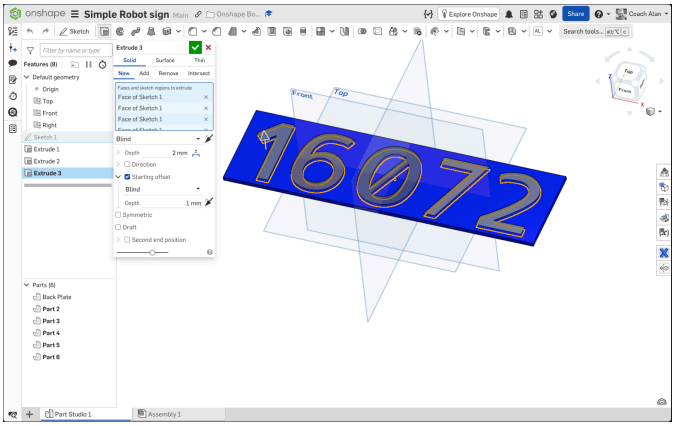
Now, go in and edit the sketch. Select Extrude (or press Shift+E) and select everything except the numbers. Extrude it (adding it to the existing part) with a depth of 3mm. (This will make the gap for the numbers 2mm because the first mm is in common with the back plate.)



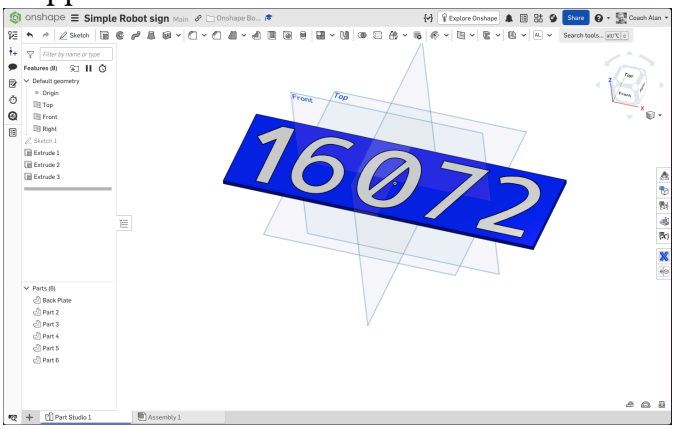
Now let's rename the Part 1 to be "Back Plate" and change the appearance (right click on the Part and Edit Appearance and select the blue color.)



Now let's go back to the sketch and select all the number parts and extrude them (Shift+E) 2mm (so they'll fit in) . We also do a starting offset of 1mm so that they'll look correct in here. Make sure you have selected New so it doesn't add it in with the existing part.



Now let's select part 2, part 3, part 4, part 5, part 6 and right click and set the appearance to be white.



Now you need to export each part like we did before in subsection 4.1.2 Make sure you print 2 back plates in blue, 2 in red, and 4 of each number. Now glue the numbers inside of the gap. Put velcro on the back of the plate and then on your robot where the plate should go.

Friendly tip - My team likes to point a color sensor at the back of the number plate so the robot knows whether it should run the red autonomous or blue autonomous. You're welcome....

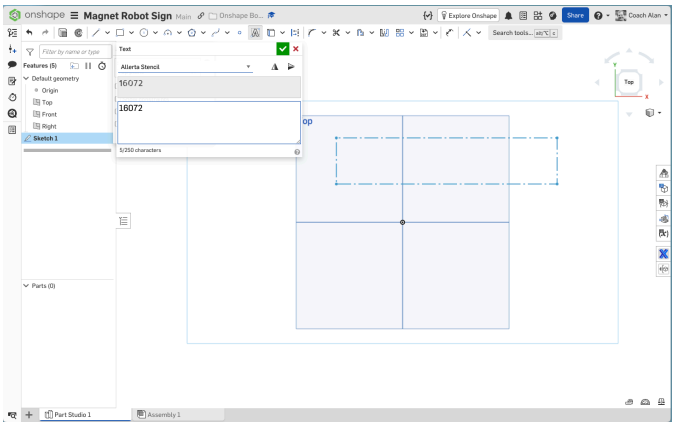
6. Magnet Robot sign

Ok, so we ended up with robot signs and we can play the game. But what if we wanted to make robot signs that screwed on and had magnets in them so that we just changed out the background color.

To do this, we start out by thinking about the design. What we want is 2 pieces. The first (backstop) that is all white and has a place that can screw in and ideally the holes are flush with the surface so the background.[1] The second is the background color that needs a place for magnets.

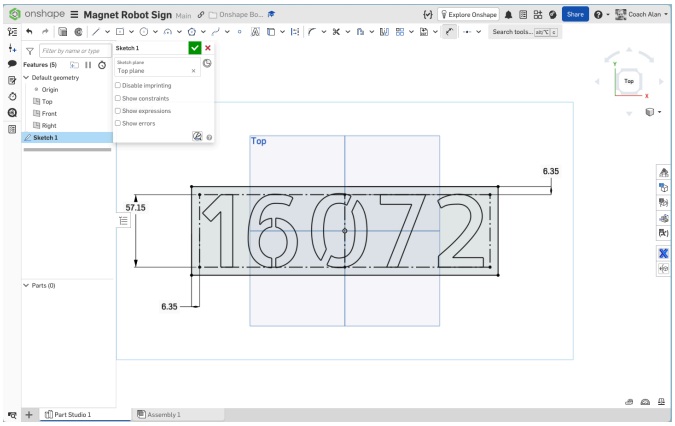
Let's start by making a new document called Magnet Robot Sign, and make a sketch on the top plane. Press N so you are looking at it like a piece of paper. (These are all from chapter 4.)

We are going to insert text (don't worry about where or the size, we'll set that later.) This time make sure the font you choose is Allerta Stencil. A stencil font has no enclosed parts. We want this so that we can have only a singular background to lift up.

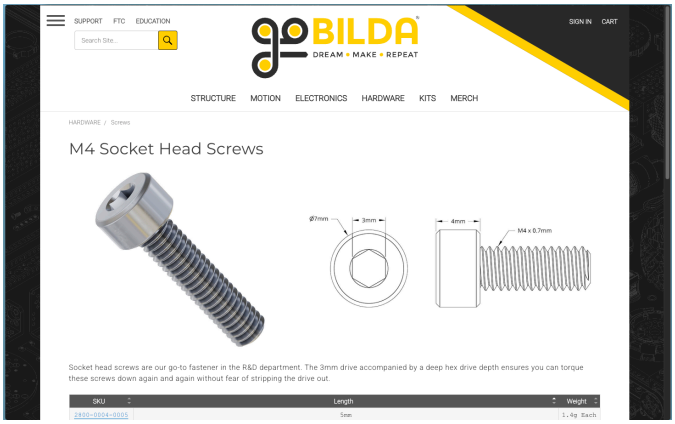


Follow the same steps from chapter 5 to get the rectangle surround it with the same border. You should now have something that looks like:

[1] Those of you that are familiar with woodworking might be thinking of countersinking . This is similar but since the screw heads are cylinders instead of angles, we call them counterboring

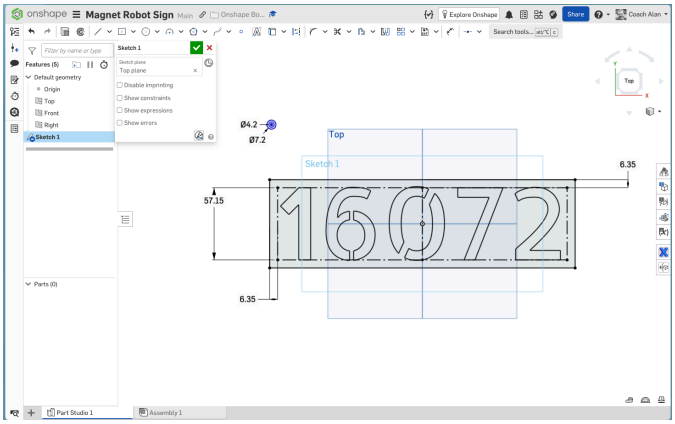


(I did move the measurements to make them easier to read. To do that, just click on them and drag them to where you want them.) . Now in this case, we want the bottom to be thick enough to hide the screw head. We go look at the data sheet for the screws we are using:

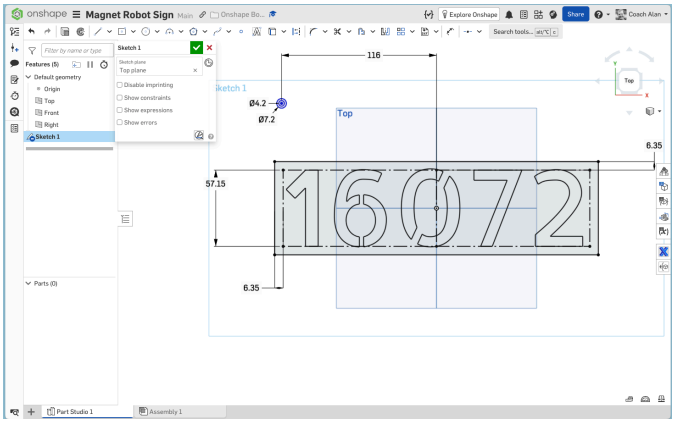


You'll notice that the OD (outer diameter) of the screw is 7mm so we need a hole that size for the head to rest in. The screw part is 4mm. Because 3d printers are not exact, we like to add a little extra to both of these so we have space. Typically .2mm is plenty.

So let's create two center point circles that share the same center point where the outer one is 7.2mm and the inner one is 4.2mm. Don't worry about exactly where to put it for now. Just make sure you stay clear of your sketch and the axis so that Onshape won't try to "help" you by creating additional constraints.

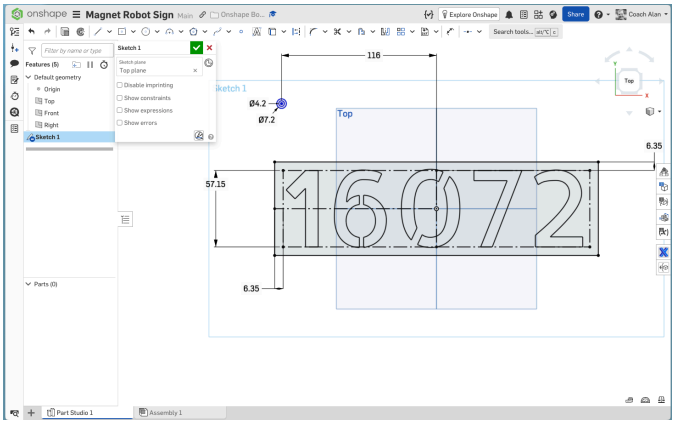


Now we are going to connect this to our robot, so we'll be happier if we do it on an 8mm pattern (GoBilda is on an 8mm pattern[2]) . To do this, let's first do a dimension from the center point to the midline and make sure it is a multiple of 4. (since we will be doing it symmetrically 4 is 1/2 of 8). We did 116 because that is close.

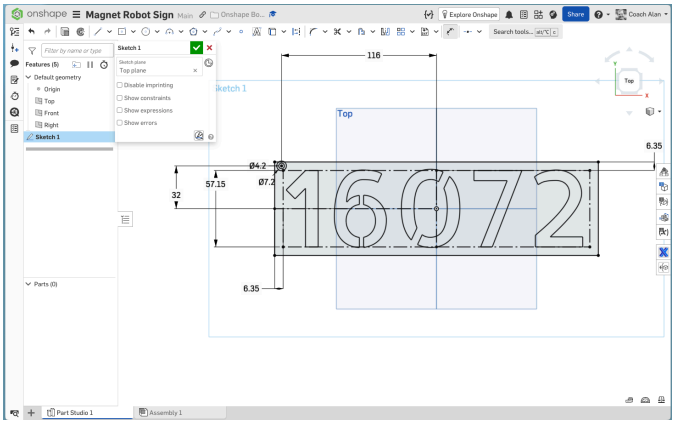


Now let's create a construction line(a construction line is a line that is just to help us making our sketch but is not part of our part. It is shown in dashes.) . To do this, select construction or press Q. Then select the line or press L. We are going to draw a line from the origin left to the outer box. Because you are connecting it to the outer box, it knows its length, because it is from the origin, it knows its starting position and Onshape guessed we wanted a horizontal line because we drew along the axis so it went ahead and gave us that constraint. It should look like this:

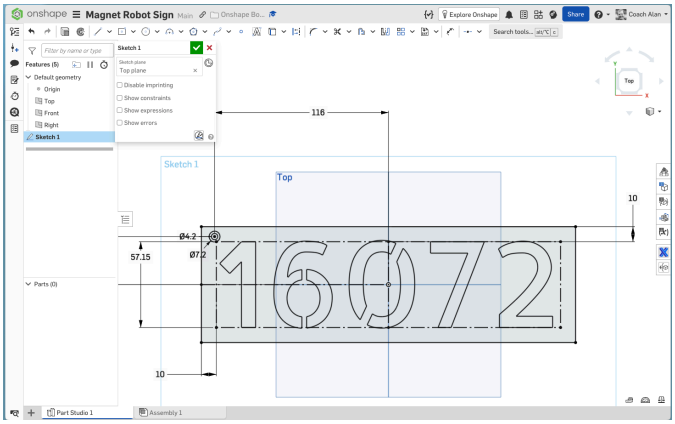
[2]So is LEGO, by the way....



Now we do a dimension of it from the center line. We'll do 32mm. You should get something like this:

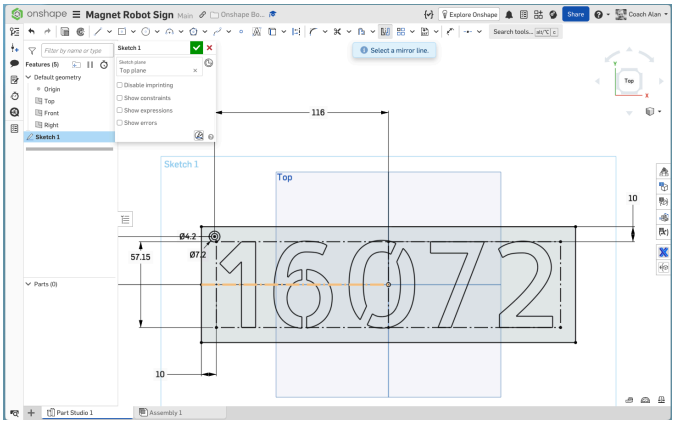


Hmm. There is a problem. Our hole goes over the edge of our sign. No big deal, let's just make the border bigger. Since we really want at least 2mm of material, let's change our borders from 6.35mm (or .25in) to 10mm.

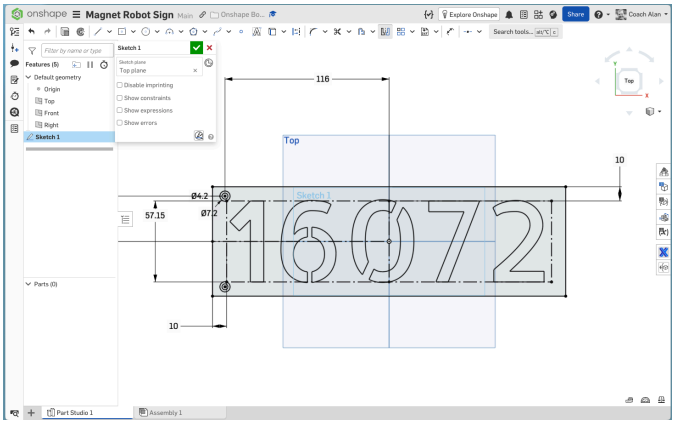


But we want four holes, and we only have one. This allows me to show one of my favorite tools. Click on the Mirror tool (looks like an “L”, a construction line, and a symmetric “L”).

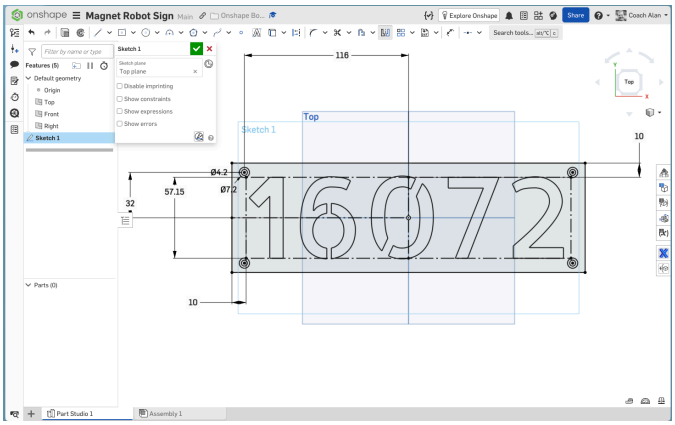
The first thing it wants is for us to select a mirror line. This is the line we will mirror around. Choose the horizontal construction line we made.



Then it will say to select the entities to mirror. Click on both the inner and outer circle. This will give you two.

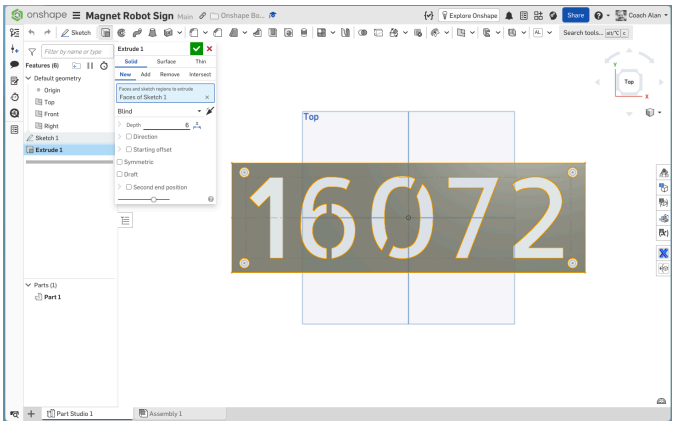


Now we do the same thing again, but this time select the vertical construction line and select all four circles. You should now get something like this:

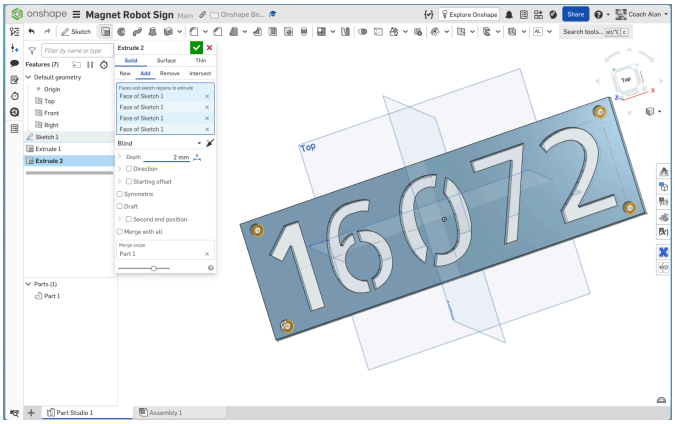


One of the really cool things is that if you change anything with the hole in the upper left (diameter, position, etc) then it will change it on all four. Lets say that you decide you want your screw hole to be 4.1mm instead, just change it in one place!!

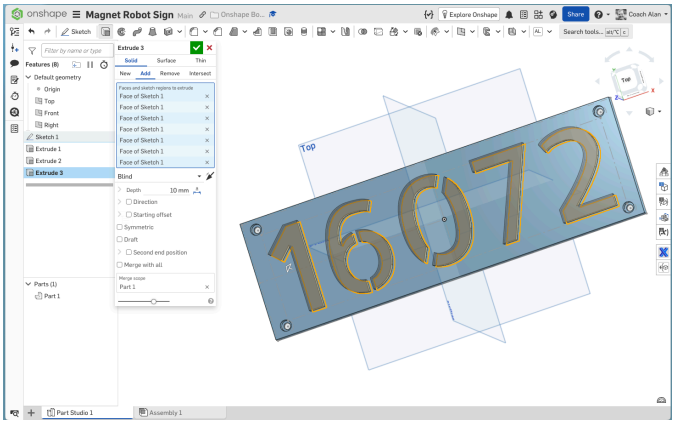
Ok, so now lets extrude (Shift+E) the entire plate except the holes by 6mm.



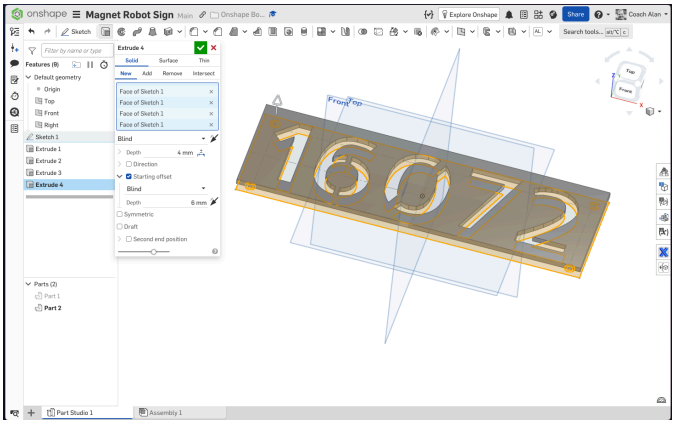
Now we'll add in the part we want for inside of the holes. Make the sketch visible (by clicking on the eye with the slash through it next to Sketch 1) and then select the outer ring of the hole. We'll extrude it by 2mm and select Add so it will make it part of Part 1. (Hint: Zooming in sometimes makes it easier to select small portions.)



Now we'll select the numbers. Since we want a place for magnets (the magnets we'll use are 5mm diameter and 2mm height) then that means we need at least 4mm. Since the plate is already 6mm, let's extrude the numbers by 10mm.

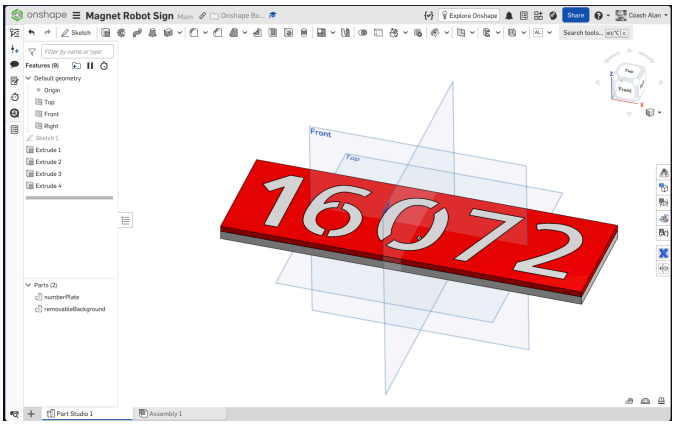


The easiest way to select the appropriate parts from the sketch is to hide "Part 1". Move your mouse over it and click the "eye" which will move it to an eye with a slash through it. Select the holes and the plate (but not the number) and extrude it for 4mm but with an offset of 6mm. (This will make the two parts look like they are correctly positioned when we show both)



Now, let's rename Part 2 to be removableBackground. (right click on Part 2 to get to Rename). Right click on it again and select edit appearance and choose red.

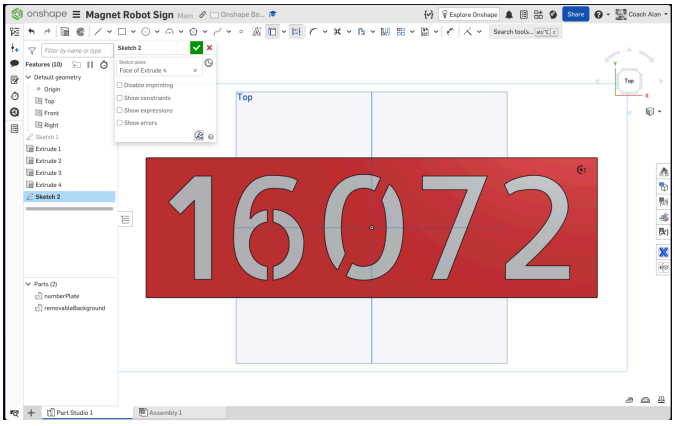
Now, we'll rename Part 1 to be numberPlate and change its appearance to be white. Make it visible and Hide Sketch 1and you should get something like this:



Now we need to make the holes for the magnets. Since the screws are made out of steel magnets will stick to them, so let's have the magnets over the holes.

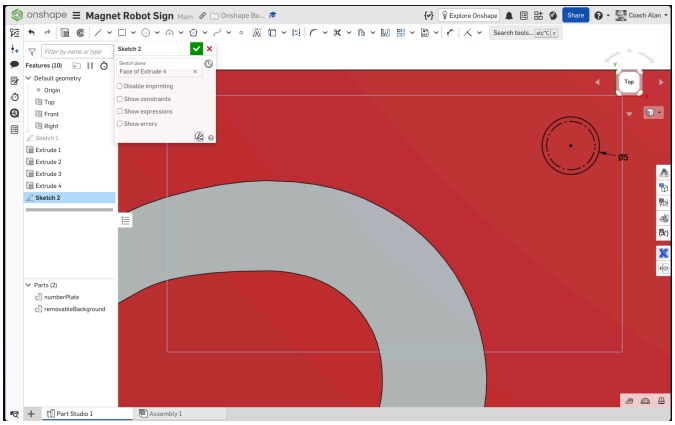
Start by creating a sketch but instead of clicking on "Top Plane" click on the top surface of the removableBackground. (Remember to press N so we can work on it top down)

The best way to reference something else is to use "Use". You can either select it from the top or press U. Press N again and you are now looking at the back of the number plate. Select the construction line (Q) and then click on the upper left hand hole. Press N again and you'll go back to seeing it from the top. You'll notice that the top left hole from the back side was the top right from the front side. This is fine, we'll just mirror opposite of how we did before. It should now look like this:

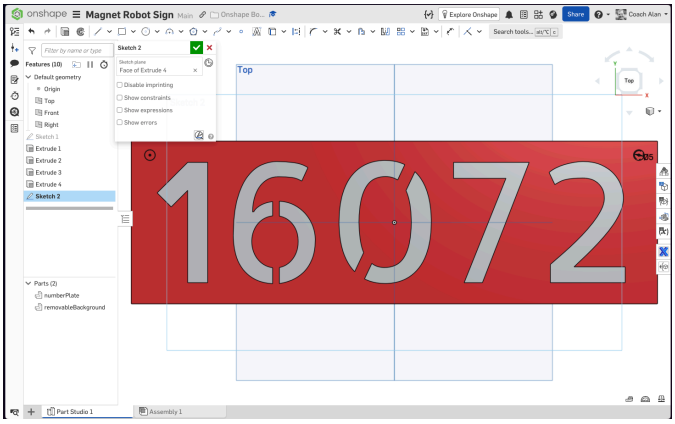


The first thing we need to do is create a 5mm circle. select the center point circle (press C) and click on the center dot. Then give it a size of 5mm. If it draws it dashed then select it and press Q to change it from a construction line to solid. This is one of those where it will likely be easier if you zoom in.

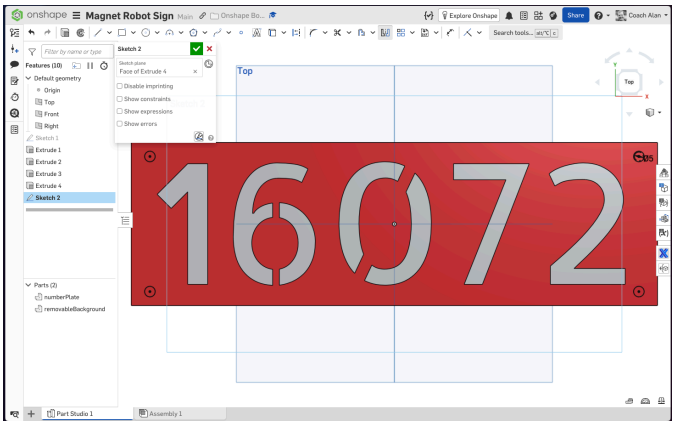
Here is what you should have.



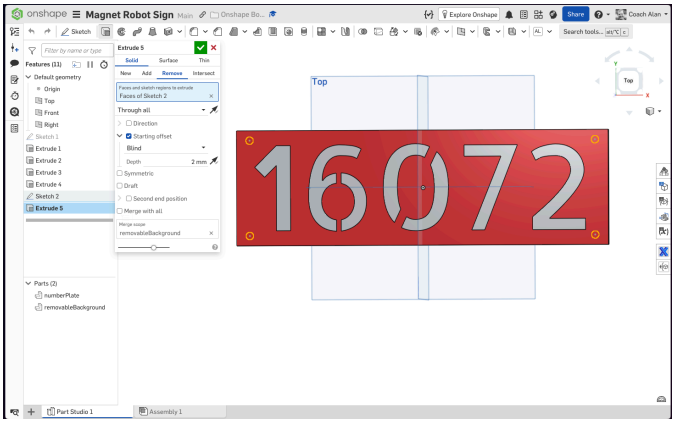
You can press F to zoom to fit. It turns out we can use the axis to mirror (we don't have to draw a construction line). So click on mirror, select the vertical axis and select the hole. (There is no need to mirror the construction circle)



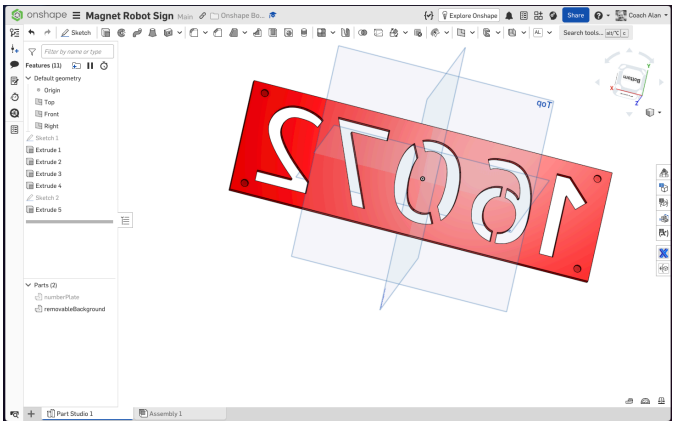
Now we'll mirror the top two holes onto the bottom. (Select Mirror, choose the horizontal axis and select the two circles.)



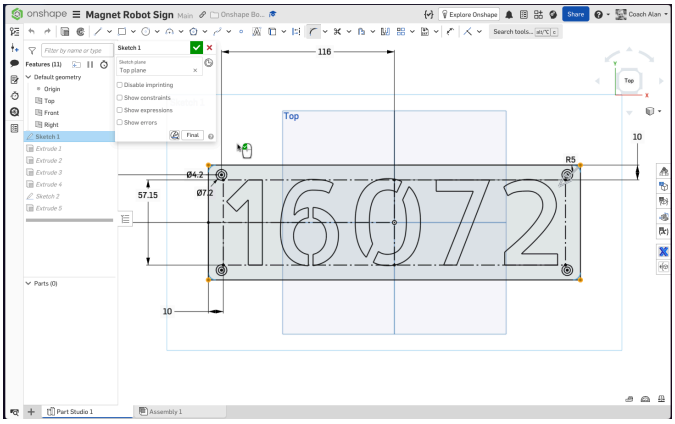
Select Extrude (Shift+E) and make sure the “Faces of Sketch 2” are selected. We’ll do a remove and this time instead of “blind” we’ll do a “Through All”. That way if we change the thickness of this part later we won’t have to change our extrude. We’ll do a starting offset of 2mm (so that the magnet isn’t visible) and select a merge scope of the “removableBackground” so the extrude knows which part we are removing from. If we did numberPlate here as well it would also make the hole go through it which isn’t what we want. Make sure the arrows here are going the correct direction.



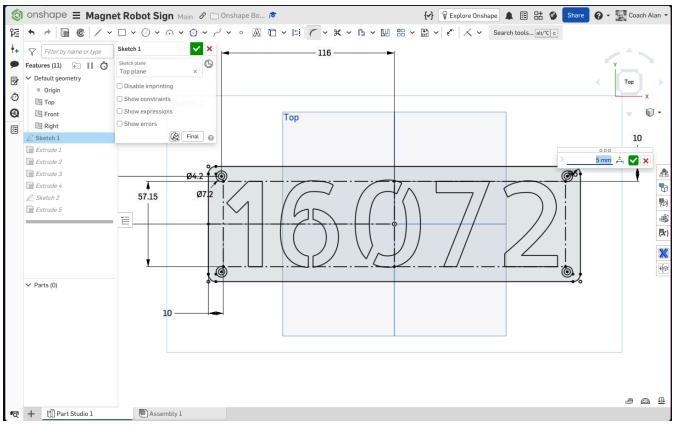
If you hide the numberPlate and turn it around you’ll see the places for our magnets.



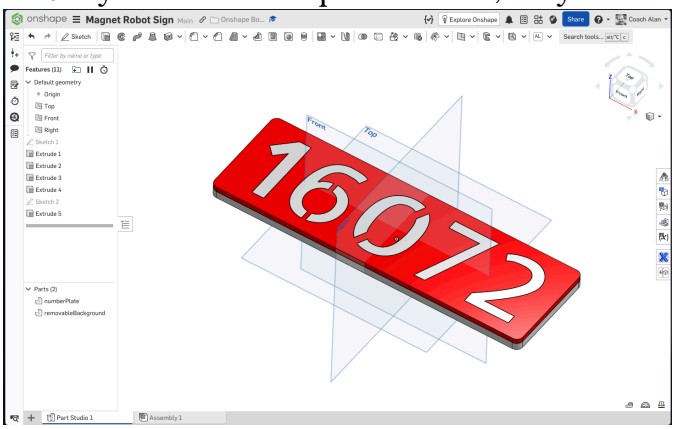
One of the cool things about everything based off of a sketch is that we can just make a change there and it affects everything. Right click on Sketch 1 and select Edit. Hit N twice so that it looks like we are seeing it from top down. Now select Fillet (Shift+F) and click on the four corners.



Right click on an empty space and it will ask you to set the radius of the fillet. The default of 5mm works well here so just accept it by pressing the green check.



Accept your sketch (Green check mark) and you'll see that it affected your parts. If you make both parts visible, they'll look like this:



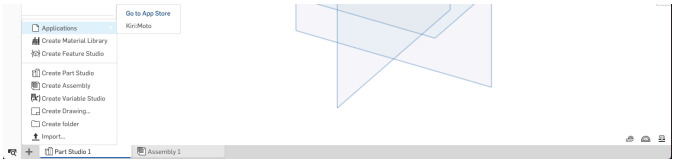
Not only does fillet look cool, it also makes your parts less likely to break because they don't have sharp edges.

Now you need to export each part like we did before in subsection 4.1.2 Make sure you print 2 back plates in white, 2 removableBackgrounds in red, and 2 in blue. Now glue the magnets into the holes on the removableBackgrounds (you'll need 16 magnets in total. You can easily get 5mm x 2mm magnets on Amazon)

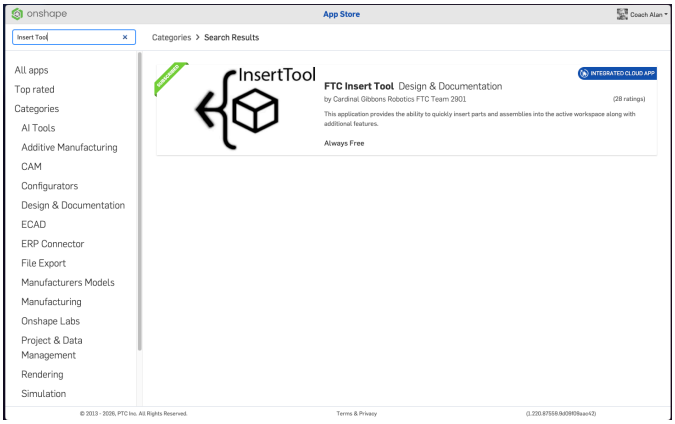
7. Cover end of U-Channel

Some years there are game elements on the field that could get stuck in your robot. (The samples in Into the Deep (2024-2025) are a great example.) . Since there was a possession limit, getting these pieces stuck was “no good”. So let’s create something that just seals the empty front of our U-channel and is slanted so that pieces get pushed to the side (like a cow catcher on a train).

Create a new document called “U-Channel cover”. Next we want to make sure you have the Insert Tool[1]. Press on the + in the lower left, then Applications then Go to App Store.



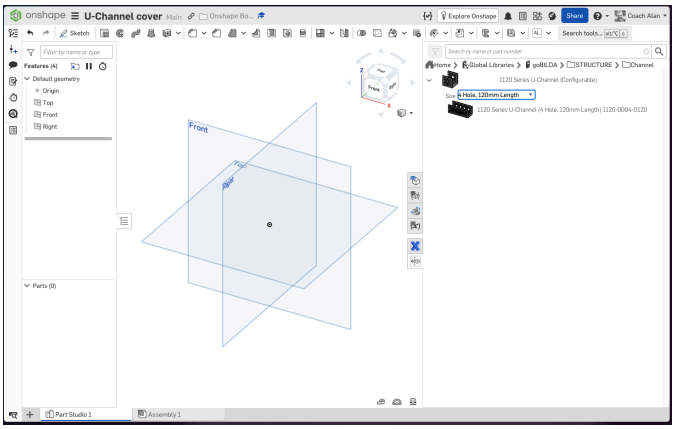
Now in the app store, type in Insert Tool and press Enter.



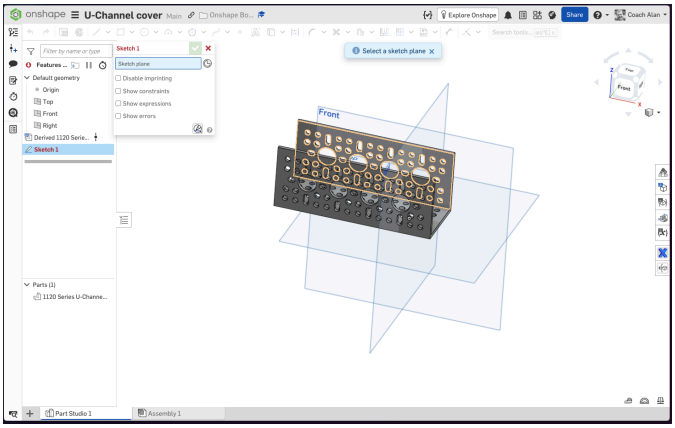
If yours doesn’t say “Subscribed” then click on it and subscribe.

Click on the Insert tool, Then Home -> Global Libraries -> goBILDA -> STRUCTURE -> Channel. Select the 1120 Series U-Channel (Configurable) and change its size to 4 Hole. It should now look like this:

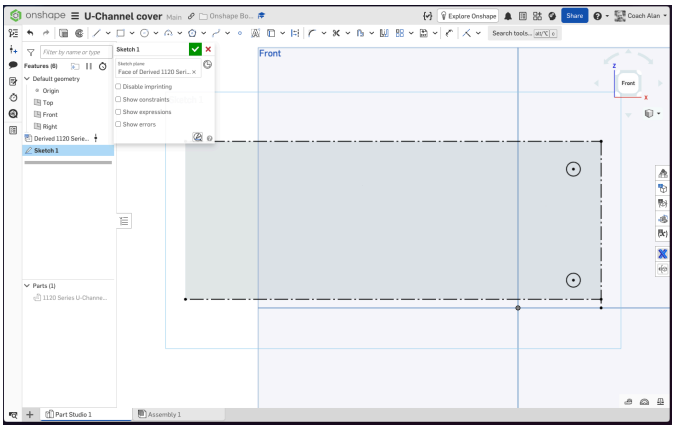
[1] Provided by Cardinal Gibbons Robotics FTC Team 2901



When we click on the channel, it will “derive” into our part studio. We can then use it in our sketches. (If you are in an assembly, then you can insert it just like your own custom parts. We’ll do that starting in Part III.) Click on the insert tool again to hide it and now lets create a sketch on the inner wall of the U-Channel. (Shift+S)

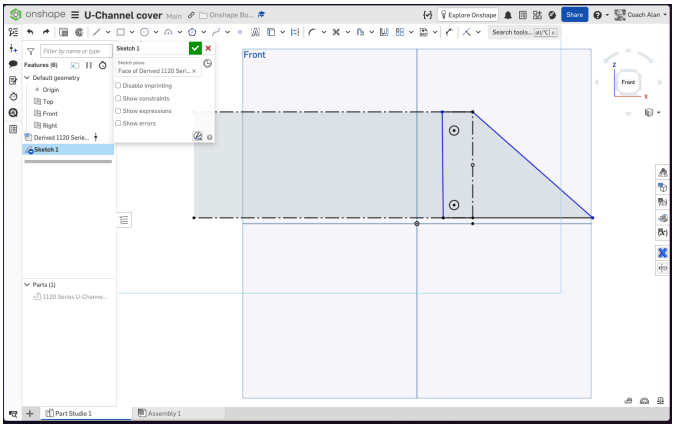


Much like the last chapter, we are going to “use” the holes we want it to go through. So select “use” (U) and click on the two holes. It will help us to also have construction lines showing the top bottom and edge of the U-channel so select “construction line” (Q) and click on the top, bottom, and right edge of the u-channel. Now go hide the derived part (select under parts and click on the eye). Your sketch should now look like:

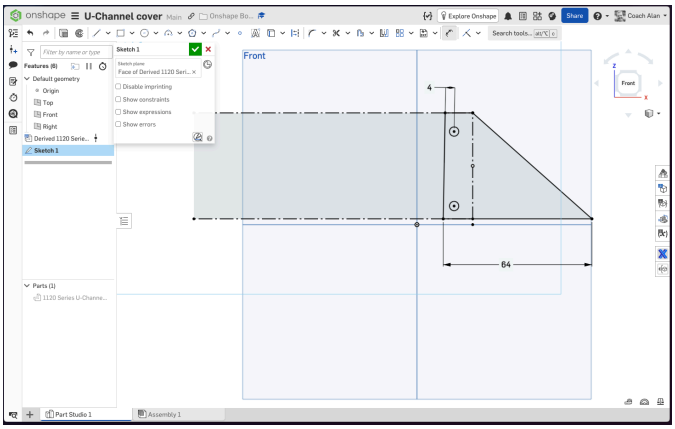


If you have a gap between the gray and your bottom line, it probably means that you selected the outside of the material inside of the edge on the inside.

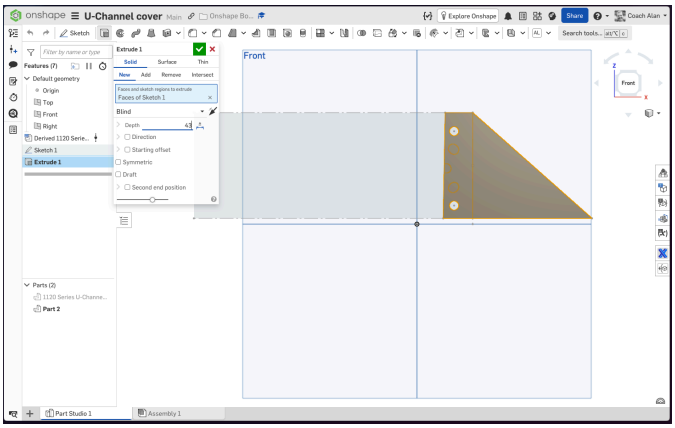
Now use the Line tool (L) to draw four lines.



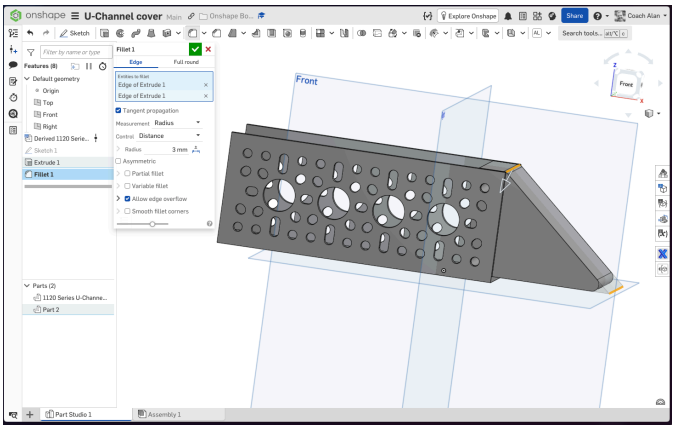
Now select Dimension (D) and make the vertical line 4mm away from the center of the hole. (I chose 4mm because there are 8mm between holes on the Gobilda pattern so it is half. But you could choose a bigger number if you felt like there wasn't enough material here.) Then make the length of the long line 64mm. Your sketch should now look like this (I moved the 4mm outside of the arrows so I could read it.)



Now let's Extrude (Shift+E). We are going to make the depth 43mm. (You can look that up on GoBilda's website to see that is the empty amount of space between walls in U-Channel.) and accept it.



You can show the U-Channel again and it is pretty obvious we have some sharp edges on our part that are exposed. This time instead of using a sketch fillet we'll just use a normal fillet. Choose the fillet item (press Shift+F) and then click on the two sharp edges. (You can do all edges, but I like to not do them on the surface I am 3d printing on because then you have some unsupported area on your print.)



To 3d print this part, just export Part 2. (Yes, you probably want to rename it.)

This is one of three ways to connect a part. In it we make a hole all the way through the part and we need a long screw to go through it. This works and is often a pretty easy way to do it. However there are two other ways.

1. Make the part slightly smaller than a screw and have the screw “bite” into the plastic. This can be a win for easy, but be aware that it limits how many times you can remove and reinsert the part before the screw doesn’t hold anymore. This would involve changing the size of the hole from 4.2 to perhaps 3.8mm for the diameter.

2. Use brass inserts. If you aren’t familiar with brass inserts, they are very cool. They allow you to put a set of screw threads into your 3d printed part. This allows you to screw into it just as if it was tapped. The ones we typically use (M4x4), we set a diameter hole of 5.6mm and if it isn’t a through hole we make it 5mm deep. Then you put the brass insert in the hole with a soldering iron (heated) on the inside and press on the brass insert until it sinks into the part.

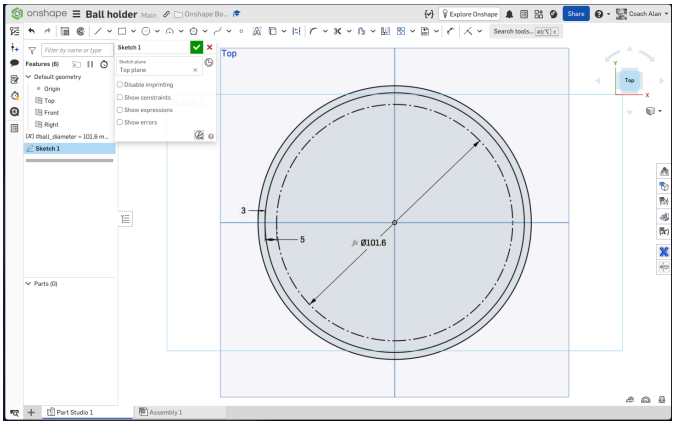
8. Ball holder

Several years FIRST has had game pieces that are spherical. This is actually great because they are relatively easier to design mechanisms for as they are the same no matter how they are rotated.

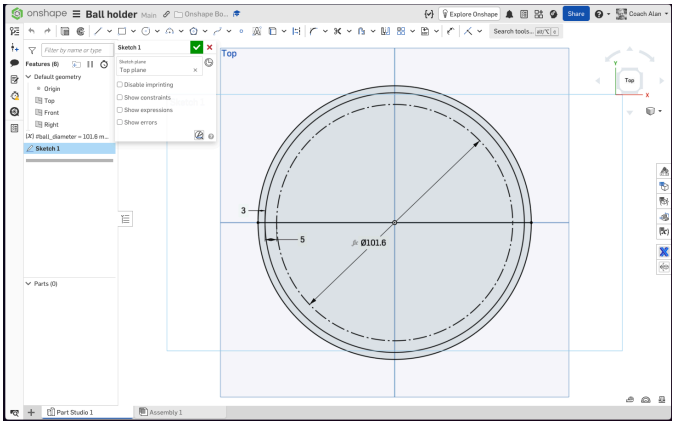
So let's imagine that we are trying to make a basket to carry a 4" diameter ball in. While we could just make it a square basket close to the size or a cylinder, we will use this opportunity to talk about revolving sketches.

Create a new document called "Ball holder" and make a sketch on the Top plane (Shift+S) and press N so we are looking at it top down.

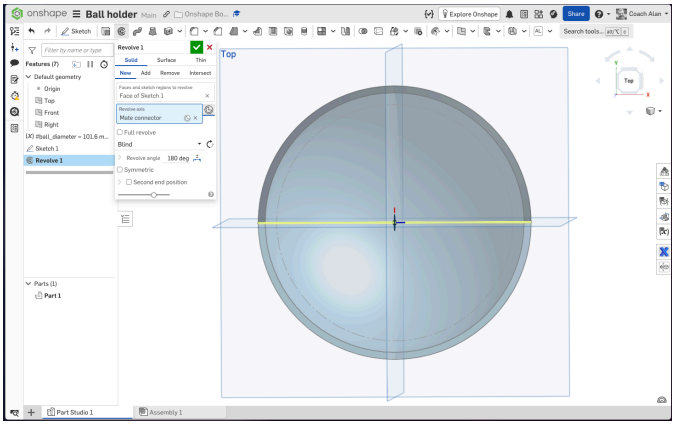
Start by making a circle the diameter of the ball as a construction line. (C for center point circle, Q for construction line). Put in #ball_diameter as the diameter and define it as a new variable that is 4in. We then want a circle (not construction line, so press Q to turn off construction line) that is offset from it (O for offset) by 5mm (to give some tolerance) and then another circle with an offset of 3mm (thickness of shell we want). It should look like this (I moved the offset amounts to make them easier to read)



Because a revolve will take a part of our sketch and revolve it around an axis, we need a line dividing the circle in half. So let's make a horizontal line from one edge of the circles to the other. (Select Line or L) and you'll end up with a sketch like this:

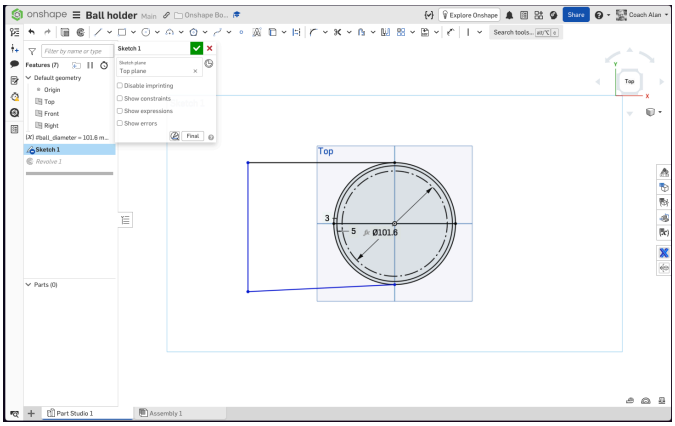


We select Revolve (Shift+W[1]), select the bottom half of our circle and select the axis as our mate connector. Note that we change from a full revolve to having a revolve angle of 180 (because we don't want a sphere, but only half of one.) . It should look like this:

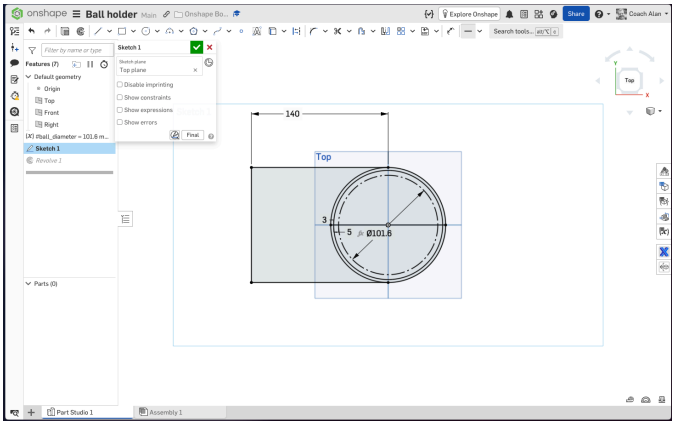


Select the green check to accept. This is pretty cool, but we still need a way to attach it to our robot. So right click on Sketch 1 and select "Edit..." (if you played with rotating it around to see it, you'll want to press N to get it back to being top down. Let's draw 3 lines.

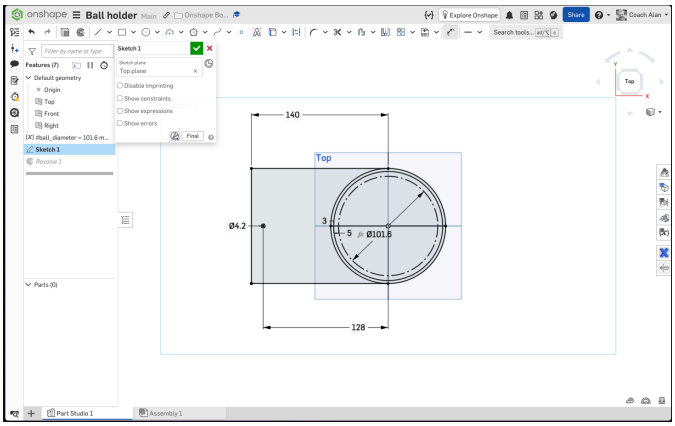
[1] I have no idea why the default keyboard shortcut for revolve is Shift+W



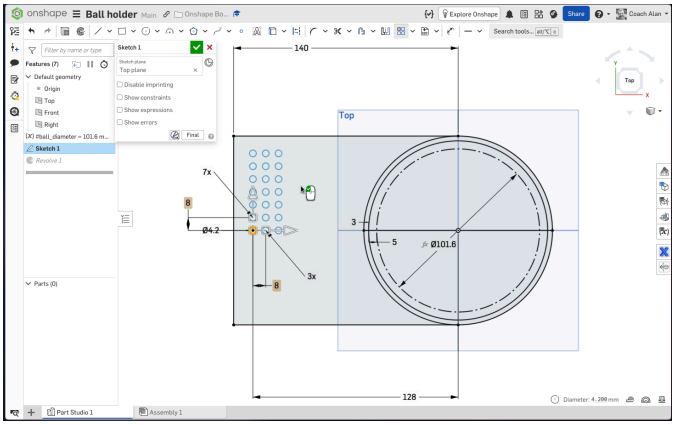
Let's apply the horizontal constraint (H) to the bottom line and a dimension of 140mm to the top line. For mine this made everything black (constrained.) . You may have to work with the various constraints to get yours all black. This is what it should look like:



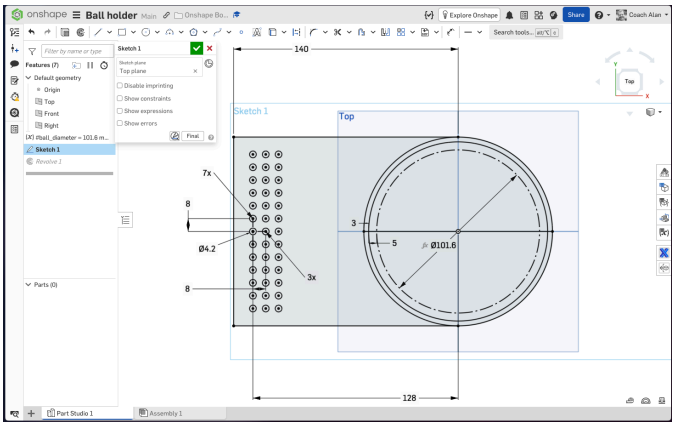
We need some holes on here to make it easy to attach. So select center point circle (C) and give it a size of 4.2mm and put it on the axis. Then give a dimension of 128 from the center of our new circle to the origin. It should look like:



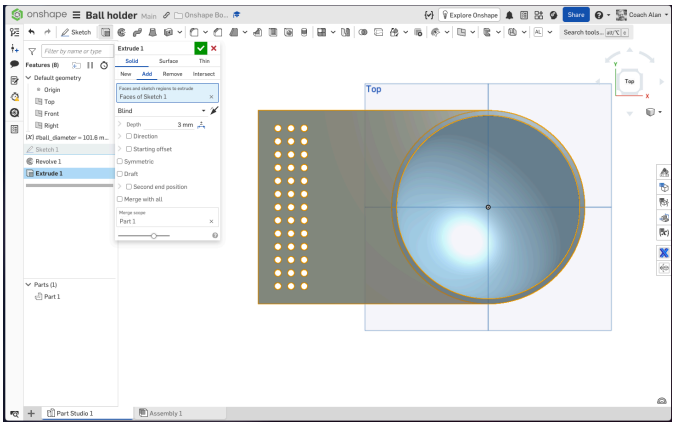
Now we are going to make a field of holes so we'll have lots of choices. Remember that GoBilda is on an 8mm pattern. We are going to use one of my favorite tools which is sketch pattern to show how to make a field of holes. Select Linear Pattern and then click on our circle. Change the vertical from 1x to 7x and the horizontal from 1x to 3x and change both of the amounts from 25 to 8mm. It should look like this (I moved the 8mm so you could read the number)



Press outside of the drawing area with the left mouse button to accept. Now select mirror and select the horizontal axis as your mirror line and use the select rectangle to select all of the holes (except the row on the axis). You'll get something like this: (I moved the 8mm and the diameter of the hole measurement so we could see them)

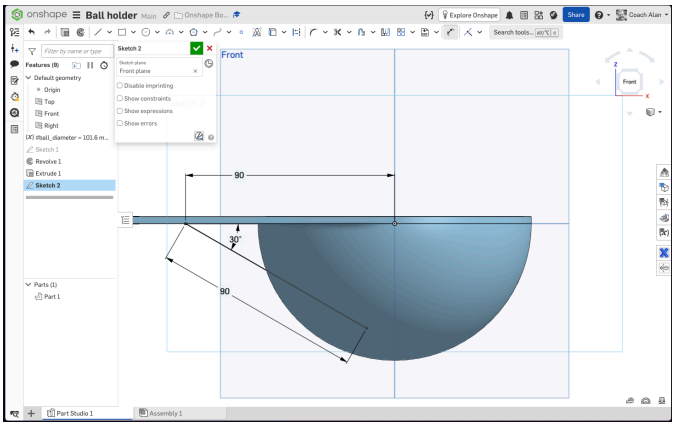


Now let's extrude that by 3mm (Shift+E) and add it to our part.



Now you might look at this and say that you are afraid it will break where the ball catcher piece meets the flat piece. It would be good if we could add a rib to connect those two.

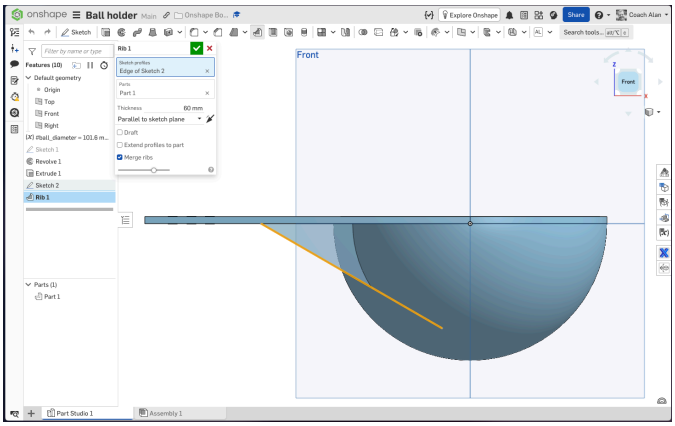
To do this. Let's create a new sketch (Shift+S) on the Front plane and hit N to make it so we are looking top down. Let's draw two lines on this sketch even though we only need one because it makes it easier to constrain. You'll notice that I made a line 90mm long coming out from the origin on the axis, and then I made a line that was also 90mm long (It doesn't really matter, it just needs to go significantly into the sphere.) . I made the second line come out at a 30 degree angle from the first. To do this, select Dimension (D) and click on both lines and enter 30. It should look like this:



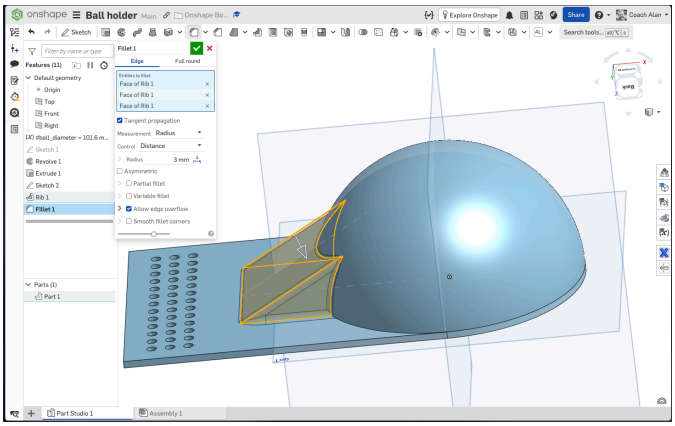
Now accept the sketch (clicking on the green check mark)

Select Rib and click on your line for the sketch profile and Part 1 for the parts. Change the Thickness to 60 (I am completely making that number up.) . You'll have to change the arrow next to Parallel to sketch plane if your rib ends up inside your sphere.

You should now have a part you can be proud of that looks like this:



This is a great example where you probably want to apply fillets to make the part stronger but you can't in a sketch because the edges weren't created there. Select Fillet (Shift+F) and then select the three faces of the rib. (You could also select all of the edges individually but I think this is easier) . It will look like:

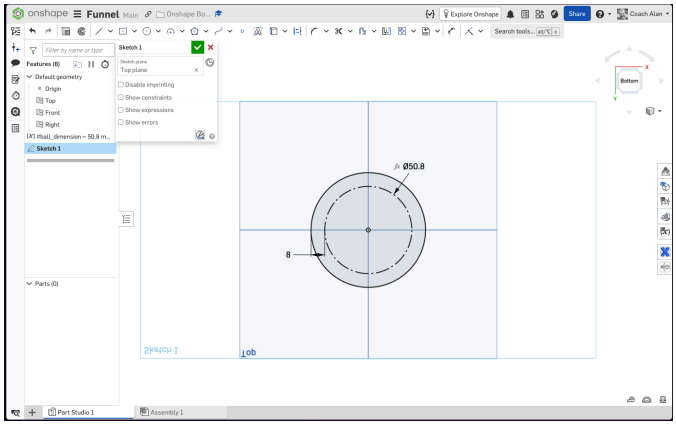


Accept it (pressing the green check) and you can send it to your 3d printer by exporting the Part.

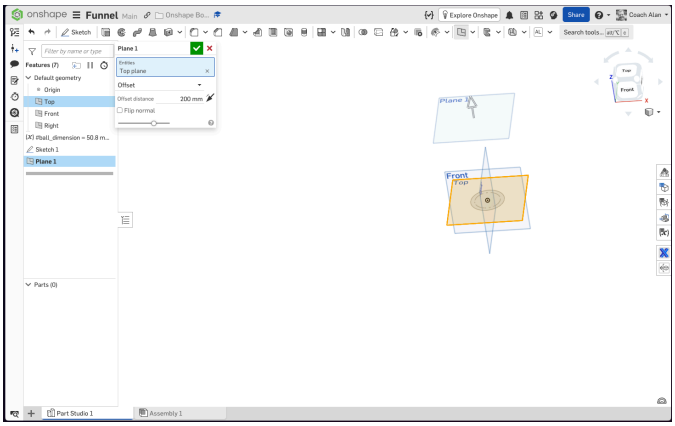
9. Funnel

Let's imagine that we want a way to create a funnel. Perhaps a game is all about catching balls but we want them to end up in a certain place. For this particular game the balls are 2 inches in diameter. We will do just like we did in chapter 8 where we make a circle with a construction line the size of the ball and then we'll use offset to make a larger circle. (In this case we do 8mm because we just need the outside of the material. This one is 5mm for tolerance + 3mm for wall thickness.)

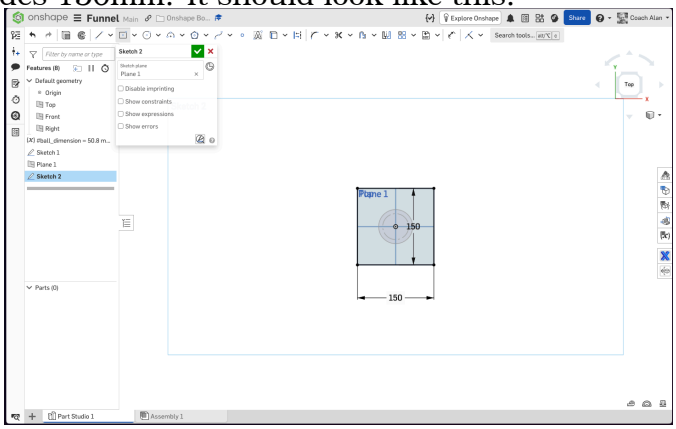
It should look like this:



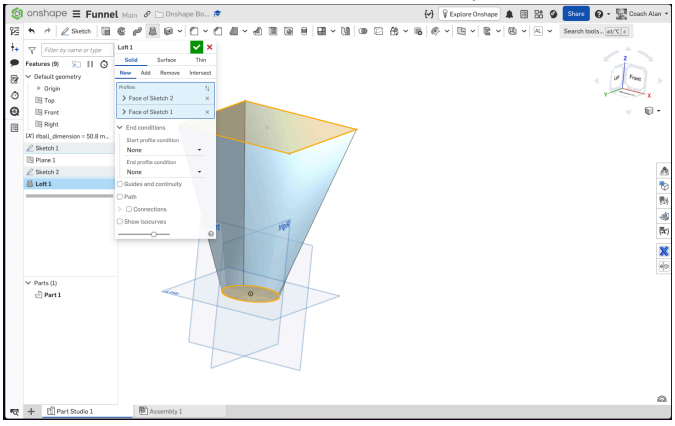
Accept it. Now we are going to create a new plane that is offset 200mm from the top plane. To do that, click on the Plane Icon, set the Entity as Top Plane with an Offset distance of 200mm. If you zoom it out, you can see it as we create it.



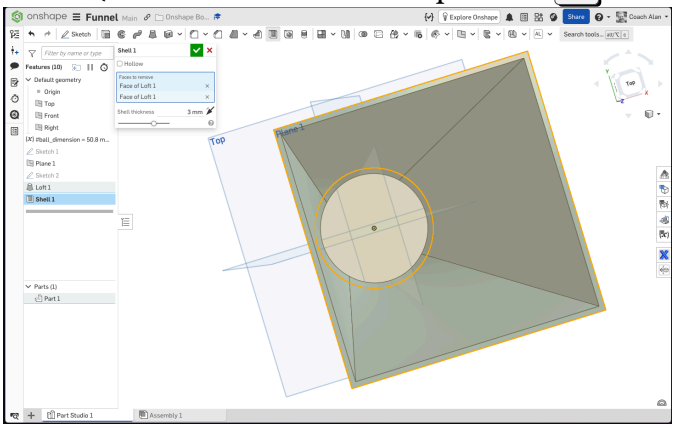
Accept it (green checkmark) then Create a sketch (Shift+S) selecting our new Plane 1 as the Sketch Plane. Create a Center Point Rectangle (R) with both sides 150mm. It should look like this:



Accept it and then select Loft selecting the face of the two sketches. This will make a solid that connects the two. It should look like this:



Accept it (green checkmark). Well, we need it to be hollow to be of any use. So select Shell and select both the top and bottom face of the loft. It should look like this (I turned it around and pressed F to zoom to fit.):



Once again, we need some way to attach this to our robot. But in this case, we'll get you to try it yourself using what we did at the end of chapter 8. You could make it at the top or bottom of the funnel (or both)

Part III.

Modeling your robot

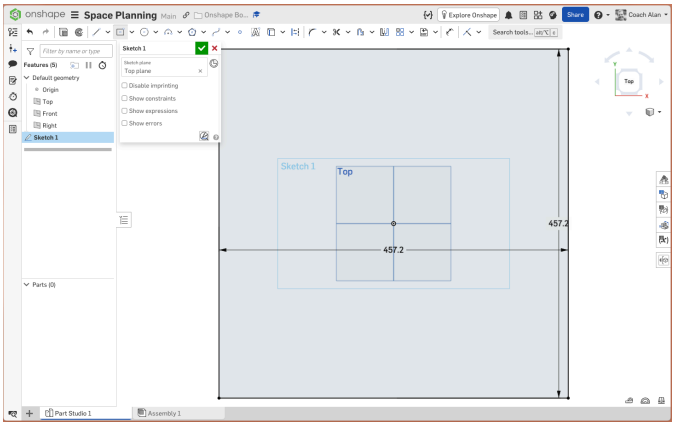
10. Design your robot - Space Planning

For this book, we'll design a relatively simple robot that can intake and shoot the "pollen" that is a game element for FTC BIOBUZZ season. (A roughly 70mm diameter sphere with holes in it.) It is important to know that I am doing this robot as an exercise in CAD. I have not built it and this robot is not designed for a particular game so it is not guaranteed that this robot will be useful for you.

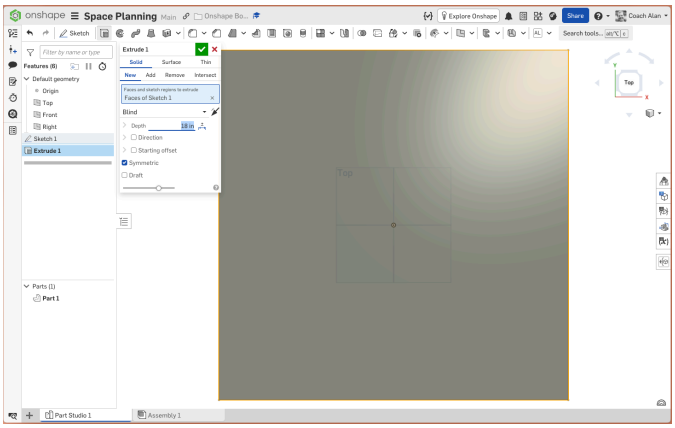
To make this robot, we'll need a mechanism to drive, a mechanism to shoot, and a mechanism to intake.

While we will iterate on design as we learn more, I recommend that we start with an idea and do rough space planning. Space Planning is the idea that you allocate how much space various things on the robot are allowed. This allows multiple people on the team to work on CAD at the same time.

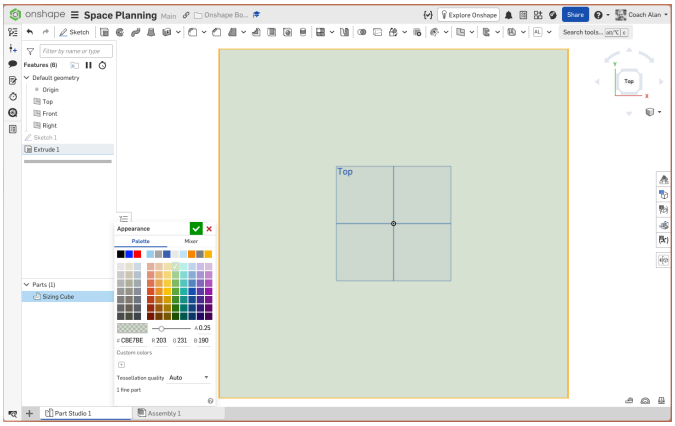
Make a new document called "Space Planning". In the Part Studio 1, create a sketch (Shift+S) and then N so we are looking at it top down. Create a center point rectangle (R) that is 18in x 18in. After you have done this, press F for zoom to fit. It should look like this. (Since our units are in mm, it shows is as 457.2mm which is the same as 18in.)



We are now going to Extrude (Shift+E) this with a depth of 18in and we are going to mark Symmetric. This will give us an 18x18x18 cube with our origin at the center of the cube. (Note that if you aren't in the depth entry, it will show it as mm)



After we accept it (pressing on green check) let's rename our Part 1 to Sizing Cube (by right clicking on it and selecting rename.) Next we'll change its appearance by right clicking on it and selecting Edit Appearance... We'll choose a light green and make it translucent by setting its Alpha (shown as A) to around 0.25 . (An Alpha of 1 is completely opaque, and an Alpha of 0 is completely transparent.) . Accept it by pressing the green check.



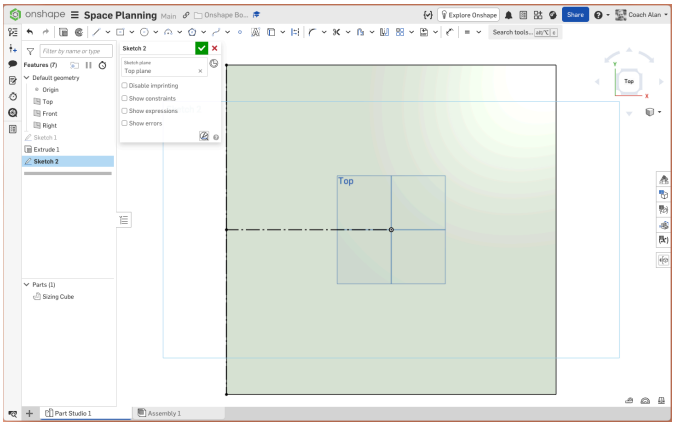
This will help us stay honest by making sure our designs fit within the sizing cube. (We can hide and show it with the eye next to the Sizing Cube part.)

10.1. Space planning for Drive train

Make a new sketch (Shift+S) with our Top plane (Hint: You can select the Top plane from default Geometry). Use (U) the left side of our sizing cube as a construction line (Q). If you forget to select construction line before you use it, you can select on the line and choose construction line and it will make it a

contruction line.

Now select line (L) and construction (Q) and draw a line from the origin to the middle of the construction line. It should now look like:



We'll now make a rectangle for our left channel + wheel space. Use the center point rectangle (R) and make it 440mm high by 94mm wide with its center on the line we drew earlier. Use Dimension (D) to make its center 168 from the origin. (We made it 440 because that is the length of the 17 hole U-Channel with 104mm Grip Force wheels in the next to last hole on the front and back.)

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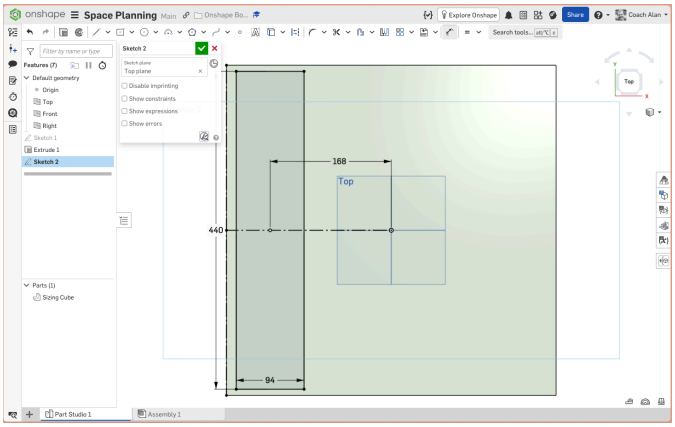
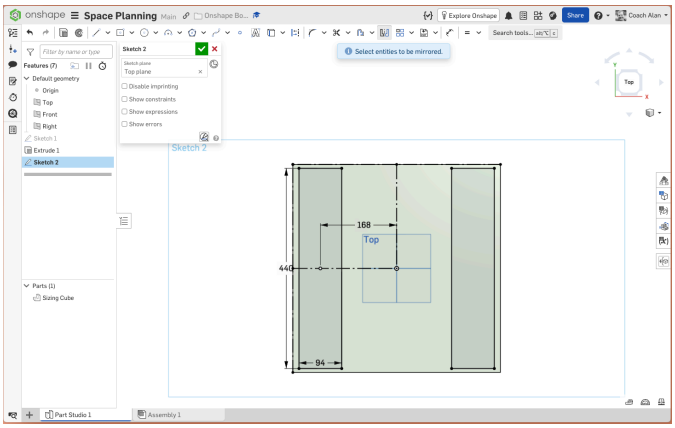
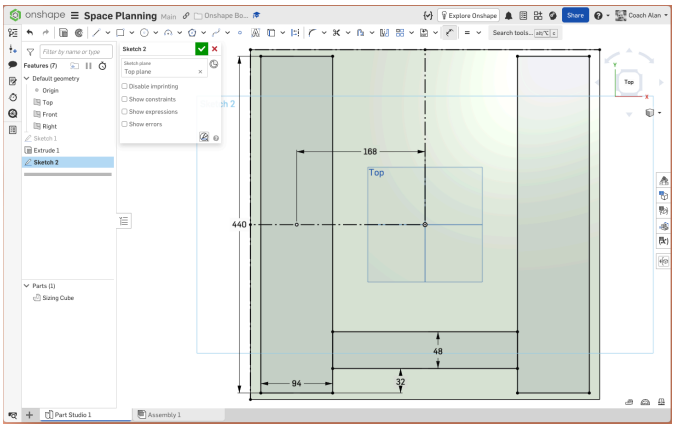


We need two of these so we'll want to use Mirror. But before we can do that, we need to make a symmetry line. We'll first Use (U) the top of our sizing cube (use Q to make a construction line) and then we'll use Line (L) and Construction (Q) to draw a vertical line from the origin. Now use Mirror, select the vertical line as your mirror line and select the four sides of the rectangle as your entities to be mirrored. It should now look like this:

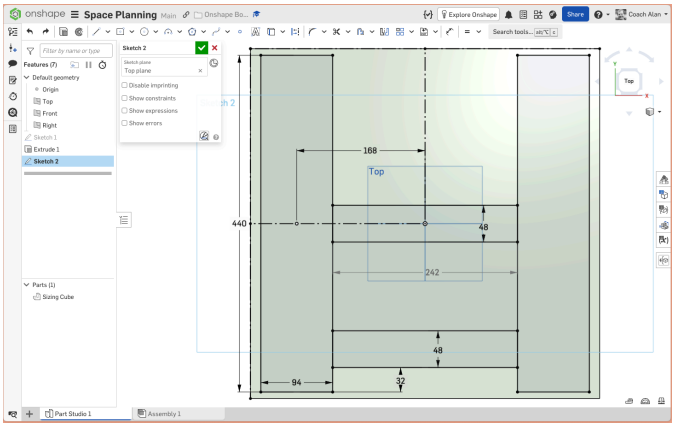


Now, we'll use the corner rectangle (G) to draw our back piece. Draw it as a rectangle between the two boxes and then use Dimension (D) to make it 48mm wide and 32 mm from the back. (48 mm is because that is the width of U-Channel, and 32mm because that is where the connecting piece is on a standard GoBilda chassis.) . It should now look like this:

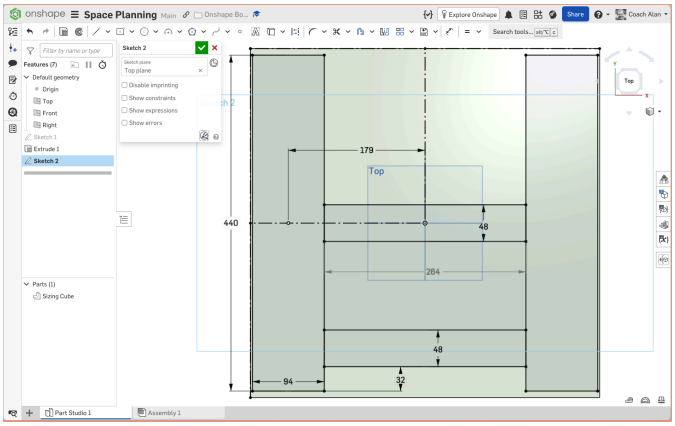


Let's add one more cross piece in the center. It is easiest to do this with a center point rectangle (R) where the center point is on the origin, it is 48mm wide and connects both rectangles. (If you move the edges until you see the coincident constraint marker show up (looks like a slanted T) then Onshape will know they are to be connected.)

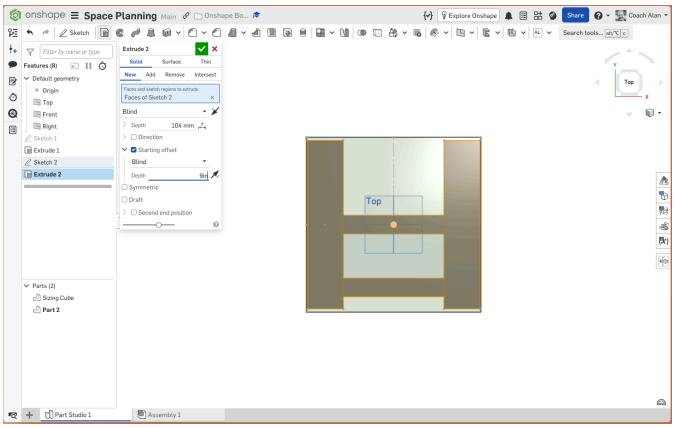
At this point, let's add a dimension between the two insides. This will show up as grey because it is a "driven" dimension. (In other words it is determined by other things.) .



We see that there are 242mm between the two. But GoBilda doesn't make a U-Channel that is 242mm long. At this point we have a choice. We can either go for slightly smaller (a 9 hole is 240mm) or larger (a 10 hole is 264mm). It looks like we have some extra room, so we'll increase the dimension from 168 to 179. (because 264-242 = 22, and half of that is 11) and see if it still fits. It does but barely. This is the kind of thing where CAD is fantastic because it helped us identify a problem with our design early.



Now we'll extrude (Shift+E). Make sure you make "New" and Extrude 104mm (because that is the height of the wheel which is the tallest part). Also, set our starting offset to be 9in off so that we are coming from the ground.



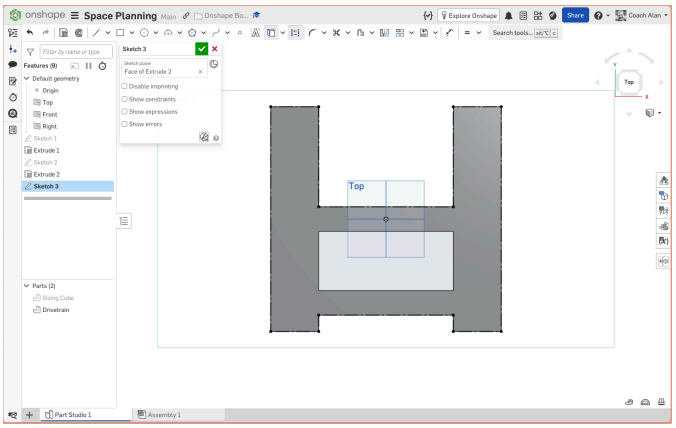
You'll notice that this is not very precise. We could model the wheels to be more exact, but at this point we are just trying to give ourselves a rough guess and do it quickly.

Why did we do the drivetrain first? Basically because if we can't drive then our robot is useless. So it is the most important thing to get right. What should we do next? From experience, we typically do the intake next since it also needs to be by the ground. (But I could accept the argument that you should do the shooter next because if you can intake but not shoot then what good is that.)

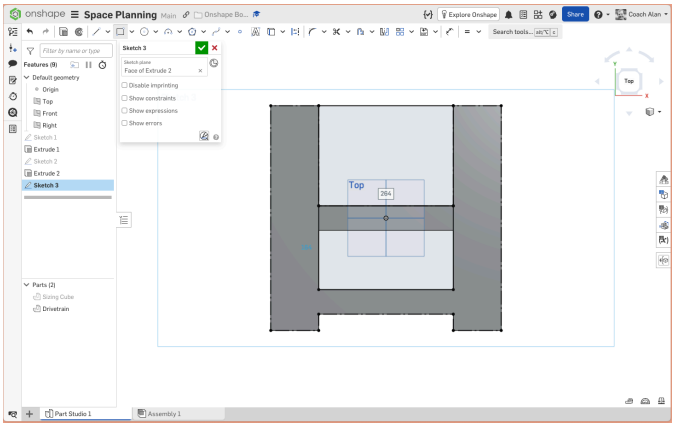
10.2. Space planning for intake

Let's start by hiding our sizing cube. (Click the eye when you are over the sizing cube in the parts list) Let's give intake all of the space in front between the channels up to 9 inches high and the space between the cross piece and back piece up to 9 inches high.

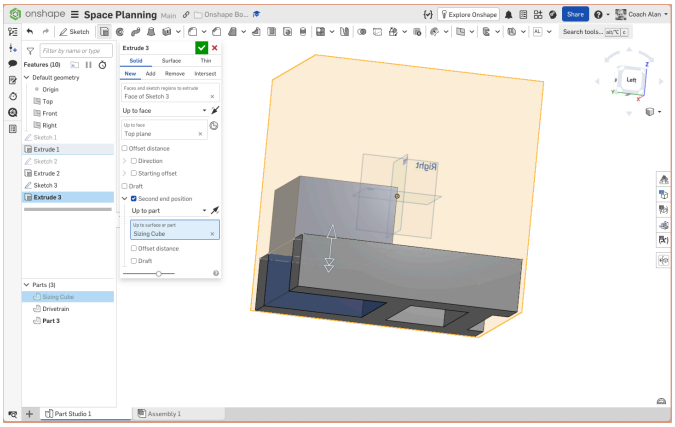
So create a sketch (Shift+S) on the top of the drivetrain. You can select "Use" (U) and construction line (Q) then click on the drivetrain and it will put the outline of the drivetrain on our sketch.



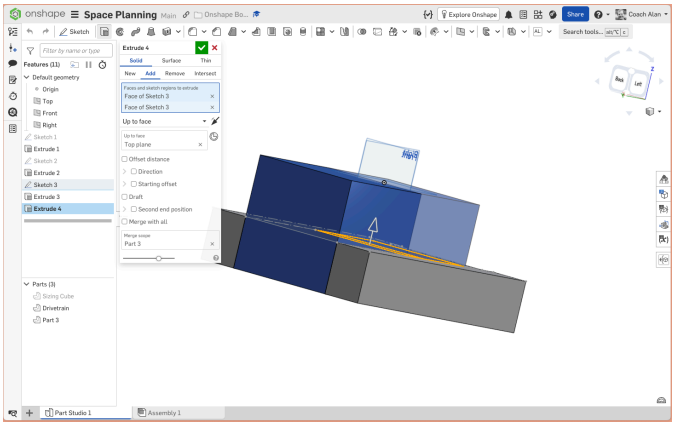
Draw two corner rectangles (g). One that is from the inner upper left to the lower right of the front area. Then a second one that is next to it back to the back cross piece. It should now look like:



Extrude (Shift+E) the top one only (New part) up to the face (top plane) and with a second end position down to part (select Sizing Cube). It should look like this (I rotated to make it look more obvious)



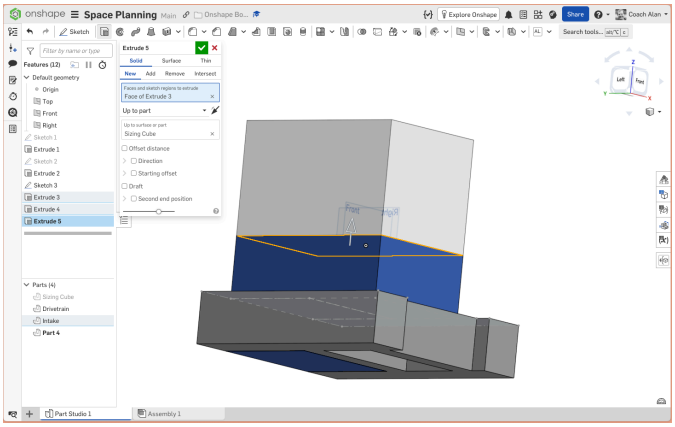
Accept it and then do a second extrude of the other two rectangles (add) - You'll have to make the sketch visible in order to see the rectangles.



Accept it and rename this part "Intake"

10.3. Space planning for shooter

For the shooter, we'll simply give it all the space above the intake. Instead of making another sketch, we'll click on the top face of the intake and do an extrude up to the sizing cube. Make sure you do this as New so that you'll get another part.

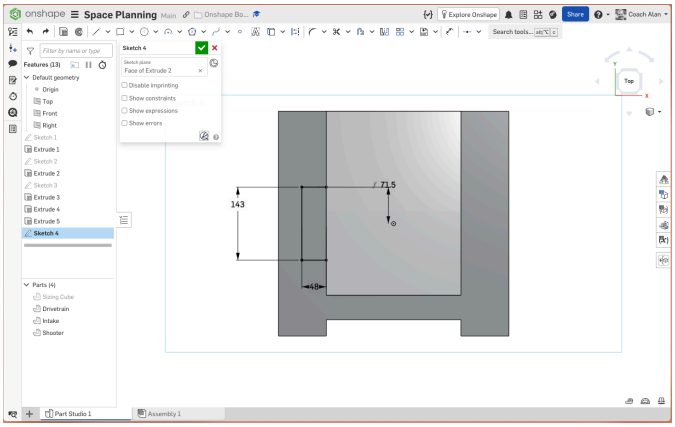


Accept this and rename it as Shooter.

10.4. Space planning for electronics and battery

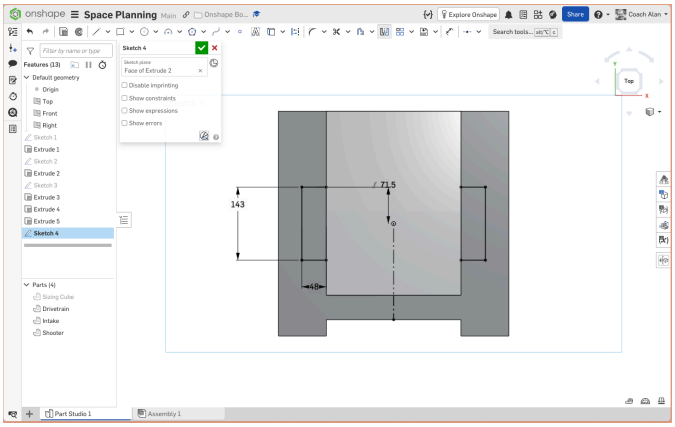
Reserving room for the electronics and battery is critical. With our current design, let's plan to put the battery on the back cross piece and the control hub on the left side and the expansion hub on the right side.

Start by making a new Sketch (Shift+S) and make a corner rectangle (g) that is on the inside of the channel and is 143mm tall (from the REV site), 48mm wide (The hubs are 29.5mm tall but I like to round up to a U-channel for making a 3d printed mount and room for cables). Then use Dimension (D) to make the top = 143/2 from the origin. (You'll notice a f to show there is a formula. I like to leave the math in there instead of calculating it outside so I can see why.)



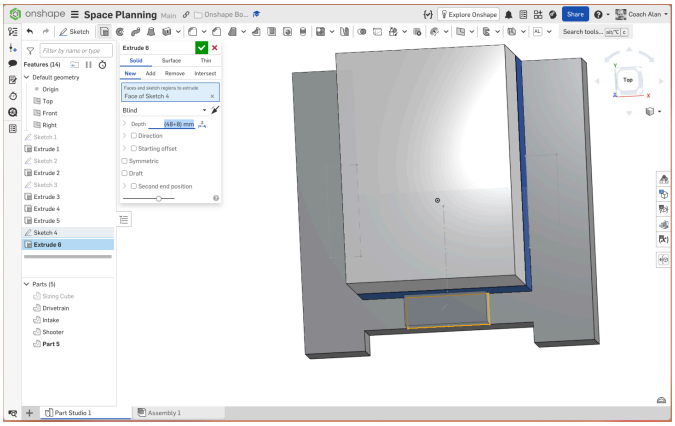
Then let's make a construction line from the origin down to the bottom that we can use as a mirror line to make our second hub. Here is what it will look

like after you have made your two hub rectangles.



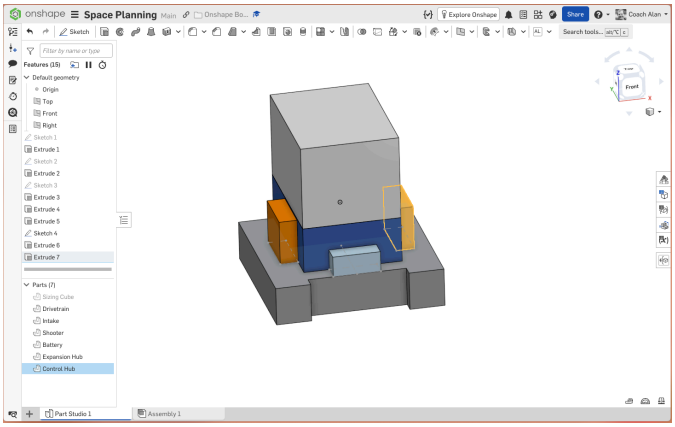
For the battery, I use the Gobilda Nested battery and battery mount which means I need to reserve 128mm by 48mm for it. I center it since it is one of the heaviest things on the robot.

Now let's extrude these parts. The battery needs to be 48 (for battery) + 8 (for mount). So extrude it (Shift+E) and then name it as battery.



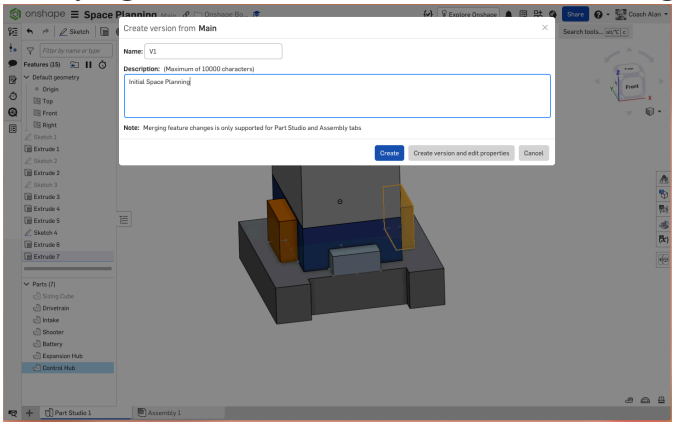
You'll need to make Sketch 4 visible and then extrude the two hubs. While they are 103mm high, I go ahead and do 120 to reserve myself some wire space.

You now have a space planning for your robot. It should look like this:



10.5. Some comments

I like to go press the “Create Version” (upper left - 2nd one down that looks like a dot with a plus sign). This will make it so we can compare to it later easier than trying to remember when it was and finding the version.



The one thing we know is that this plan will change. But at least now we can have different people on our team go work on the different pieces.

Hopefully you’ll see that it isn’t that much work to do the space planning but that it helps you start to think about where things are going to go on your robot.

11. Mecanum Drivetrain

Start with design. While the standard GoBilda chassis is pretty good, it can make it difficult to assemble your own mechanisms because of all of the space motors take up inside the chassis.

So we'll start with a mecanum chassis that has the motors sticking up so we have the maximum amount of space for adding things. (You'll notice that we just decided to violate our space planning. But this space wasn't being used by anyone. In an ideal world, we would go update the space planning so someone else doesn't also decide to use it. I'll leave that as an exercise for the reader.)

Why would we use CAD to put together standard parts instead of just building it? A few reasons:

1. It turns out that it is much faster to use CAD to put the parts together

2. It allows you to figure out what parts you will need when you might not have all of them.

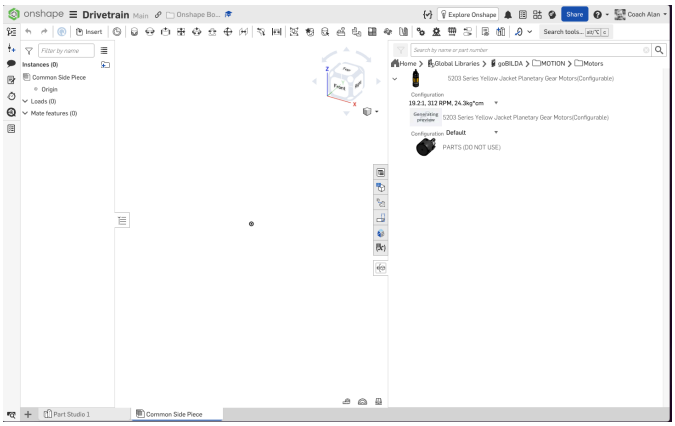
3. It allows you to find problems earlier

I admitted earlier that I was not a mechanical engineer. While it will make mechanical engineers very nervous, I don't model every part and every mate. I do just enough that will help me to make sure I know what I am building and what parts I need to buy. So I typically leave shims and screws out. This can be dangerous because we can forget to leave clearance for screw heads.

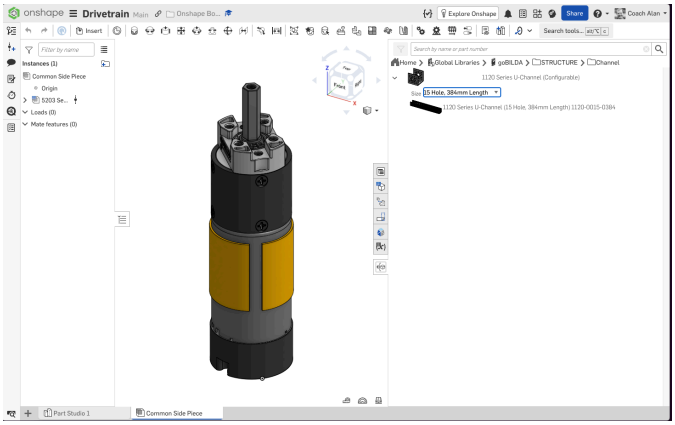
We make pretty heavy use of subassemblies because they make it easier to have multiple people work, to make changes in smaller parts, and also it speeds up Onshape.

So make a new document called "Drivetrain". In that, rename your "Assembly 1" to "Common Side Piece". (To rename, you can right click on the tab and select Rename...). The first part we are going to make is the common side piece that we'll need one for the left and right sides of our robot.

We are going to use insert tool and insert the motor first and select the configuration to be the 19.2:1 motor.

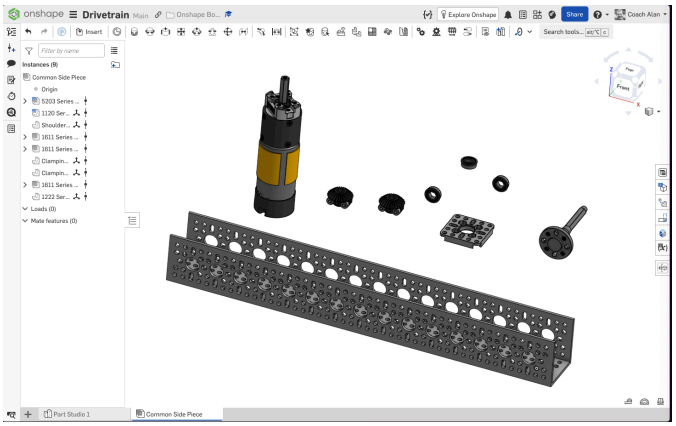


Next we'll insert a 15 hole u-channel.

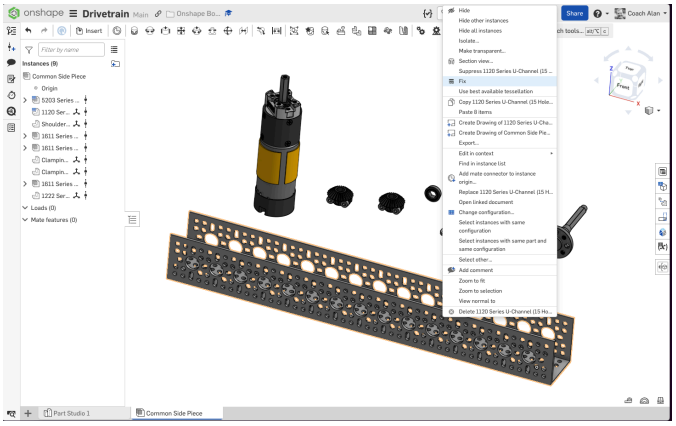


One of the things that makes these different is that they are configurations. As a hint, if you search on a GoBilda part number and don't find it (and it isn't a brand new part), then look to see if it is in here as a configurable part. The menu tree here (mostly) matches the GoBilda site.

Now, we'll need the Shouldered 8mm REX Hub-Shaft (you can search for part 2110-0608-0001) and 2 8mm REX ID x 14mm OD (1611-0514-4008), 2 Clamping Steel Miter Gears (2320-4008-0024) and 1 6mm ID x 14mm OD bearing (1611-0514-0006) and a flat pattern mount (1222-0001-0001). When you need 2 of something, you can either insert it twice or copy and paste it out of the instances list. We can now close the insert tool and you'll notice all of these parts in our instance list and that they are all on top of each other in our CAD. I spread them out so they aren't overlapping. You should have 9 in your instances.

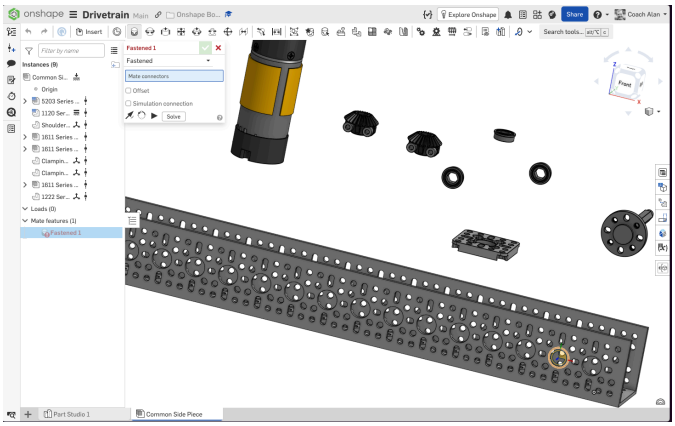


The next thing I do is pick the major piece (in this case the U-Channel) and "Fix" its place so it can't move. To do this right click on the U-Channel and select Fix. It is tempting to Fix multiple parts but **DO NOT DO THIS!!** You want to use the mates so that you can use the power of CAD.

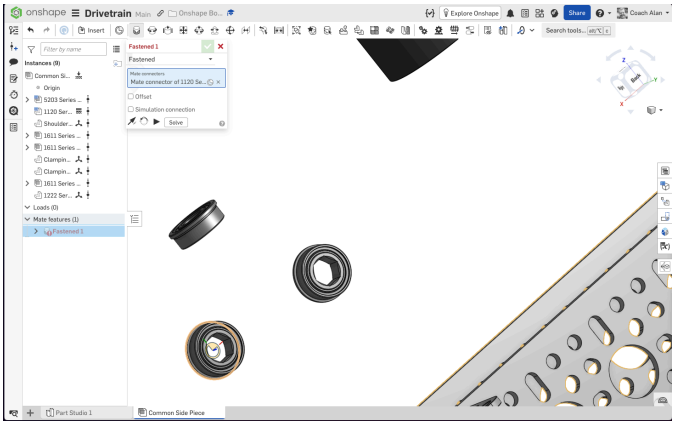


Next, we are going to put the bearings in the U-Channel.

To do this, select fasten mate (M) . First, select the second hole from the right on the U-Channel. You'll notice that a mate connector will show up with the blue pointing out. The three lines are the three axis of the mate connector. The blue is the Z (pointing out).



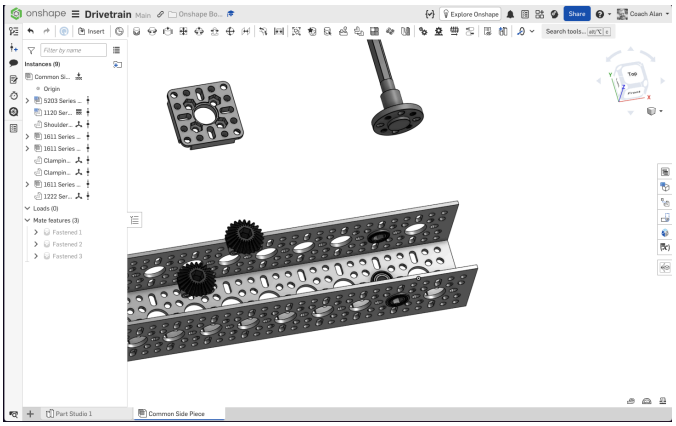
After you select it, you'll have to spin around in 3D to select the outer edge of the bearing from the back. (To move your parts in 3D, you can hold the right mouse button down (start when it isn't over any part) and it will move the view cube (and your view).) The reason why you want the outer edge from the back is because you want the outer edge to be "fastened" to the outer part of the hole on the U-Channel. This is one of the ways that CAD can be dangerous. Once you say it is "fastened" in CAD, it will stay together even though we probably won't glue the bearing in.



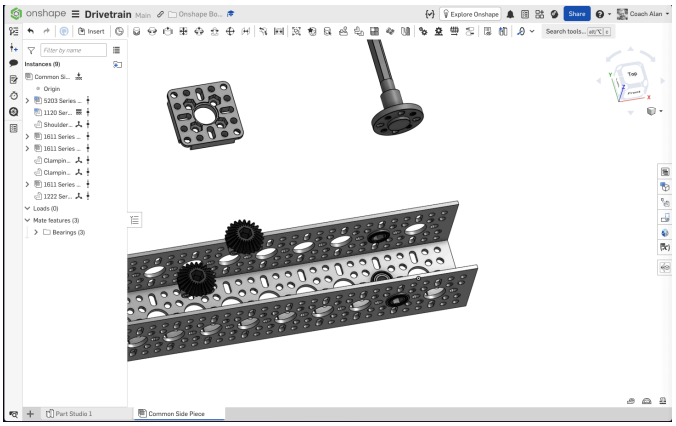
After we select it and press the green check mark, we will see that it has snapped into the U-Channel. If it is backwards (ie, the bearing is pointed out, then we can flip it with the arrow that flips the primary axis.) . If you have already accepted it, you can go back and edit it.

You'll notice that when you accept a fasten mate, it automatically creates another one for you.

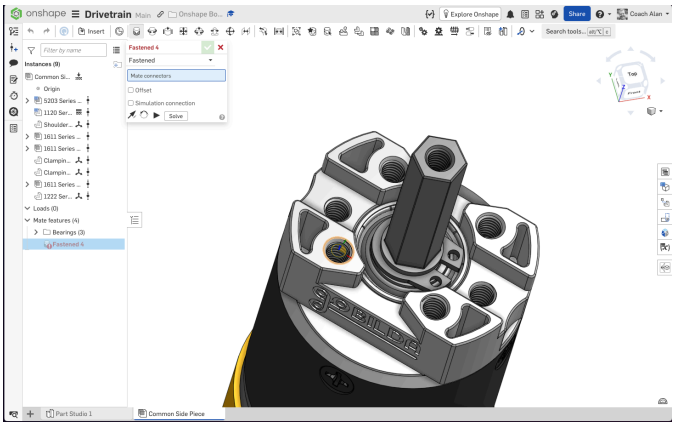
Now lets put the other 8mm bearing in the second hole on the top and then the 6mm ID bearing in the second hole on the back.



You can select all three mates in the mate features list, right click and then add to folder. We'll name the folder bearings. This isn't strictly necessary but it makes it easier for us to find things later if needed.

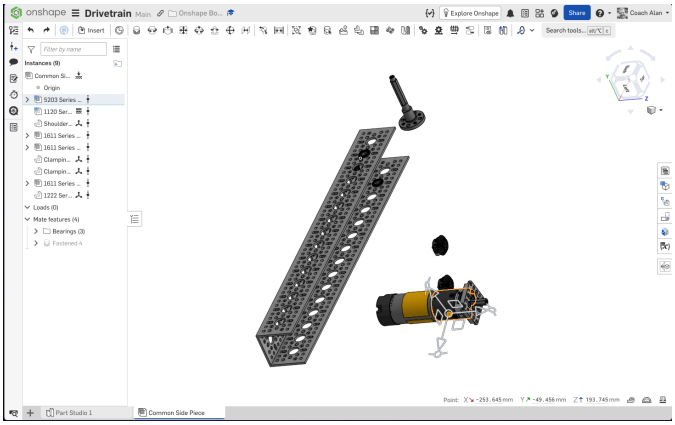


Next, we are going to attach the motor to the pattern mount. Press F to zoom to fit so you can see everything. Then create a fasten mate (M) with one of the holes diagonal on the motor and then the back of the pattern mount.

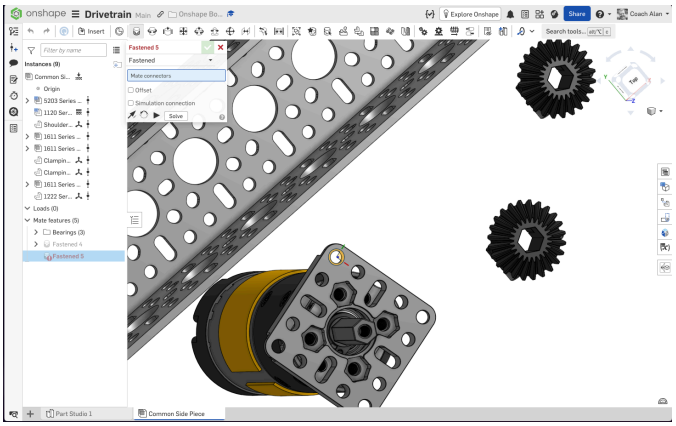


(You can press solve before pressing the green check to see if it is right. If you didn't pick the correct hole, you can see if pressing the circle next to the arrow will work as it will change your secondary axis.) .

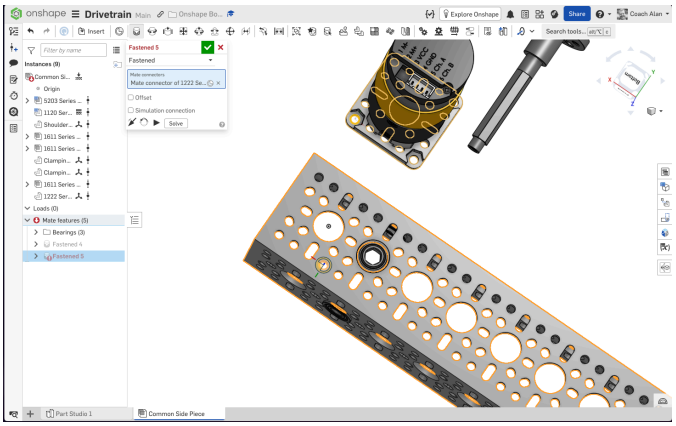
You can drag and move parts around that aren't connected (or fixed) by clicking on them (when you don't have a dialog up like a fastened mate that is looking for a mate connector). You'll see some strange handles come up that allow you to move it or rotate it in all three dimensions.



Next, we are going to attach the pattern mount to the U-Channel. (This will bring the motor with it) . In this case, we'll select the upper right hole of the pattern mount.

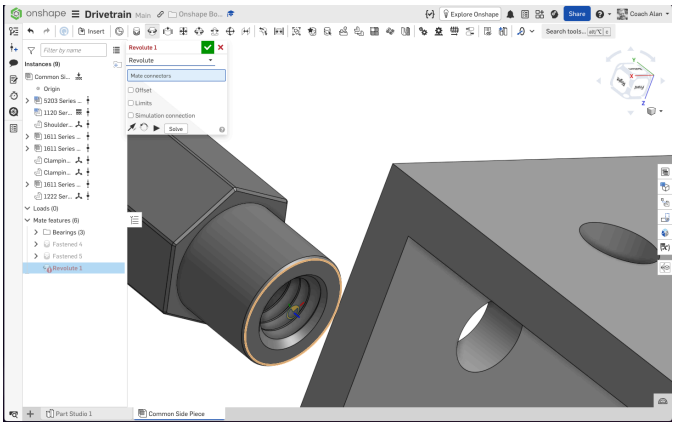


Then we'll select on the u-channel.

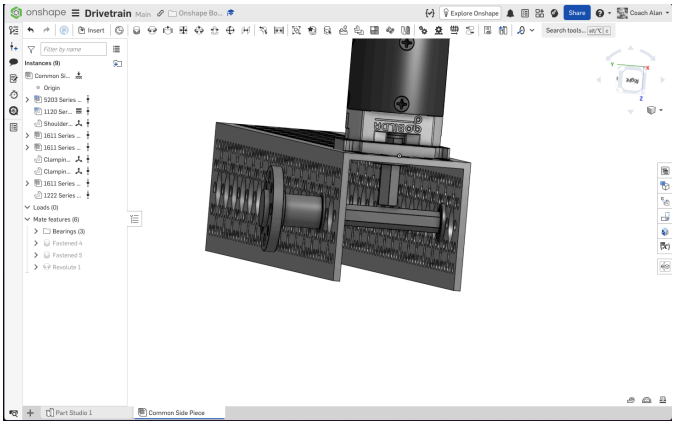


Next, we are going to put the shaft in the bearing. Since the shaft can rotate in the bearing, we are going to use a revolute mate. (You can either press m and change the type of mate with the drop down, or select the revolute mate icon from the top.)

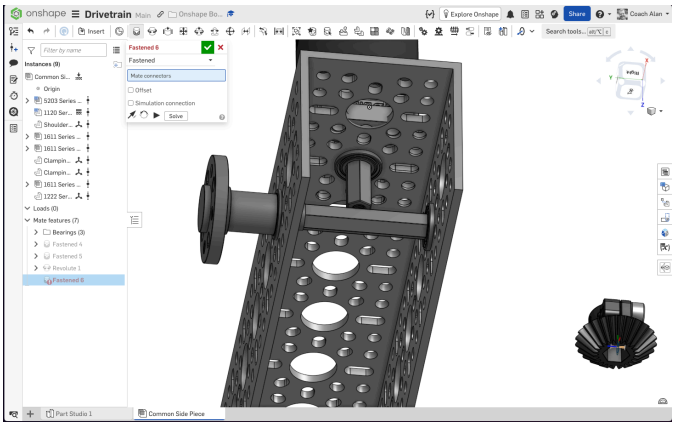
Select the outer part of the circular bit of the shaft that will fit in the smaller bearing.



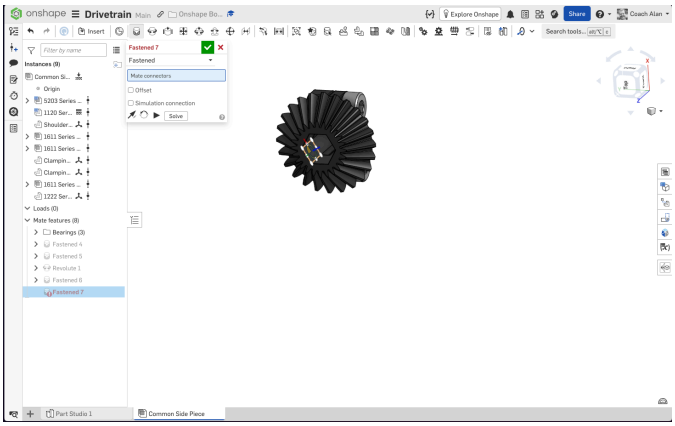
Now select the inner part of the small bearing. After you accept it, you should have something that looks like this. (after rotating and zooming so you can see this.)



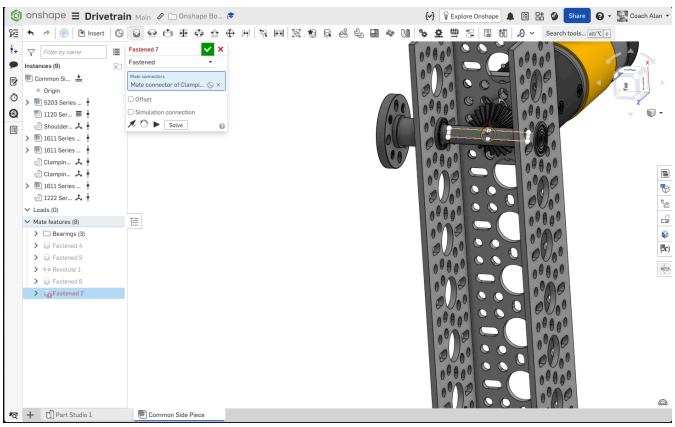
Next, we are going to put the first miter gear on the motor shaft. This should be a fasten mate because the miter gear rotates with the shaft. (as opposed to the shaft in the bearing where the shaft rotates inside the bearing.) . Select a front edge of the hex hole on the miter gear and then the front edge of the shaft.



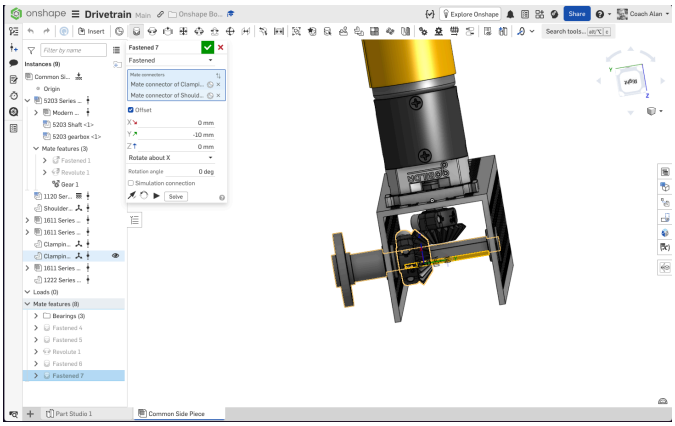
Next, we are going to put the other miter gear on the shoulder shaft. I find this one tricky. To start with pick the inner side of one of the sides inside the miter gear.



Now for the other half, pick the shaft face.

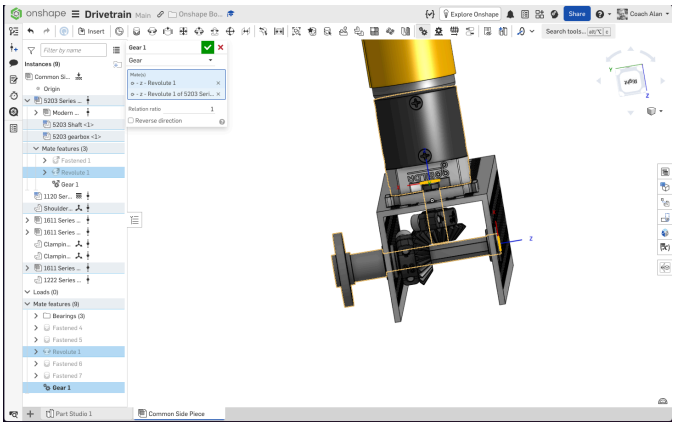


You'll notice that this aligns the center of both faces which isn't really what we want. So we select offset, and choose a Y offset of 10mm so it looks right.



Next, we are going to put a gear relation so we can make the shaft move. (To be honest, I probably wouldn't do this normally but it does show how we can do this so that we can test out things working together.) .

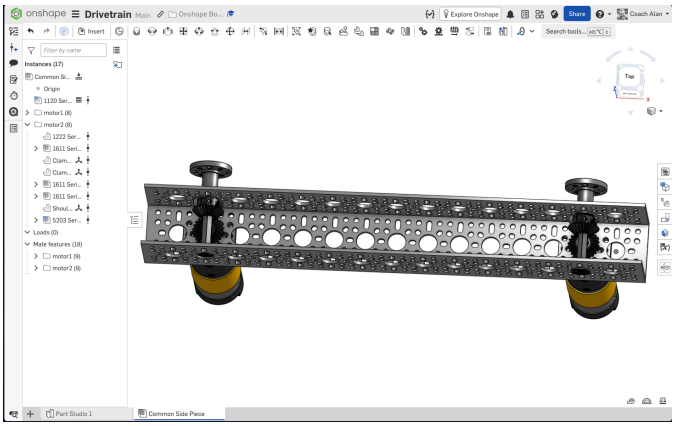
Select Gear Relation and then select our Revolute 1 and the Revolute in the mate features of the motor (if you accidentally select the one inside the motor it won't account for the gearbox so it will be off by the speed of the gearbox.)



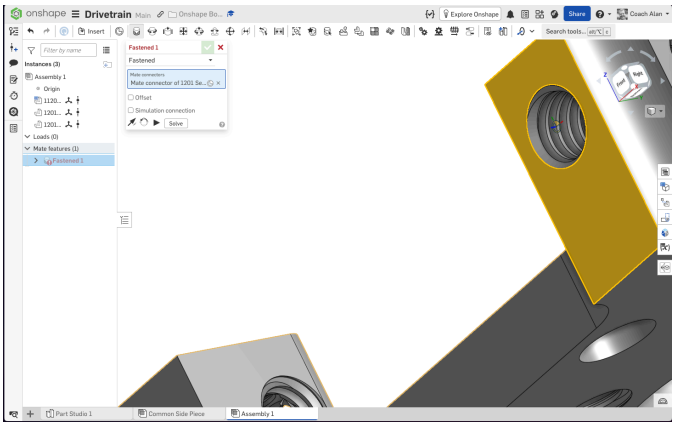
Accept it with the green check mark. Now you can right click on our revolute 1, select "Animate..." and then when the dialog comes up press the play button and you can watch all of the things move that should.

I went and made a folder (by selecting all the mates) and called the folder motor1 so I could keep them organized.

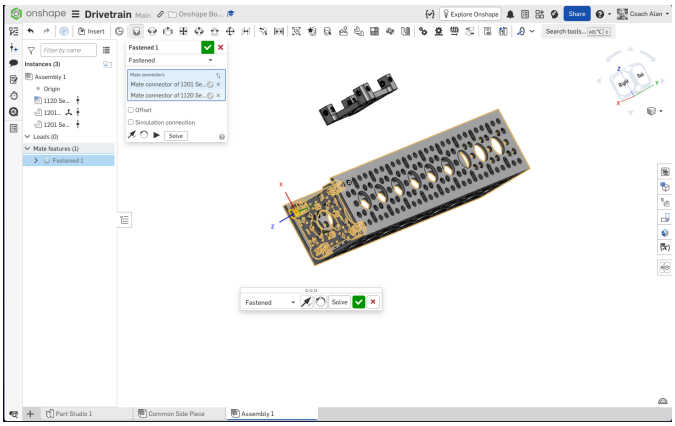
Now it is your turn to add a second one just like this but to the other end. (second from the end). Obviously you won't need a second U-channel but you'll need everything else. It should look like this:



Now let's go make a second assembly (press on the plus side on the lower left) and name it "Main Assembly". Into it, we are going to use the FTC Insert Tool and insert a 9-hole U-channel (under Structure/Channel/Configurable) and two quad block holes. Use the fasten mate (selecting the outer left of the quad block and the hole on the upper inside left of the u-channel.) . Here is a trick for things like tapped holes that are tricky to get the right mate connector. Put your mouse over on the face and press and hold Shift and it will keep it to points on that face without jumping to other nearby points.

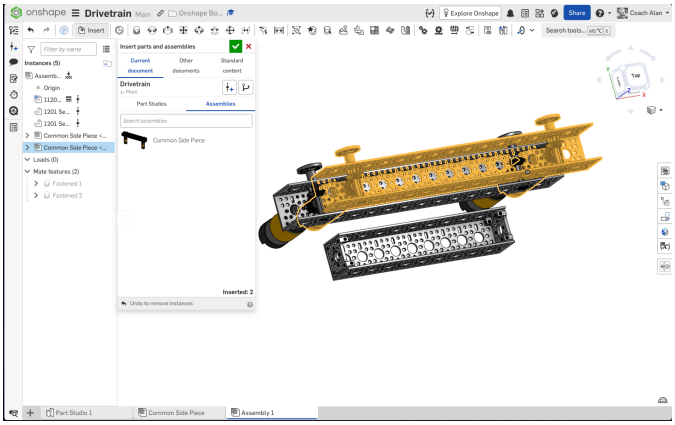


I had to click on the circle a few times to get it to rotate correctly for me.

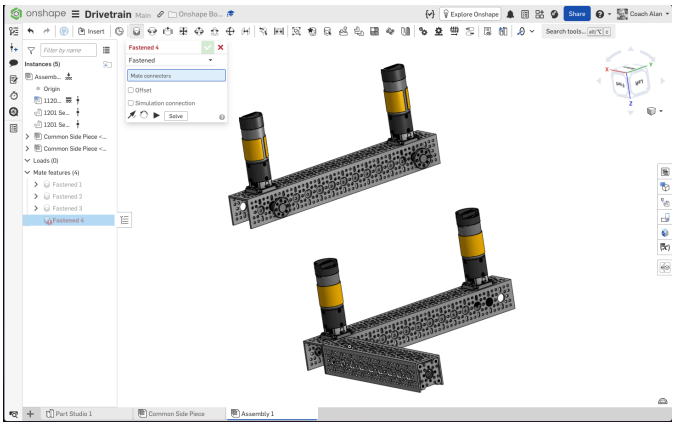


Now go do it for the other side.

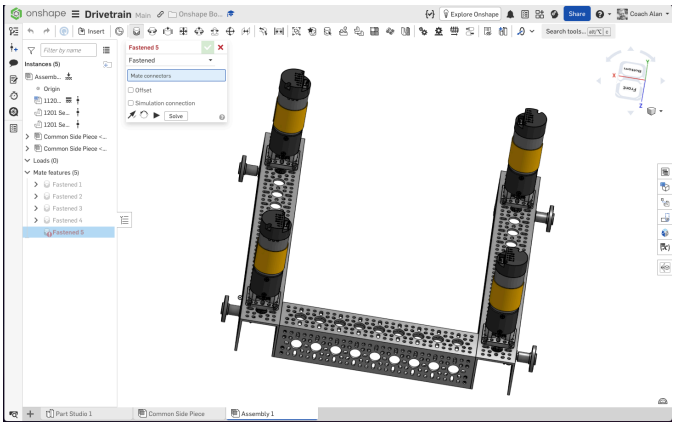
Right click on the U-Channel and select “Fix” so we won’t have it moving on us. Now, let’s insert our common side piece twice.



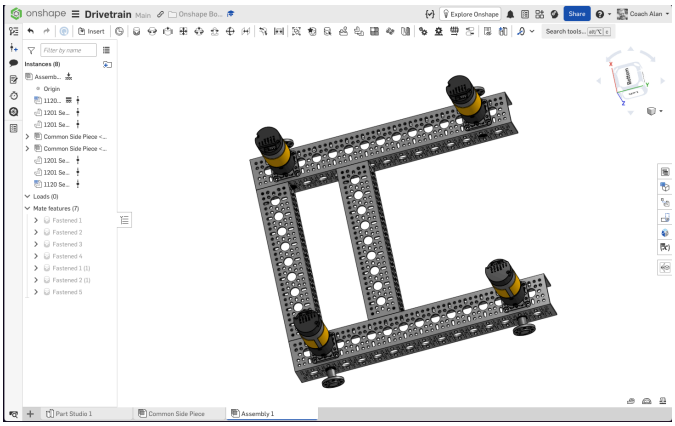
Now, let’s do a fasten mate between the quad block on our cross piece that we just made and the very back (make sure the hubs are on the outside) . It should look like this:



Let's go connect our second common side piece to the back as well.

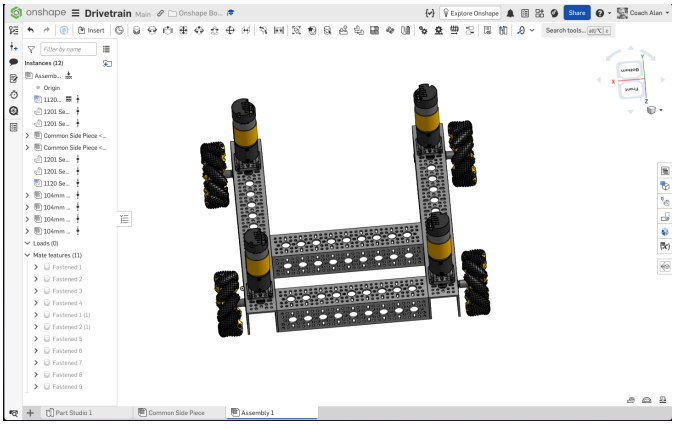


Instead of doing this twice, select our three new parts under assembly (Ctrl+C on PC, Cmd+C on Mac) and then paste (Ctrl+V on PC, Cmd+V on Mac). Now make a fasten mate between the new piece (on the block mount) and one of the common side pieces so it looks like this.



Some people would say you should have a fasten mount on both sides and they are probably right. But I would normally skip it.

Now go back into the FTC Insert Tool and insert 2 mecanum wheels (104mm) with left slant and 2 with right slant. Use a fasten mate to connect each wheel to its corresponding hub (the disc at the end of the shoulder shaft). The mecanum rollers should make an X as you look at it from the top. (ie, all of them should point toward the center of the robot.) . It should look like this:



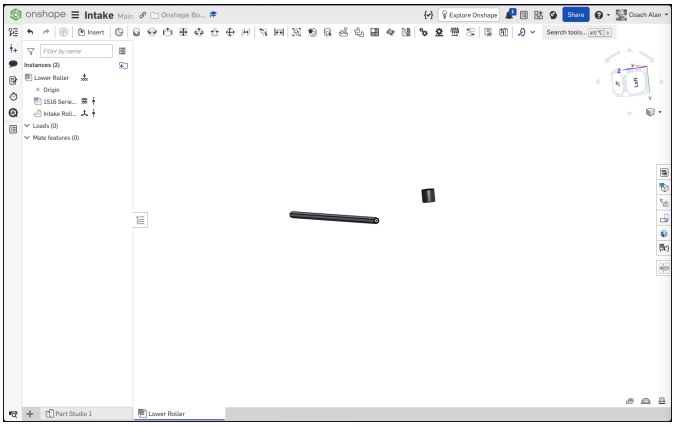
If you actually did all of these steps and it looks like this then congratulations! You now know most of what you need to use CAD for FTC. I promise you will get faster as you do it. If you didn't do all of these steps, I encourage you to go back and do them. It will help cement the steps in your head better than reading it a million times would.

12. Intake

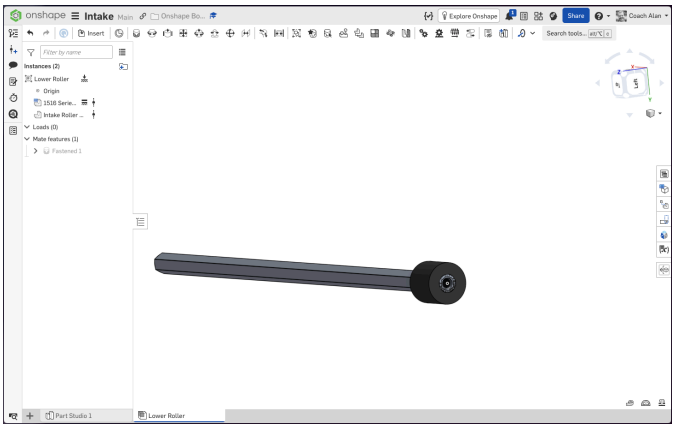
Just like we did earlier, we need to make sure we know what we are designing before we start. We will need a rubber piece going across to help the ball come up off the mat. This can be stationary. Then we'll need a piece that is spinning with compliant wheels on it that can move up and down. We'll call these the lower roller and the upper roller.

For the lower intake assembly, we want an 8mm REX standoff (240mm long) with Intake Rollers on them. We will need 15 intake rollers, but before you go and insert all of them from the insert tool let's show a better way.

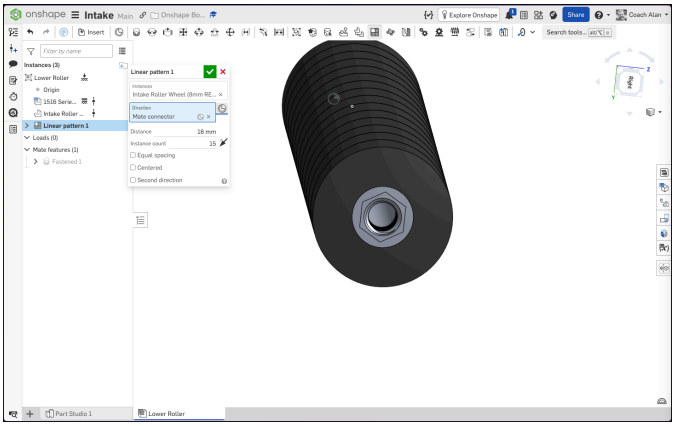
First, create a new document called "Intake". Then rename "Assembly 1" to "Lower Roller". Use the FTC insert tool to insert a 1516 Standoff (240mm) and an Intake Roller (8mm REX ID). Go ahead and Fix the Standoff (Right click on it and select Fix). This should look like:



Now, we are going to fasten the roller to the end of the shaft. So we create a fasten mate (m) and select the mate connectors. I think the easiest way to do this is to move my mouse on the surface until I see the mate connector I want (the one in the middle) appear and then I press and hold Shift and go and click on that one. That keeps Onshape from changing which mate connectors are visible. After I have done that, I should get something that looks like this. (Remember that you can press F to zoom to Fit)



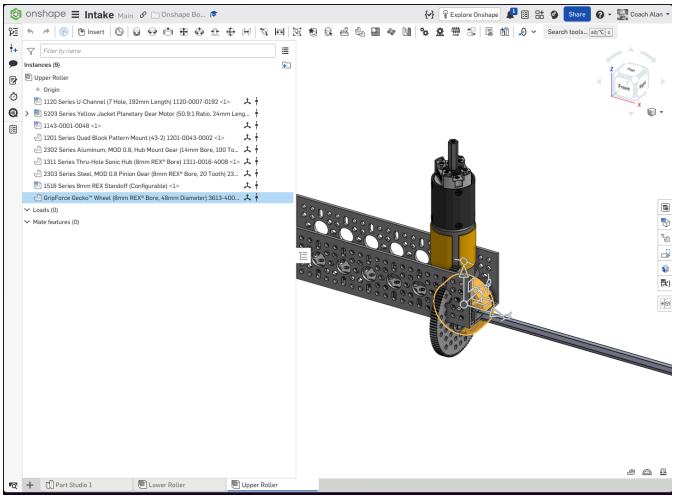
Next, we are going to select Assembly Linear Pattern (looks like four blocks with the upper left one clear.), select our Intake Roller Wheel for the instance. Then select the direction and click on the part of the roller that is where you want the others to show up. Change Distance to 16mm and the Instance count to 15. You'll get something like this:



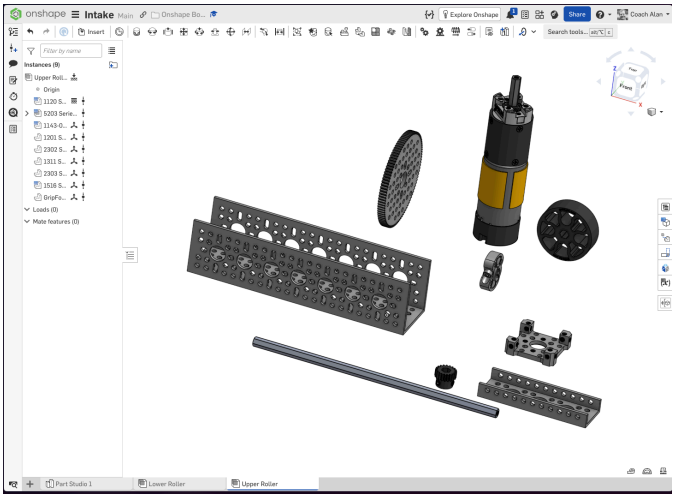
Now go press on the + (lower left) and create Assembly. Rename it "Upper Roller". For this part, we want a motor inside of an u-channel with a big gear connecting to a smaller gear below it that will drive the shaft with the compliant wheels on it. This entire thing will need to be able to rotate with the balls and connect to our chassis.

Let's start by going to FTC insert and grabbing some of the parts we know we'll need. Let's grab a 7 hole U-Channel, a 50.9 Gear ratio 5203 motor, a 3 hole mini low side U-Channel (1143-0001-xxxx), a quad pattern block (1201-0043-0002), a 100 tooth hub gear (2302-0014-0100), a sonic hub (1311-0016-4008), a 20 tooth REX gear (2303-4008-0020), a 216mm 8mm REX standoff, and a 48mm Grip Force Gecko Wheel (3613-4008-0048).

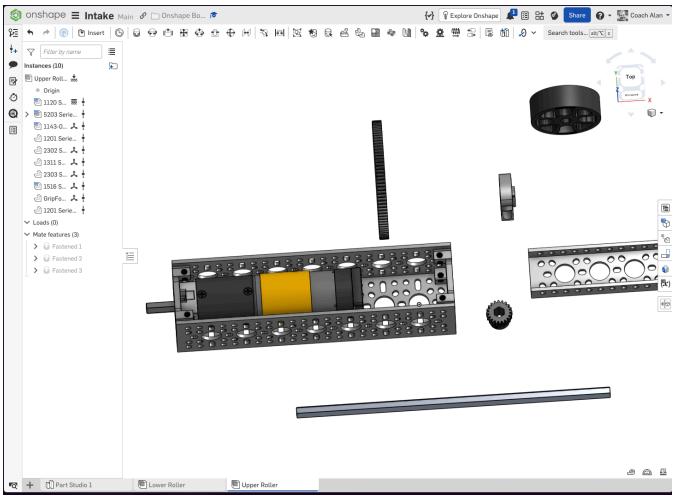
They will all be together in the middle of the screen like this. (I expanded the instances part so you could see what to get from the FTC Insert tool.)



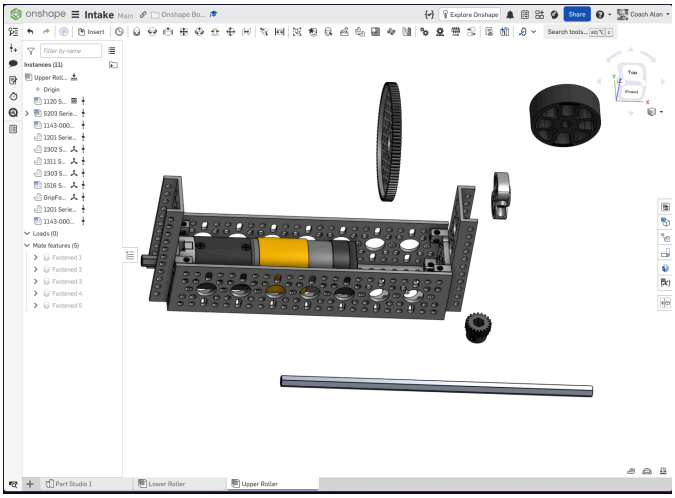
Next, I spread them out so I can see all of them and I fix the U-Channel (by right clicking on it and selecting Fix)



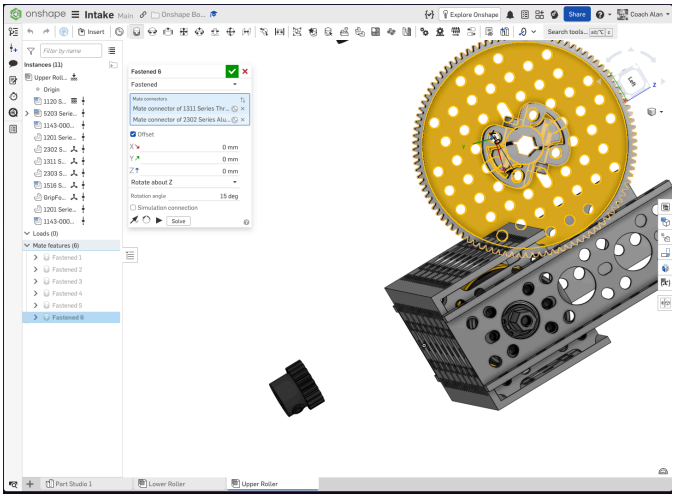
I then put the quad block on the motor by using a fasten mate. Don't forget to connect the top of the motor to the underside of the quad block. I then select the quad block and copy/paste to get another instance. Now I am going to use the fasten mate to connect each quad blocks to the channel. (The one already connected to the motor should bring the motor along with it when you press Solve or the green check.) Figuring out how it is easiest to connect pieces together comes with practice. You'll find yourself getting it wrong a lot before you get it right. This is what it should look like now.



Now we'll make a second copy of the 3 hole mini-u channel and connect both of them to the sides using fasten mates.



Next, we'll put a sonic hub on the inside of the 100 tooth gear. (For me to get it to line up, I had to rotate the sonic hub by 15 degrees in the offset around the Z axis.)



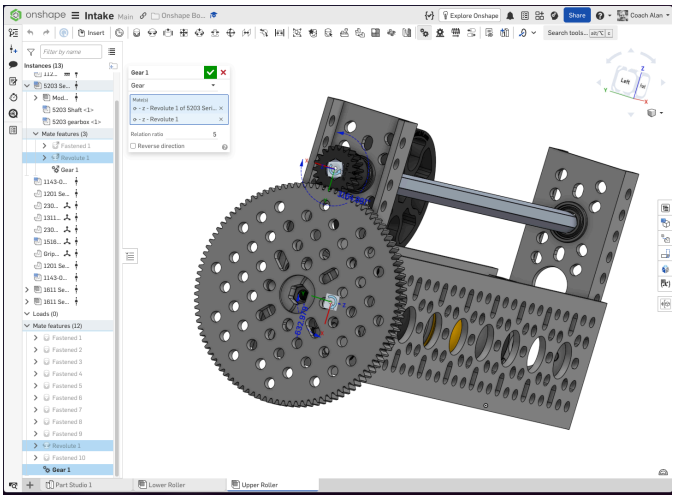
Now we'll go put that on the shaft. Before doing the next step, we realize we need two REX flanged bearings but we forget to get one[1] so let's go back to the FTC insert tool and grab two. (1611-0514-4008) You can either click on it twice in the insert tool or you can insert one and then copy/paste. Your choice. Now put the two bearings in the top hole of the mini low side U with fasten mates. (With the flange on the inside) . I find the easiest way to do this is to grab the back of the flange and fasten mate it to the hole in the U channel.

Now put the shaft in the bearings with a revolute mate. (This allows the shaft to spin. I find the easiest way is to use the middle for both and then use an offset in the Z to make it stick out the amount I need). Put the small gear on the shaft with a fasten mate.

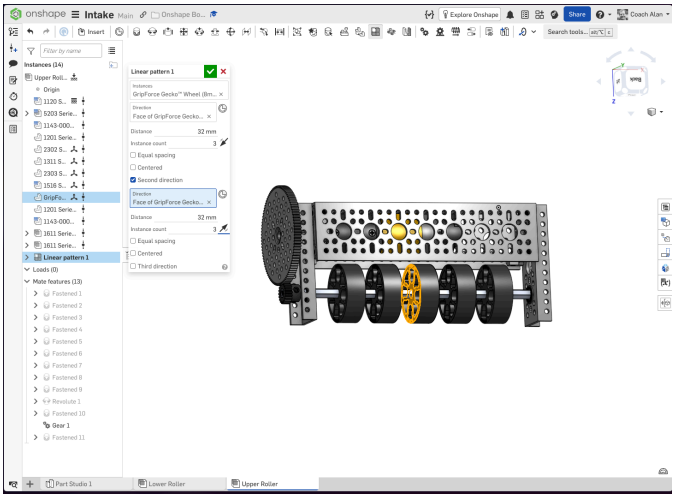
Now create a Gear Relation between the revolute of the motor and the revolute of the shaft with a relation ratio of 5. (Because 100:20 is the same as 5:1)

Everything should look like this:

[1] I thought about going back and doing it in the first step, but I thought it might be helpful to know that forgetting a part is common and no big deal. You just go get it.



Now let's put the gecko wheel on the shaft with a fasten mate. I used an offset to put it exactly in the center. Now let's use Linear pattern.

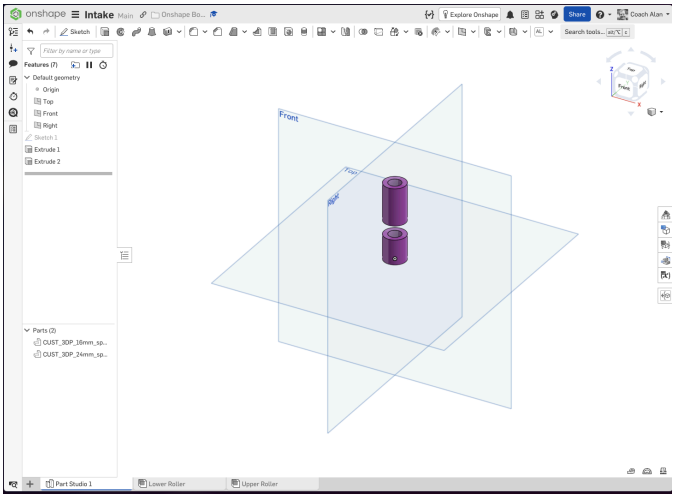


This should bother you. There is nothing to keep these gecko wheels from sliding on the shaft. Luckily we know how to make spacers - (If you don't remember, then refresh chapter 4) . We'll go to the part studio and make 2 spacers. One that is 16mm (for in between the wheels) and 24mm (for the edges). While you could be fancy and make them have a hex cutout since they are just for axially constraining, we'll just make them out of 8mm diameter circles with a 3mm offset for the outer circle.

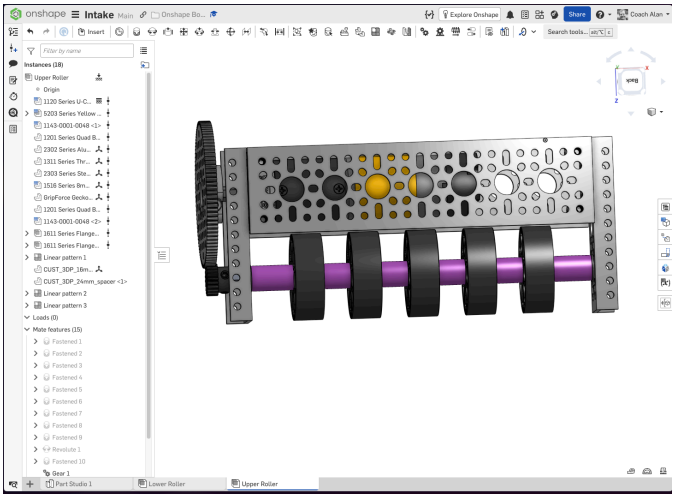
I suggest making all of your custom parts an unusual color so it is obvious looking at the CAD that they are custom and also naming them an identifier that means they are custom and then how they are to be manufactured and then

their description. (My team starts custom parts with QQ, but I used CUST here)

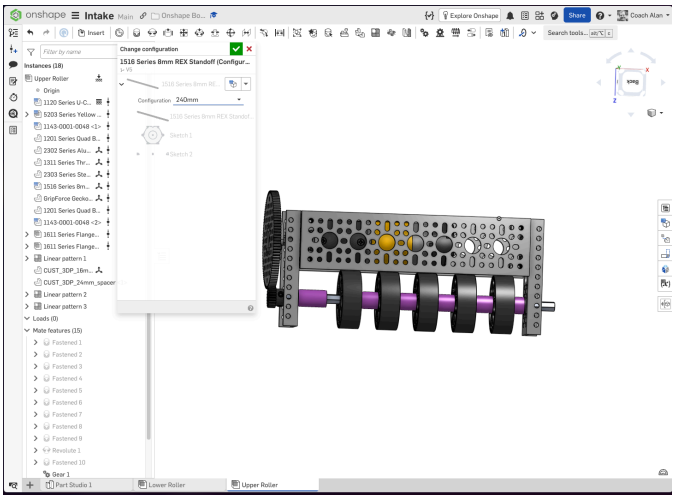
Here is what my two spacer parts look like:



Now we'll go put them in the assembly. (Use Insert and Current Document and Part Studio) . I used the fasten mate for one of each and then the linear pattern to make the rest.



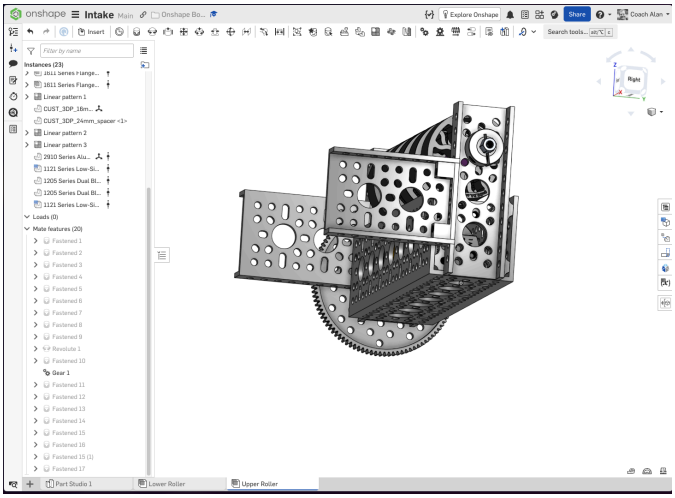
You probably noticed that we don't have anything constraining our axle on the other side from the gear. And we don't have room to add a shaft collar. Right click on the 1516 standoff and choose change configuration. Now change it to be 240mm long.



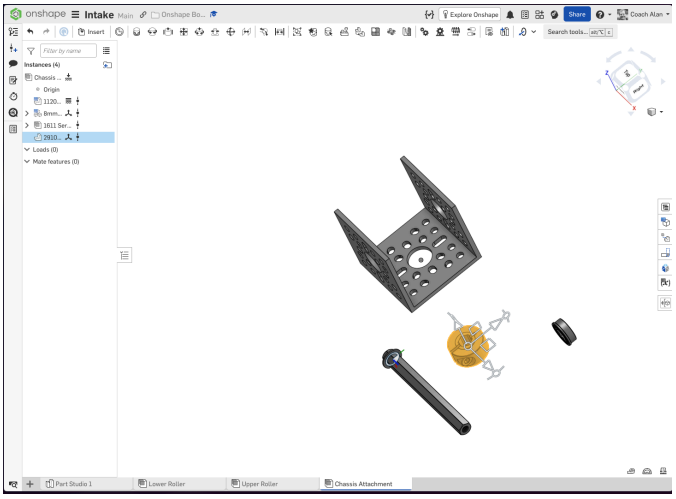
You'll see that this messed up my spacing because of the way I did the mates. No big deal, I can move my mouse over each mate feature and it will show me which parts it affects. I'll go into the one with my middle Gecko wheel and change the offset to be zero. (You may not need to do this depending on what you did.)

Now I'll go back to the FTC insert tool and insert a REX shaft bearing (2910-1020-4008) and use a fasten mate to put it on the end.

Let's add the parts we'll need to mount it now. Add a 2 hole low-side U-channel and a dual block mount (1205-0001-0003). I fastened the dual block mount to the U-Channel and then copied and pasted both of them and then did a fasten mate from the dual block mount to our existing roller (on both sides). This should now look like this:

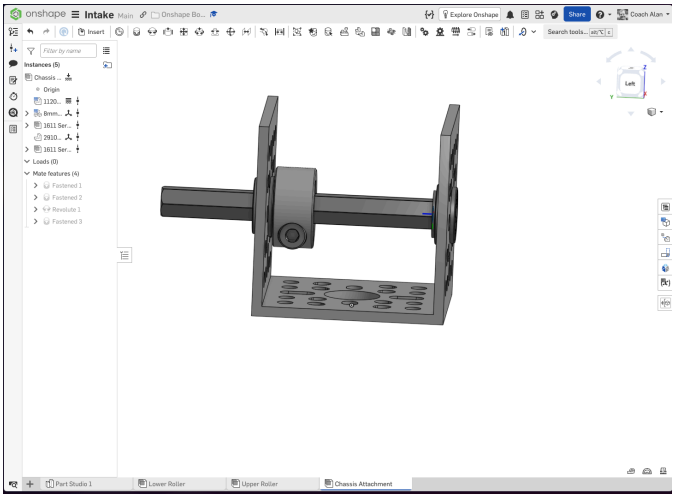


Now to attach it we are going to make a new assembly since there are two of them and then we won't have to do work twice. So press the + sign and rename it Chassis Attachment. Here we go into FTC insert tool and insert a 1 hole u-channel, an 8mm REX Shaft with E-Clip (72mm), a 1611 Flanged Ball Bearing for REX (1611-0514-4008), a REX shaft collar (2910-1020-4008) . I move everything away from the part I am going to fix (in this case the U-Channel) and then I fix the U-Channel. It should look like this.



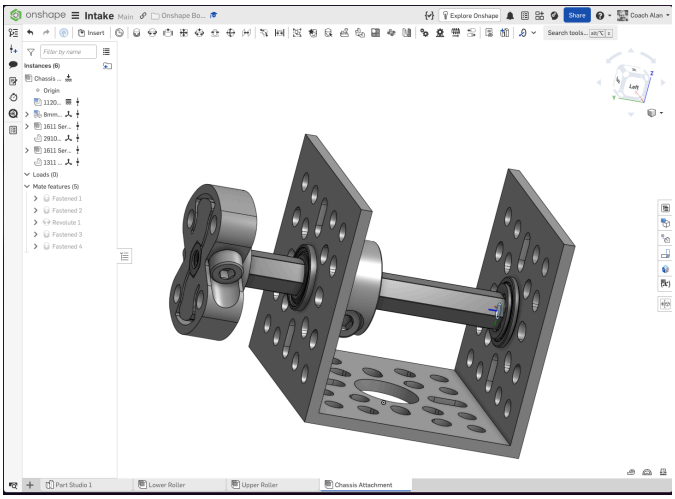
Now go put the two bearings in the two holes opposite each other with fasten mates. Then we'll put the shaft in (with a revolute mate) with the e-clip on the inside of the outer wall.

Something like this:

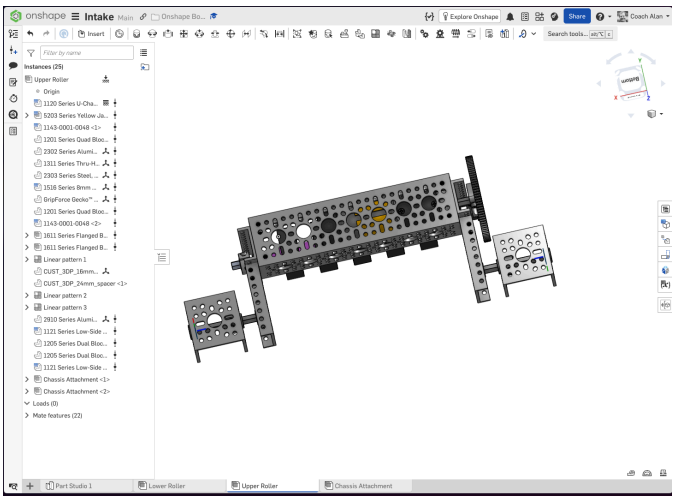


Let's add a sonic hub (1311-0016-4008) to the end of the shaft to be able to

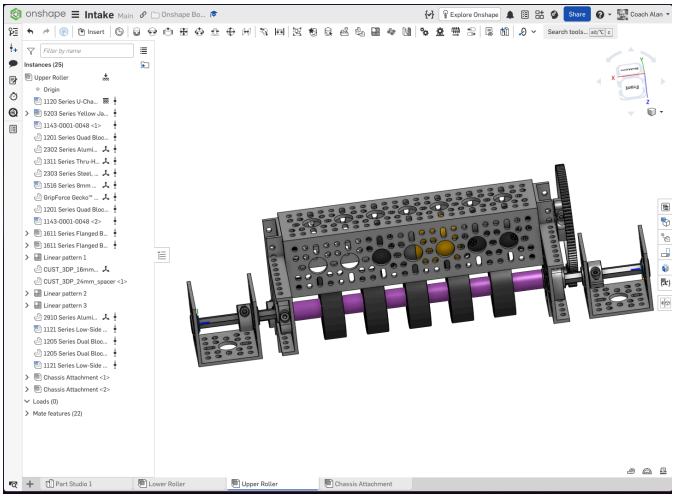
attach to our main upper roller. Go get it out of FTC insert tool and then fasten it to the end of the shaft. It should look like this:



Now we will go back to our Upper Roller and add in two of these and fix the sonic hubs to the mini low side U-Channel. This will look like this:



If you have setup all the revolute mates correctly, you should be able to rotate the two pieces on the side so that it will look like it needs to on the chassis. This is what mine looks like:



We have now made the two pieces needed for our intake!

13. Shooter

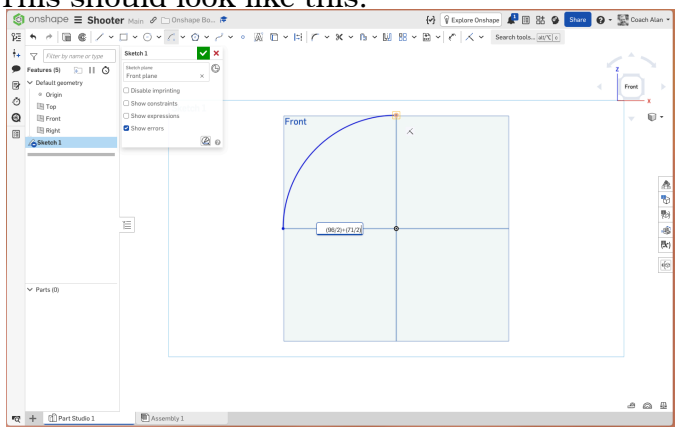
Just like we did earlier, we need to make sure we know what we are designing before we start. We need a ramp piece for the ball to go on and we need a motor to turn a wheel that will transfer its motion to the ball.

Let's use a 96mm Rhino wheel from Gobilda. (This was an arbitrary choice because I have one. It doesn't mean it is the best choice. You can spend a lot of time debating what to design. You are better off designing and building something and then changing things that you think will improve it.)

First, create a new document called "Shooter".

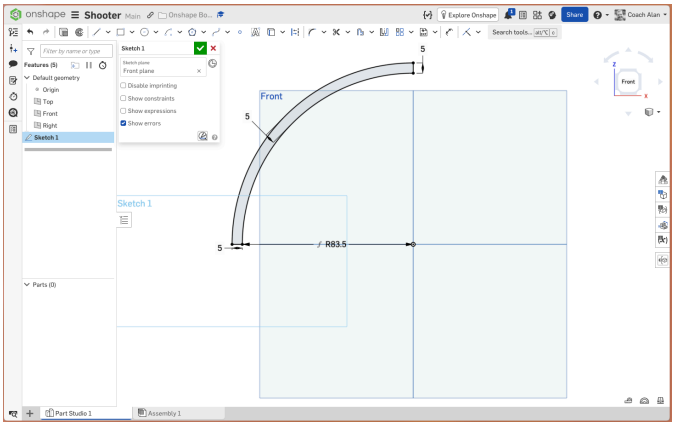
13.1. Making the shooter ramp

In the Part Studio 1, create a sketch on the Front plane. Press N to make it normal. Since the curve is the most critical thing to get right, we'll do it first. We are going to make a Center Point Arc that goes from the vertical plane line to the horizontal plane line. For the radius, we will use the ability of Onshape to do math for us and enter $\frac{96}{2} + \frac{71}{2}$ (71 came from measuring the "pollen" diameter) . This should look like this:



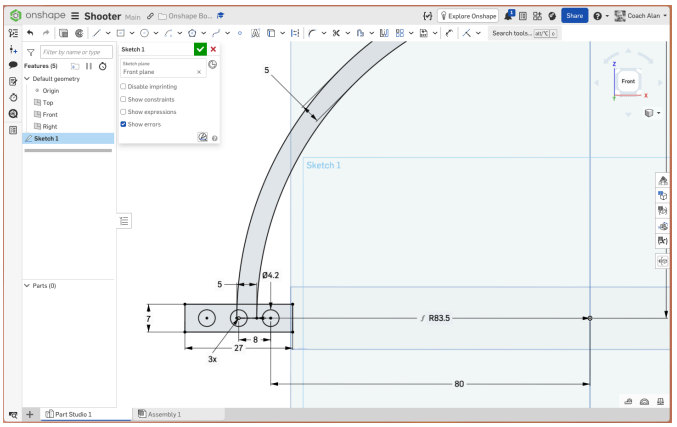
Next, we'll do an offset (O) of 5mm to the outside. (I made this a little thicker than my normal 3 because it will have the stress of a ball on it. That might not be necessary, but I wanted to explain my thoughts. We then add a line to both ends (make sure it is a vertical line at the top and a horizontal line at the

bottom.) . Add a dimension to both the bottom line and the line on the top right of 5mm. This will have our sketch be fully constrained. It should look like this: (you'll notice that I move dimensions to make them easily readable.)

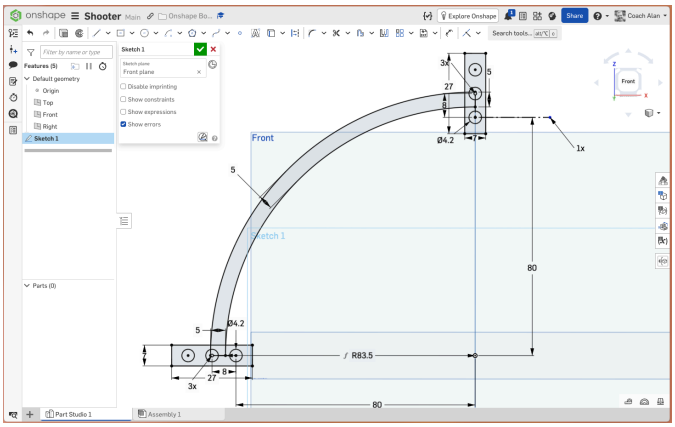


Well, we need a way to mount our ramp. We know that we will be connecting to Gobilda channel which is on an 8mm grid. Because we know we can extrude parts of our sketch later and only the middle of the ball is the full size (it is a sphere so the edges won't fully touch the ramp in the back), let's make 3 holes at the bottom (80mm, 88mm, and 96mm away from the center point.) We'll make the holes 4.2mm in diameter and we'll make sure it is in a rectangle that 7mm high and 27mm wide so there is enough plastic around it it won't break. (I made a center circle and then a linear pattern for the other 2 and then I did a center point rectangle based around the middle one.)

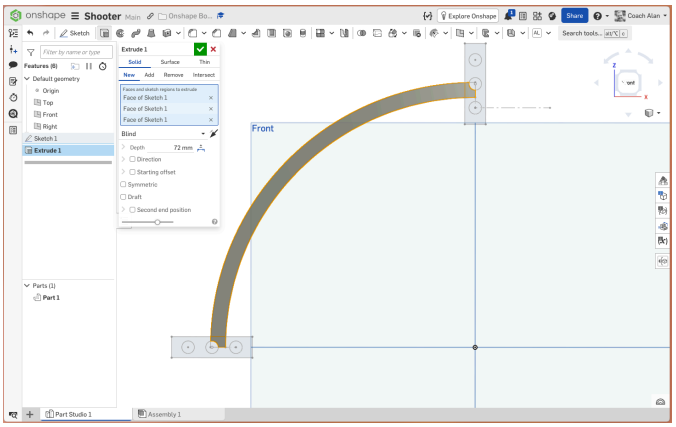
It should look similar to this:



Now, let's do a similar thing for the top. 3 holes (80mm, 88mm, and 96mm above the center point each hole 4.2mm in diameter, and inside a rectangle that is 7mm wide and 27mm tall) . It should look like this:

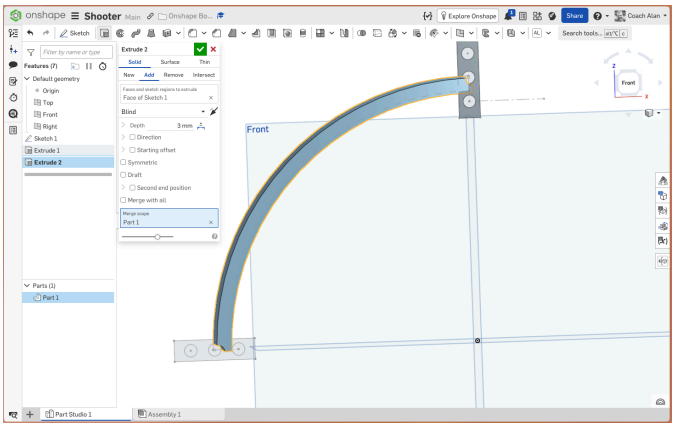


Now we'll extrude our ramp portion (but not our rectangles) to be 72mm (blind).

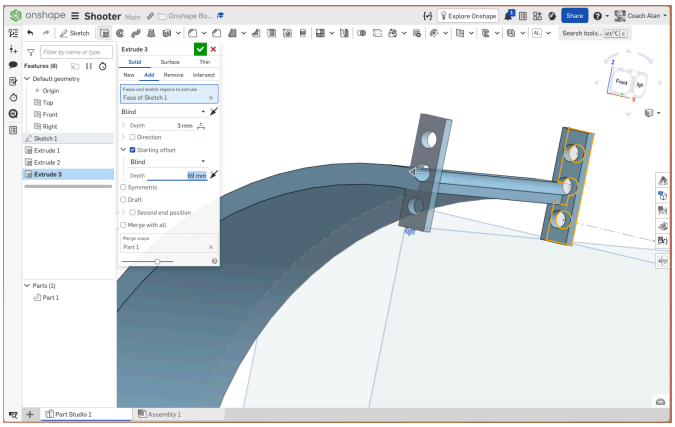


Now, we are going to extrude each of the rectangles (minus the holes) to be on either end. To do this, we'll need to do 2 extrudes for each one. I'll demonstrate with the top one and then you do it with the bottom one.

For the top one, make the Sketch 1 visible (click on the eyeball when you go over it) Then select on the rectangle and do an extrude add of 3mm. It should look like this:

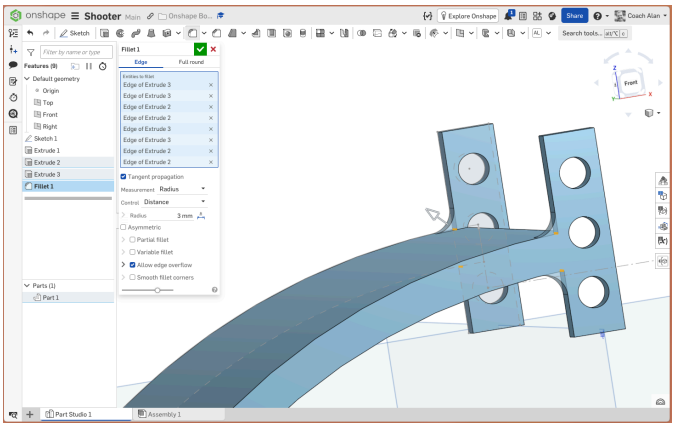


Then for our second one, we do an extrude add of 3mm but with an offset of 69mm . (The reason why is that 72-3 is 69. You can get Onshape to do the math for you if you want.)



Now go do the same thing for the mounting holes on the bottom. (You can either do 2 extrudes again or you can add that part of the sketch to Extrude 2 and Extrude 3. That is what I did so that if I decide to change the thickness I can do it in one place.)

We are now going to add some filets so our part will be stronger. Don't forget to select all 8 edges. It should look like this:



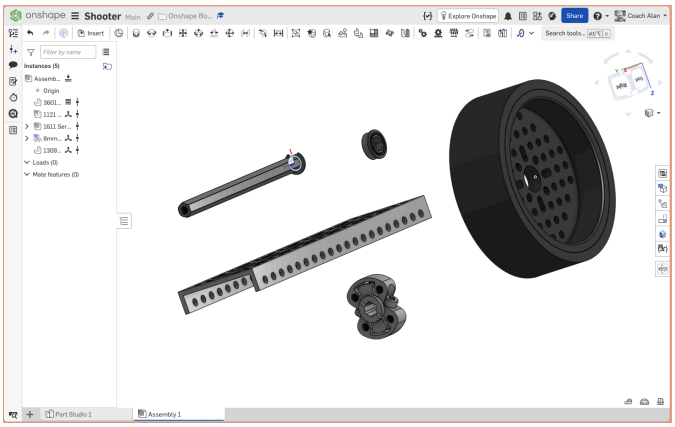
13.2. A mixed assembly (custom and COTS)

You'll often hear FTC students talk about "COTS" - this stands for "Commercial Off the Shelf" or in other words, parts you buy instead of fabricate. A lot of our assemblies will be mixed parts. While it is tempting to design everything yourself, you really only want to design and make the parts you can't buy. Commercial parts have typically had a lot more design and testing put into them so they are less likely to fail.

So now let's go to Assembly 1 and start using the FTC insert tool to insert some of the COTS parts we need. In this case, let's start with a 96mm Rhino Wheel (3601-0014-0096), a U-Channel (8 hole) (1121-0006-0168 - but be aware if you insert it by part number it isn't configurable so it is harder to make copies and change their size), a flanged ball bearing for REX (1611-0514-4008), a 8mm REX Shaft with E-Clip (112mm - 2106-4008-1120 -but again, if you insert by part number it won't be configurable), and a 8mm REX sonic hub (1309-0016-4008).

After inserting them and spreading them out, I move the Rhino Wheel to be roughly centered on the origin and then I fix it since as the main moving part, it is the most important to have right on the robot so I make it the fixed item. (Do NOT fix multiple items. It might feel like it is making your life easier, but it will become a problem later.)

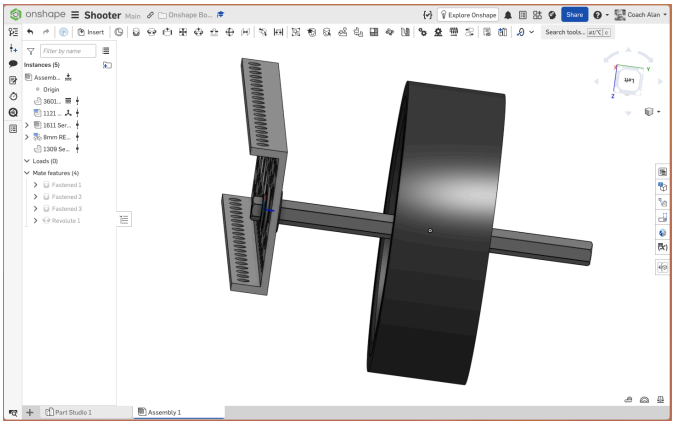
Here are my parts.



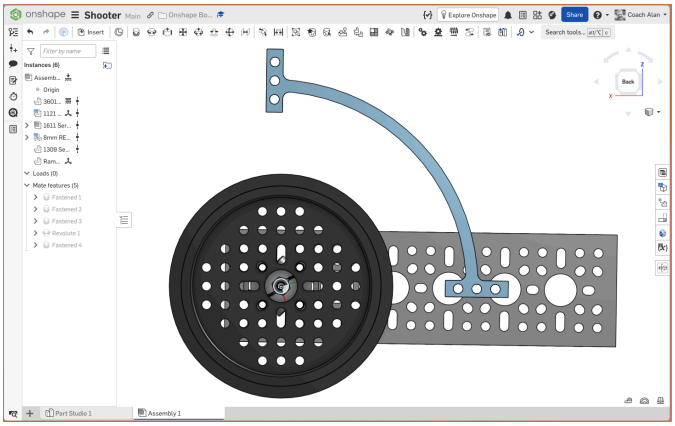
First, I attach the sonic hub to the Rhino Wheel with a fasten mate and then I attach the shaft to the sonic hub with a fasten mate. I then attach the bearing to the first hole in the low side U-channel with a fasten mate (outer flange should be on the inside of the channel) and then I attach the part of the shaft with the e-clip to the bearing with a revolute mate. (I found the easiest way to do this is to do it on the shaft with an offset of 5mm in the Z).

I am intentionally giving less and less steps directly so you can see what you know and what you need to go back and refresh. I promise it isn't *only* because I am tired of making lots of screenshots....

This should now look like:

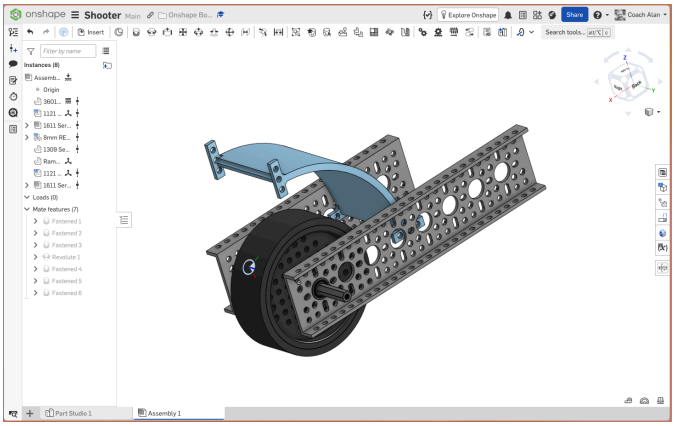


Now we are going to Insert our ramp (use the insert button) and then a fasten mate to connect the middle hole of our bottom piece of our ramp to the hole to the U-Channel. It should look like this:



Now, let's put in an 8 hole mini low-side U channel with an 8mm REX flanged bearing with the bearing on the outside and also connected to our ramp. (Instead of going back to the FTC insert tool, if you already have a part you need in the instances you can copy and paste it to get another one. For configurable parts, you can copy/paste them to get one you can change the configuration of by right clicking on it and selecting change configuration.)

It should look like this:



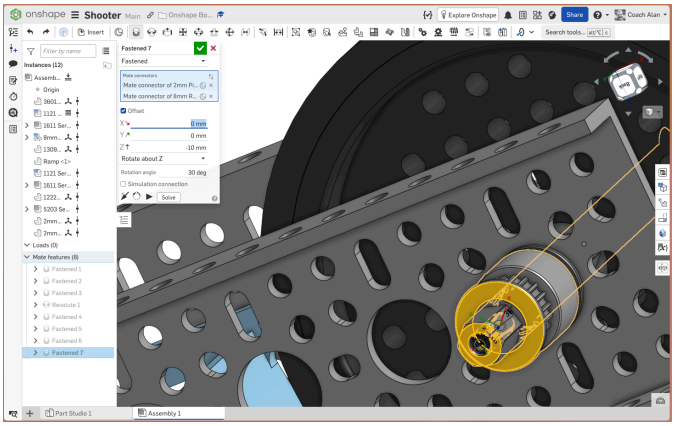
You might be wondering why we made this one longer. While it is tempting to put the motor directly on the shaft, that will make the shooter larger than the area we agreed to back when we did space planning. So we are going to put the motor on the inside and show how to use a belt.

So going back to FTC insert, let's insert a Flat pattern mount (1222-0001-0001), a 5203 Yellow Jacket Motor (1:1 - because we want fast for a shooter, 5203-2402-0001), and 2 2mm GT2 8mm REX bore Timing Pulleys (3422-4008-0020)

First, I put the pulley on the wheel shaft. I use the centers to connect them

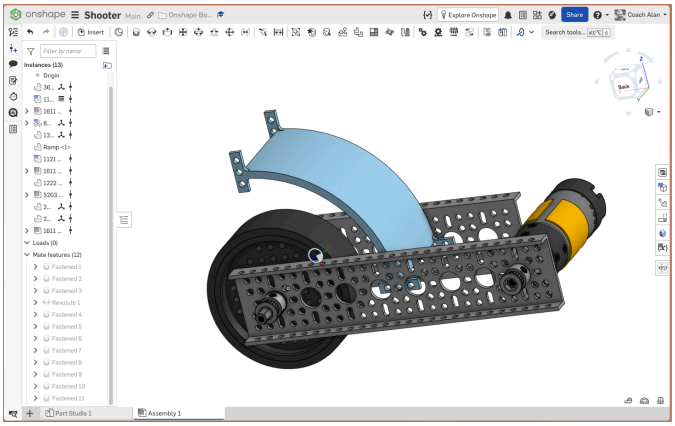
which means that I have to rotate about Z by 30 degrees to get the hex parts to line up. I also put it 10mm in because that looks like where I want it. (On the robot I would put it up against the bearing to hold it in place.)

It looks like this:

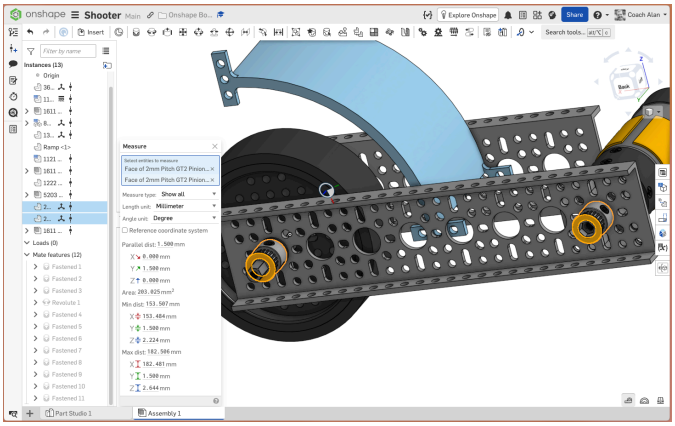


Now, we'll put a bearing in the last hole of the 8 channel with the flange on the inside and then attach a pattern mount on the motor and then mount the motor and put the timing pulley on the motor shaft. (This time, I had to use 25 degrees offset around Z and 3mm offset on Z).

This should look like this:

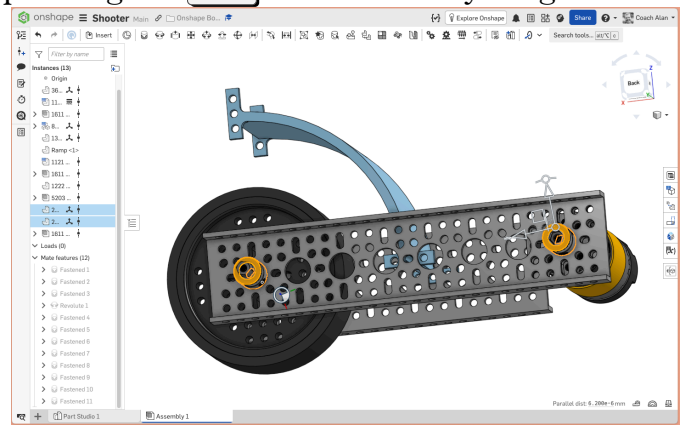


Now, let's make sure our pulleys are in line with each other. To do that we will use the measuring tool (looks like a tape measure in the bottom right). Click on the faces of the two shafts and you can see that they aren't parallel because they are 1.500mm off in the Y dimension.



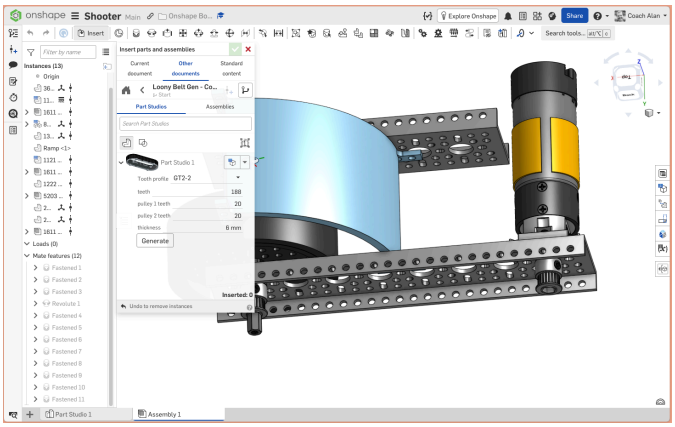
So let's change our mate with the one on the wheel so they match. Close the measure tool. Then as you move over each mate in the mate features list it will highlight the parts that are affected. Change our offset from 10mm to 8.5mm

You can just click on those two faces and you'll see the parallel distance down in the lower right. You'll see that now it is 6.2e-6mm (That is an engineer way of saying $6.2 * 10^{-6} mm$ or so small that we aren't going to worry about it.) . Tip - pressing the space deselects everything in Onshape.



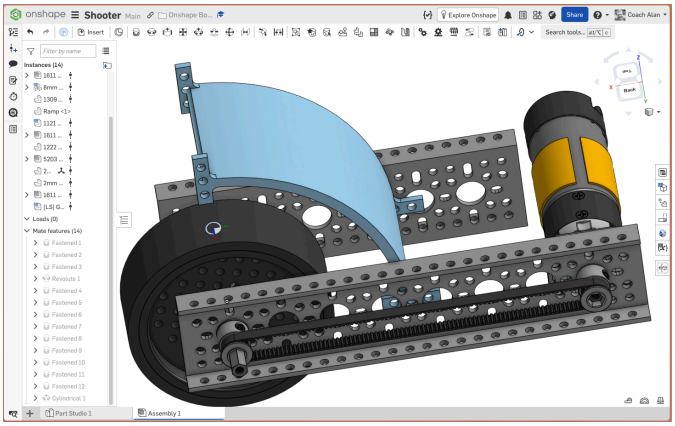
To go generate the belt, we are going to use the Loony Belt Gen[1] To do that we go to insert and other documents search for "Loony Belt Gen". You'll get a dialog box. For this we are using GT2-2, for the number of teeth we used the GoBilda belt calculator (https://www.gobilda.com/2mm-gt2-timing-belts/) with 2 20 tooth pulleys and a center to center distance of 168mm which gives us 188 for the number of belt tooth count. We then verify that GoBilda sells it in that size. (They do.) We put in 20 teeth for the 2 pulleys and 6mm for the thickness. The dialog box should look like this:

[1]From the Loony Squad - https://www.theloonysquad.com/

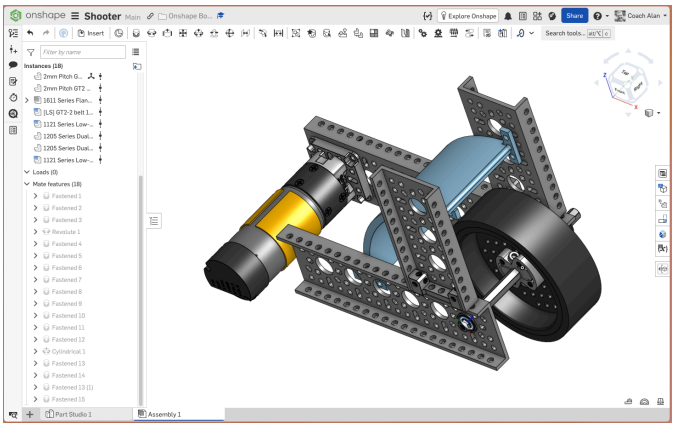


Press “Generate” and then after it is generated (you don’t have to wait for the preview to show up), click on it and it will be inserted into your document.

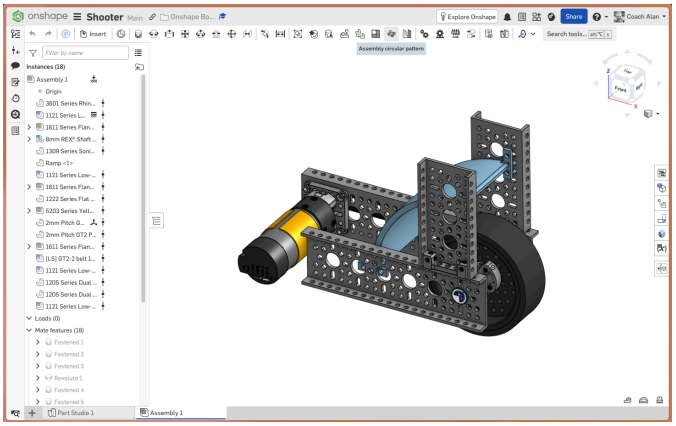
I use a Fasten mate to connect it to the pulley with the wheel and then a cylindrical mate to connect it to the one with the motor. (This isn’t really correct and it keeps us from being able to animate any more, but it does help us make sure that nothing is in the way of the belt.) . If you know the right way to connect this, please contact me and I’ll update the book and credit you. It should now look like this:



The last thing to do is add some structure to connect the top part. Let’s go to FTC insert tool and grab a 1205 series dual mount (1205-0001-0005) and a 3 hole low u-side channel. We’ll put it on both sides and you should end up with a completed shooter that looks like this:

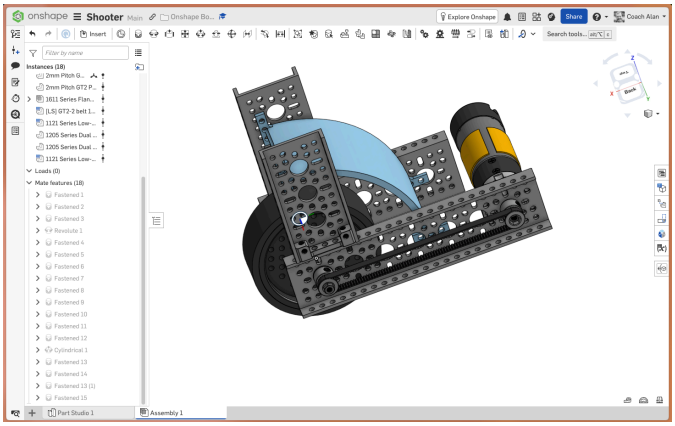


It is obvious that the wheel is not in the center and the shaft is too long. So we change the configuration of the shaft to 104mm and we make all of the changes to various offsets so that the pulley is in the right place, and the wheel is centered. You'll now end up with something like this:



It is also obvious that the low side U-Channel holes can each be reduced by one. Since in FTC space is precious, we'll reduce each of those but that will cause us to need to change a number of the mates and regenerate the belt (now 164 teeth). Go ahead and do this as it is good practice for what you will be doing a lot on your own designs. (Hint: broken mates are often because they connected to a part that isn't there anymore now that we made it shorter. Also look for fasten mates that now need to be moved.) . This should now look like this:

13. Shooter

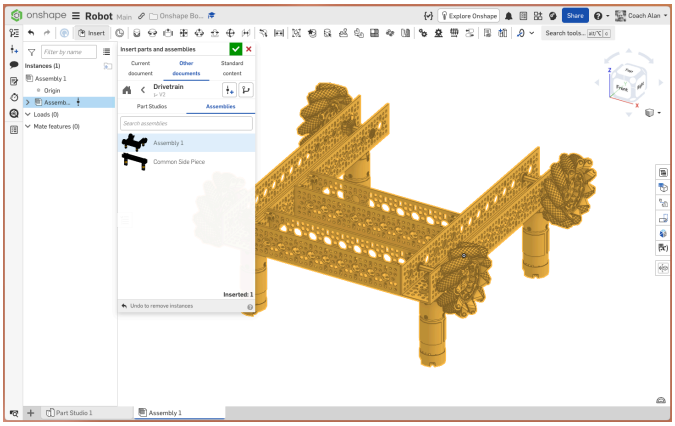

onshape
Shooter
Main
Onshape Bo...
Explore Onshape
Share
Coach Alan
Insert
Search tools...
Filter by name
Instances (18)
2mm Pitch G...
2mm Pitch GT2 P...
1611 Series Flan...
[LS] GT2-2 belt 1...
1121 Series Low-...
1205 Series Dual ...
1205 Series Dual ...
1121 Series Low-...
Loads (0)
Mate features (18)
Fastened 1
Fastened 2
Fastened 3
Revolute 1
Fastened 4
Fastened 5
Fastened 6
Fastened 7
Fastened 8
Fastened 9
Fastened 10
Fastened 11
Fastened 12
Cylindrical 1
Fastened 13
Fastened 14
Fastened 13 (1)
Fastened 15
Front
Part Studio 1
Assembly 1

14. Bringing it all together

14.1. Assembling the pieces

Create a new document called Robot (optionally you might give it a name to make it easier to identify different robot files).[1] I typically recommend that you have a folder with the season name and year that you store all of your documents in so that you can have an idea of where it came from. You can see that I have put all of the documents for this book in a folder called (Onshape Book)

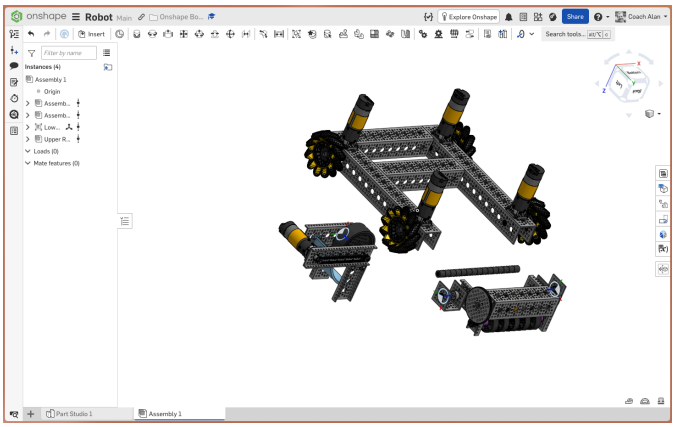
First, we'll insert the mecanum drive. Click the make new version (looks like a dot on a line with a plus sign). We want to always create a version so that we can know what version is in our main assembly. We can update the version of it later, but this allows work to continue on the sub-assemblies but have us decide when it makes it into the main robot CAD. When you are inserting an assembly, look under the assemblies tab.



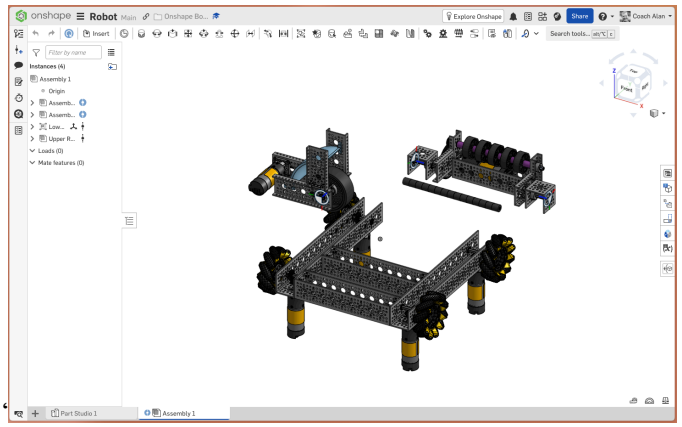
Go ahead and add the "Lower Roller and Top Roller" from the Intake file and the "Shooter" file as well. For each one make a version.

When I spread them out so they aren't overlapping I have a document like this:

[1] My team always uses the first name of a famous scientist for our robot name. One of my favorites was "Werner" which was the first name of Heisenberg (famous for the uncertainty principle) because we had no idea whether the design would work or not...



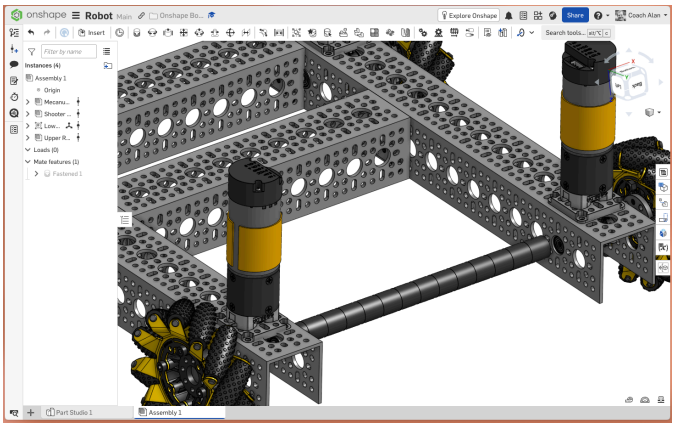
Since they are versioned, to change the name of our two “Assembly 1”s, we’ll have to go change them in their documents. After you do that, create a new version in each document. When you do that and then come back to the Robot file, you’ll see an icon showing there is a newer version of the two you just changed in the instance section. It looks like this:



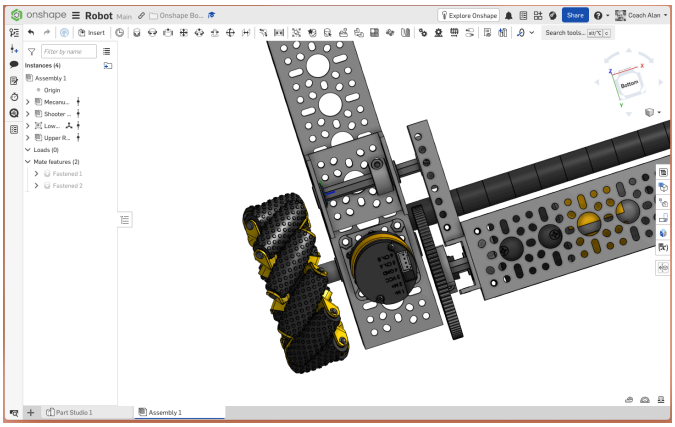
You can click on the blue icon on the top row (upper left after undo and redo) and update all to the most recent.

Now, let’s flip the mecanum drive right side up. You can click on it and then use the manipulator to rotate it.

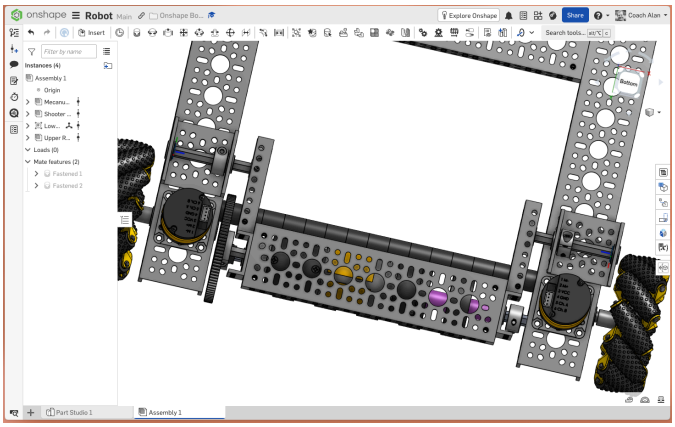
We’ll now connect the lower roller using a fasten mate. After we do that, it should look like this:



Now we'll connect the upper roller onto the right U-channel with a fasten mate. It should now look like this:



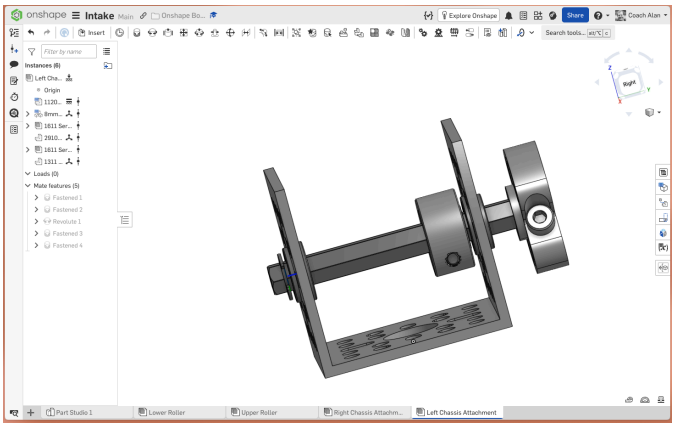
Looks good until I zoom out and see that I can't fasten the left hand side because it is off. I rotated it first, to make it flat with the U-Channel. I use my measuring tool and I am off by 6.633mm. It looks like this:



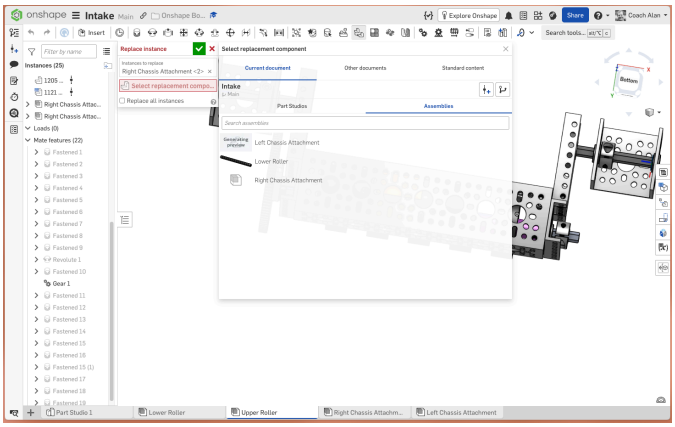
While it is tempting to say “Fine, I can see I can adjust the shaft collar when I build it”, I know that I would never be happy with that. So let’s go fix it. To do this we can either open the document directly or right-click on the Upper Roller and choose “Open Linked Document” and then click Go to Main when it tells us we can’t edit a version.

Since we only want to fix the left side and the mate is in Chassis Attachment, let’s Duplicate Chassis Attachment and rename it to “Left Chassis Attachment”. (We’ll also rename the other one to be “Right Chassis Attachment)

On the Left Chassis attachment, let’s change the revolute offset to be -5.63 (It was 1mm, so this is changing it by 6.63mm) . Then we’ll need to change the Fasten 3 offset on Y to be 12mm. (You could make it 12.63 but I don’t think that is important.)

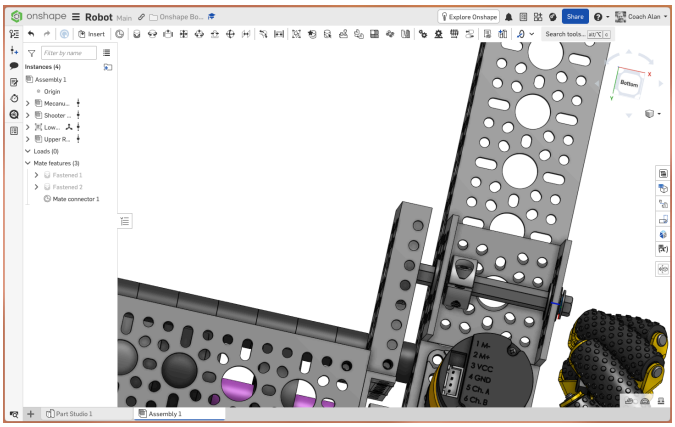


In our Upper Roller, now right click on the instance of Right Chassis Attachment that is actually on the Left, and select Replace Instances. Go and get the Left Chassis Attachment out of this document. It will look like this:



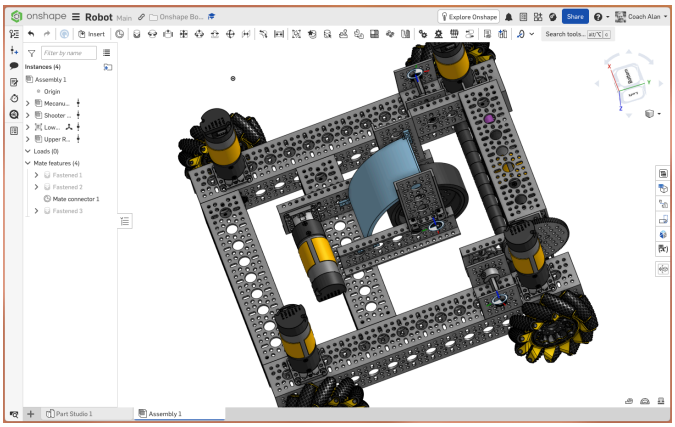
Now make a new version (Remember it is the symbol with the line and dot and + on the far left) of the intake.

Go back to our Robot and update the version of the upper roller. Now it looks like this (which is good enough)

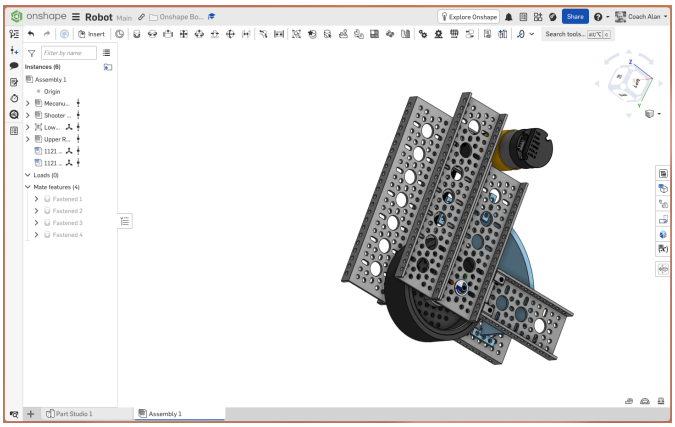


Now on to putting the shooter on. Let's affix the shooter using a fasten mate.

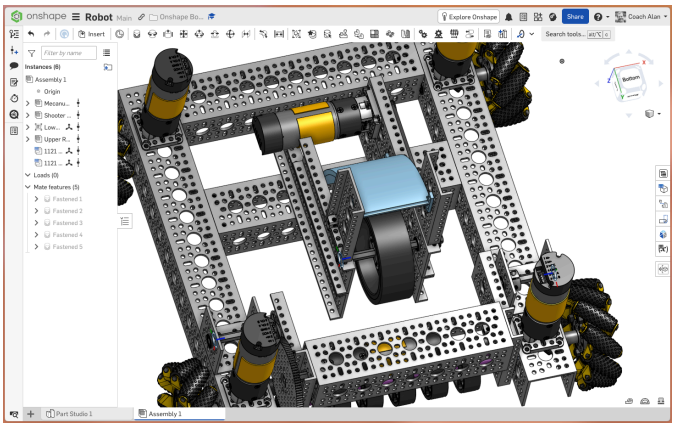
We put it in the middle on the crossbeam like this:



Hmm, we realize that this isn't high enough to have any chance of working. We could go and address this in the Shooter document, but since this is really not part of the shooter mechanism, I prefer to just add 2 7 hold mini low side U-Channel here. So let's use our old friend the Insert tool and go get them. Delete the fasten mate from the shooter to the chassis, drag it out of the way and use fasten mates to attach the new low side-u. Your shooter piece should now look like this:



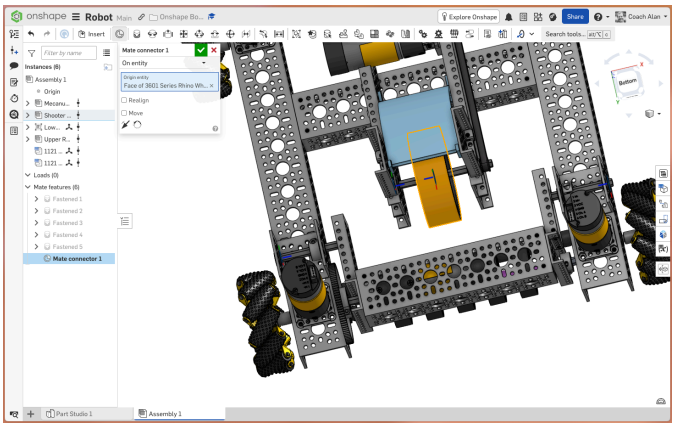
Go and attach this to the chassis. It should look like this.



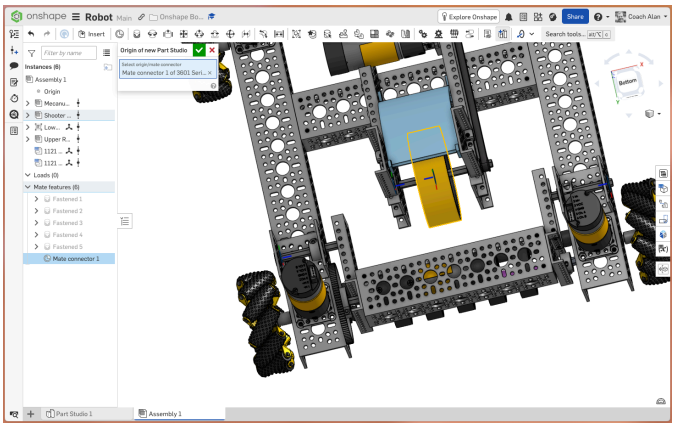
Better, but I think we will need a ramp from the intake to the shooter. I am about to show you something that is a little controversial. I think there are uses for it (like here), but there are others that believe you should always create parts directly and then insert them. So use this power for good and not evil....

We are going to create a part studio in context. First, let's go create a mate connector that we can use as our origin. We are going to use the face of the Rhino Wheel since that is a convenient place to have as the origin for our part studio.

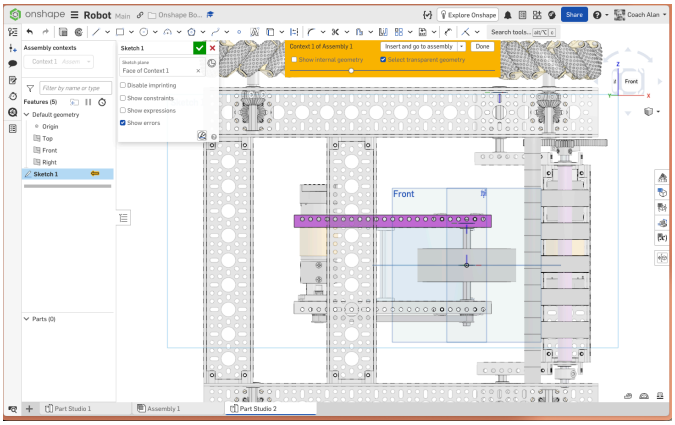
Here is what that looks like:



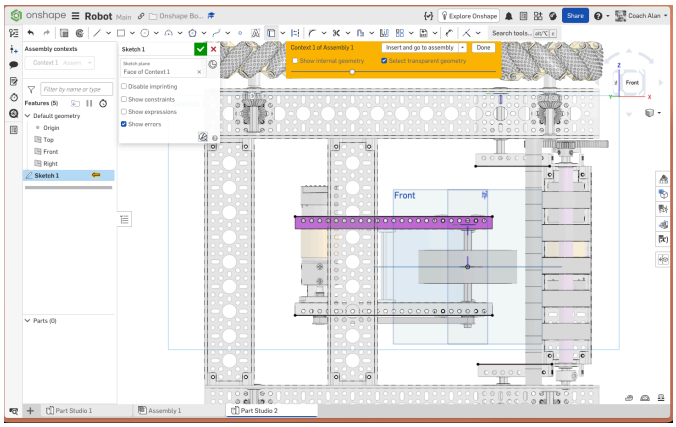
Now select "Create Part Studio in Context" (it looks like the two bars of a bar graph with a + sign) and for its origin select our mate connector. It will look like this:



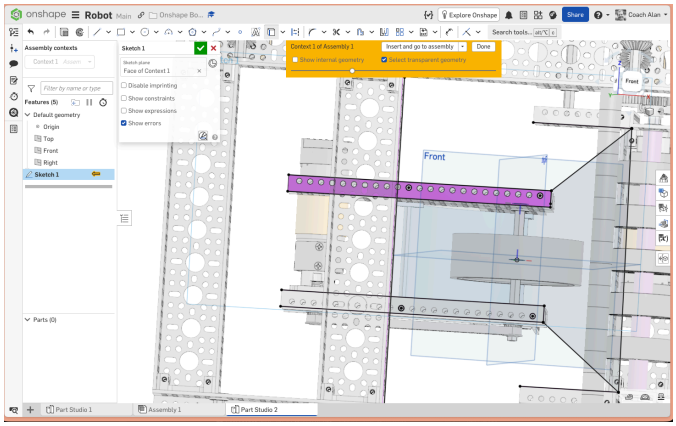
Let's create a sketch on the plane of the bottom of the left side of our shooter (just because it is a flat plane. We'll use an offset for the right height.) . This will look like this:



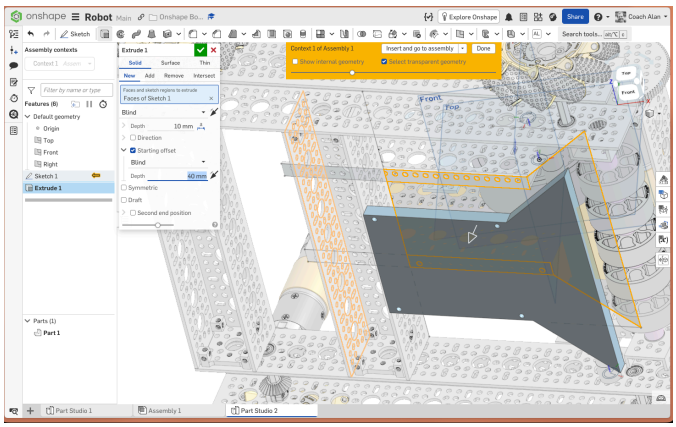
We then go and select edges of the low side-u and use the Use tool (U) to bring those into our sketch. It will look like this:



I drew some diagonal lines, decided I wanted to use more pieces so that I could have one piece for my ramp and another piece for where the channels are and four holes for how they would connect. My sketch looks like this:

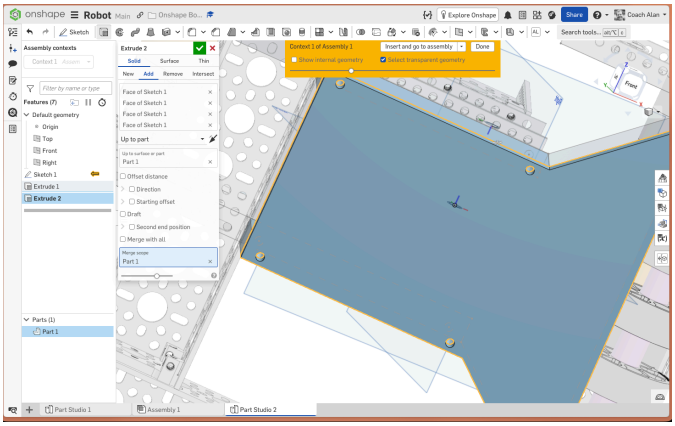


I then extruded all of the parts together with a starting offset.

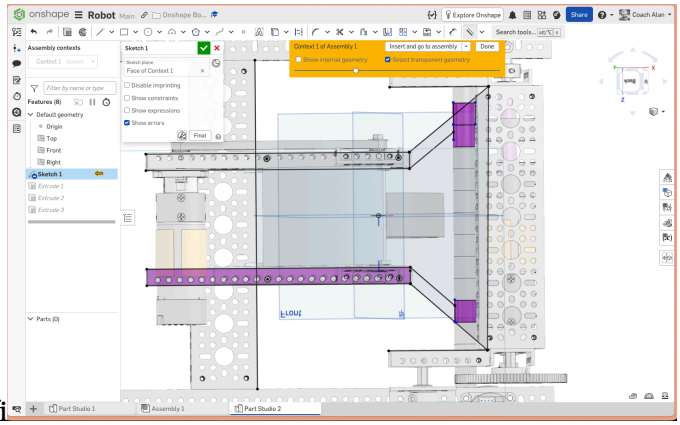


These numbers are pure guesses based off of what looks right. It wouldn't surprise me at all if when I put it on the robot I found out they needed to be adjusted.

I then extrude (as an add to the part) the side pieces and I extrude them up to the part. That way if I need to change the values of the first extrude, I won't have to change them on the second as well.



I accept it, rename the part to "intake ramp" and then realize rotating it that I need some side pieces to keep the ball from falling off the edge. I go back in and use some of the rollers and then draw a line to it. I use the "Parallel" constraint to make each line parallel to my outer line.

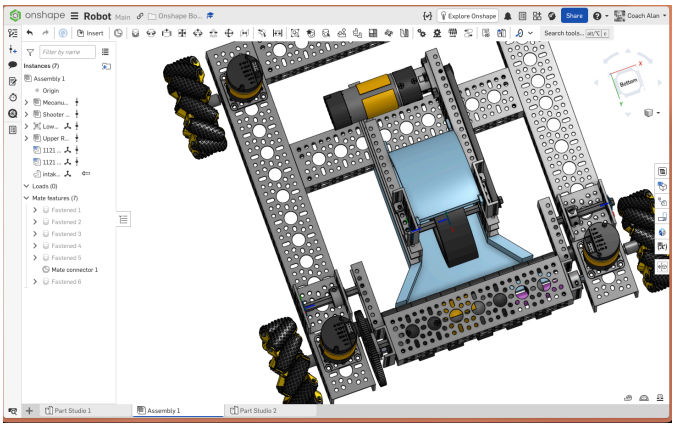


fi

Here is what the sketch looks like. I then do another extrude add just like I did for Extrude 2. (This will be Extrude 3)

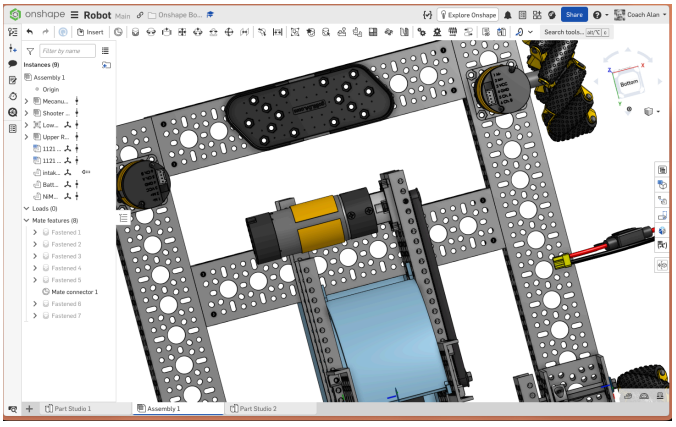
and then click Done in the box next to "Insert and go to assembly".

I then go into my assembly and insert the intake ramp and use a fasten mate to fasten it to the low-side U channel. It looks like this:

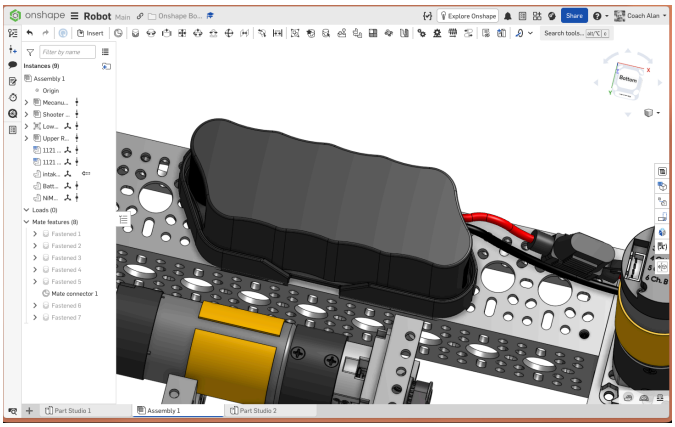


Now, let's put in our battery, our control hub, our expansion hub, and our team numbers.

Let's use our FTC Insert tool to insert the battery mount (1209-0001-0001) and the battery (3100-0012-0020). Use a fasten mate to connect the battery mount to the back crossbar and make sure it is centered. Here is what it will look like:

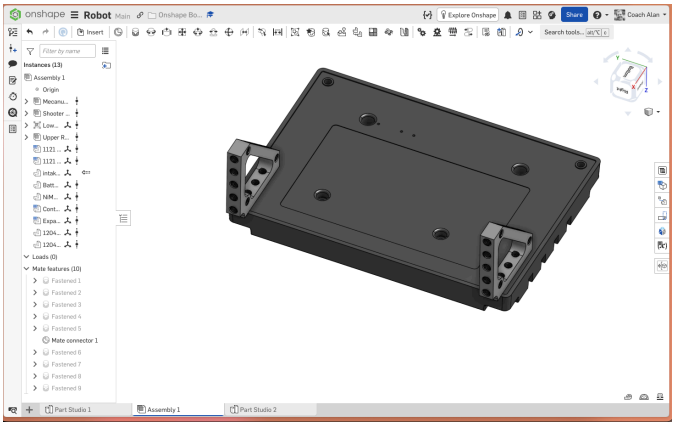


Then we'll fasten the battery to the mount. I know some people leave this out but I like to have it in because it is a critical piece of the robot. I found the easiest way to fasten it was to use the mate connector in the middle of the battery and of the mount. It will now look like this:

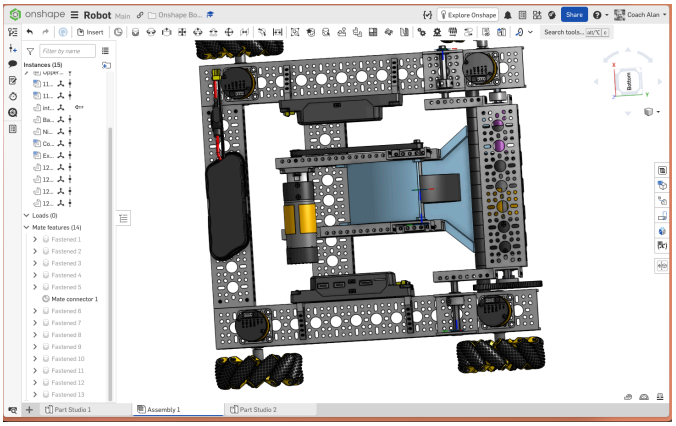


Next, we'll go to FTC Insert Tool and insert the control hub and expansion hub. Remember that these are REV parts. The Control Hub is REV-31-1152 and the Expansion Hub is REV-31-1153. To mount these, I'll use one of my favorite GoBilda parts, the Gusseted Angle Mount (1204-0002-0001). I get two of those and I fasten the next to bottom hole with the REV. (Note that on your robot you'll have to use a M3 screw here so it will be a little loose, but it will be fine. Many teams 3D print holders for their hubs. That is fine as well.)

This is what it will look like:



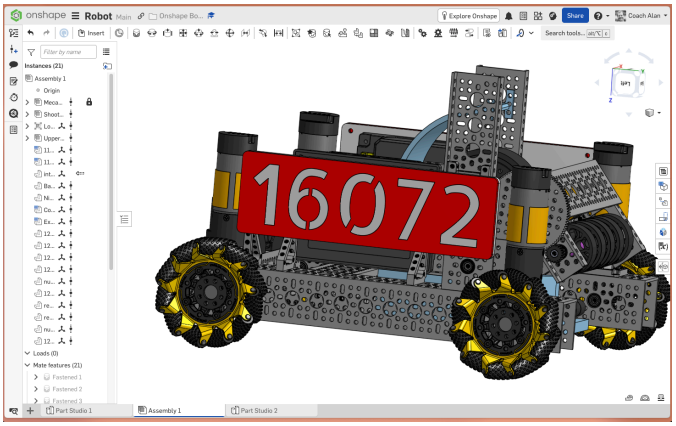
We'll do the same for both of them and then put them on the chassis. Then it should look something like this:



Now, let's go get our fancy robot signs that we made way back in chapter 6 and insert them. (I had to go make a version)

I started by fastening the two plates to each other and then I used another one of the Gusseted Angle Mounts on one side and the bracket for the upper roller on the other. (For a real robot, you probably want to have a connection on all four mounting points to make it more secure.)

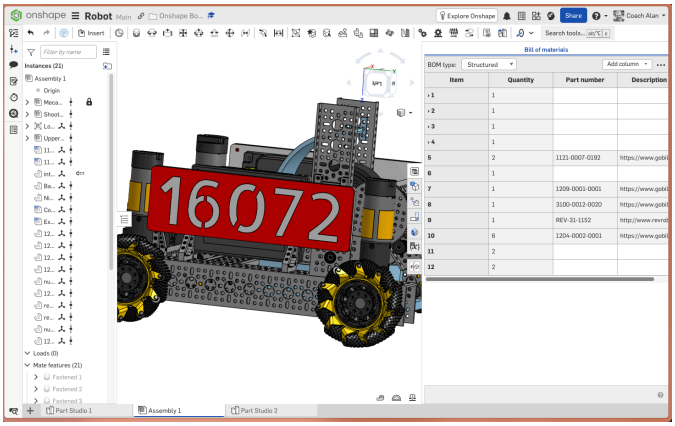
Here is what mine looks like:



14.2. Creating BOM (Bill of Materials)

One of the very cool things about putting everything in CAD, is you can then get a list of what parts your robot uses and you can check that against what you have to know what you need to order and/or fabricate.

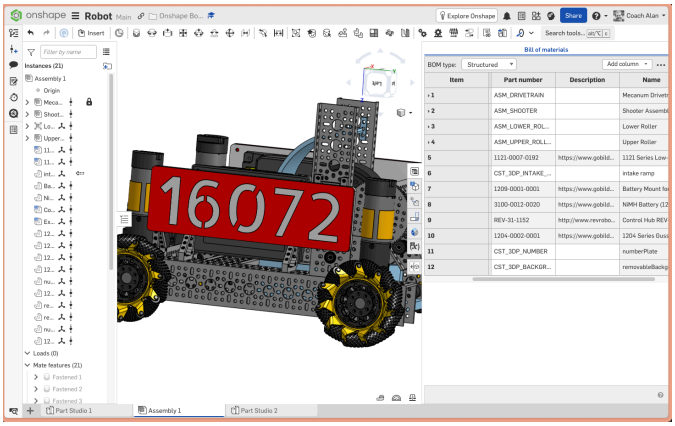
To start, click the BOM Table icon (top one on the far right) and you'll see a table that looks like it has a lot of blanks like this:



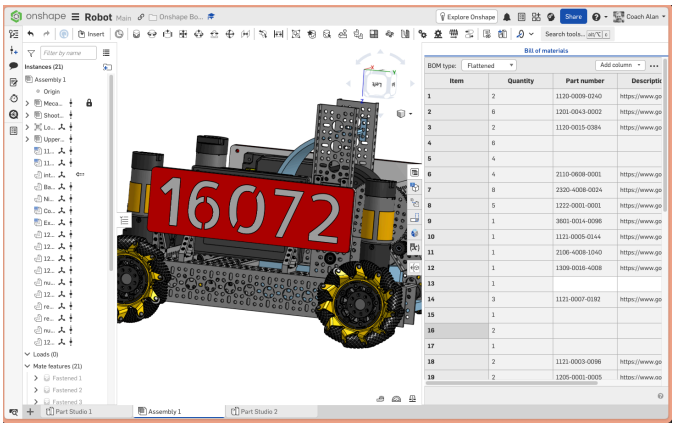
It turns out a lot of this is blank because we didn't put in Part Numbers and Descriptions as we made our Assemblies. We can either do it in each assembly and part studio (right click on assembly or the part and select Properties) or we can do it here.

Let's start by using the Add Column to Add the name so we can quickly see the name for each item. I like to start assemblies with ASM_ so they are quick to see, and for custom fabricated parts my team starts them with QQ_<manufacturing method>_ . (for example: QQ_3DP_ or QQ_LASER_ or QQ_CNC_ etc). For this book, I'll use CST (short for Custom instead).

My table now looks like this:



Hmmm, but this is only the top level. We can click on each of the arrows to see it in a structured view. Or we can use the three dot menu to expand all. While this is pretty neat, it would be helpful if we could see the parts we were using across all assemblies. Well, we can. Just change the BOM type to "Flattened" and it will combine them together to look like this:



From the three dots we can export to CSV (Comma Separated Variable) which can be imported into probably every spreadsheet program that exists. We could then go mark the ones we have and what needs to be ordered and give it to the person on our team in charge of ordering.

In addition, if you click on a row it will highlight it on the robot. (Remember that space deselects everything and when you click another row it adds it to the selection.)

14.3. Next steps

I hope this has been a useful journey for you. This is just a brief introduction to all of the functionality of Onshape. We didn't talk at all about drawings or many of the mate types or even all the constraints or sketch tools. http://learn.onshape.com has many useful lessons.

Part IV.

Appendices

A. List of Keyboard Shortcuts mentioned

You can see a full list by clicking on the ? icon in the top right and selecting Keyboard Shortcuts. But here I have listed the ones I mention in the book. (The most common ones I use.)

A.1. In 3D viewing

F	Zoom to Fit
N	Set to “Normal” - perpendicular
P	Show/Hide Planes

A.2. In Part Studio

Shift+E	Extrude
Shift+F	Fillet
Shift+S	Create Sketch

A.3. In a Sketch

C	Center point Circle
D	Dimension
Shift+E	Extrude
Shift+F	Fillet
G	Corner Point Rectangle
H	Horizontal Constraint

Shift+M	Midpoint Constraint
O	Offset
Q	Construction Line
R	Center point Rectangle
U	Use (convert/project)
Shift+W	Revolve
space	Clear selection

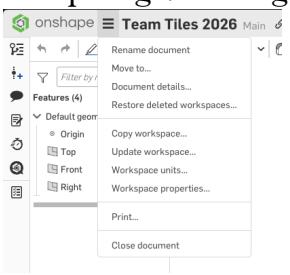
A.4. In an Assembly

m	Add a Fastened mate
shift	Lock available mates
space	Clear selection

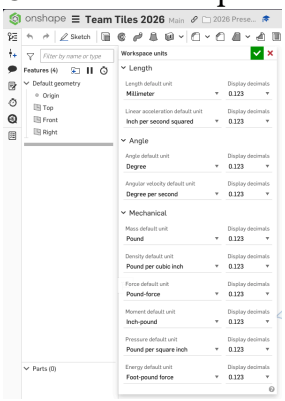
B. Setting Workspace Units

If someone else created the documents and it is set to inches, you can change the document to mm (millimeters).

To do this, click on the "hamburger" menu (≡) in the upper left after the Onshape logo. You'll get something like this:



Click on Workspace units and you'll get something like this:

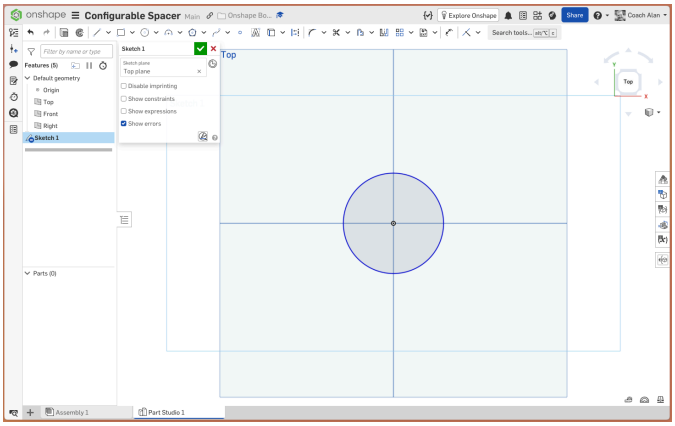


You could argue (and I would agree with you) that you should change all of the units to be metric. But I typically only change the other defaults when I need to use them since I don't use them in many documents.

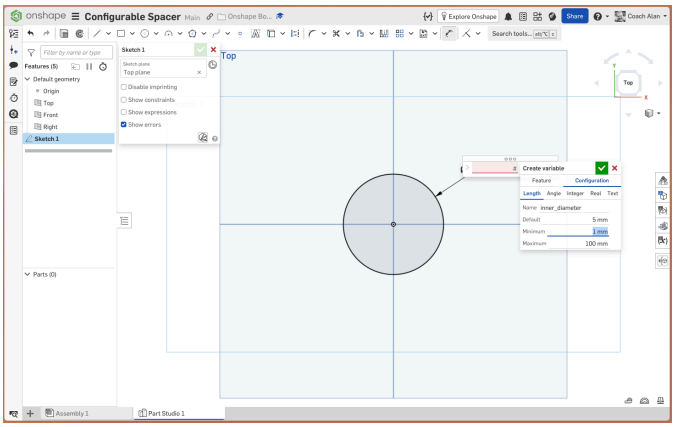
C. Configurations

We have seen how useful it is to have a part that has configurations. Well, what about making our own part that can do that.

This brief appendix will show how to do that. In order to give this as a simple example, we'll make a new document called "Configurable Spacer". In the Part Studio 1, go ahead and make a sketch on the top plane (Shift+S), make it Normal (N), and then make a center point circle on the origin (C). You should end up like this:

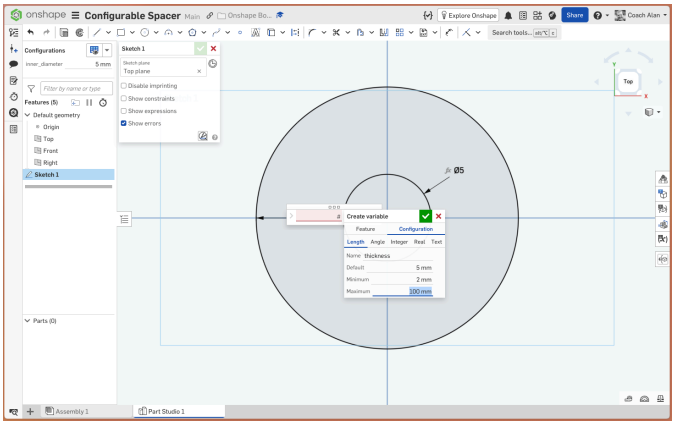


We will dimension (D) and put it on the circle and type # and select new variable. In the dialog that comes up,go from Feature to Configuration and put the values in as shown here:

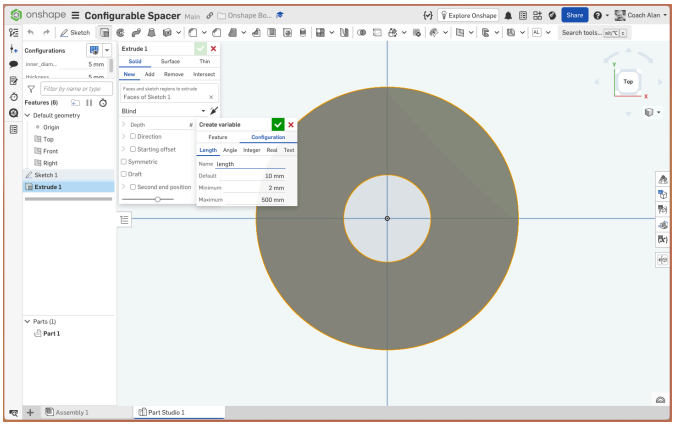


Accept this with the green check mark on this one and the next one. (It may be grayed out, don't worry about it and click it anyway.)

Then we'll do an offset (O) and we'll start typing in # and select new variable for the offset amount. This will bring us to the same dialog as before. We'll change to Configuration and put in the values here. (You can choose these values for what makes sense for your project.)



We'll accept and do an extrude (Shift+E) and for the blind amount we'll put in # and new variable. We fill in the dialog like this:



Accept both and then rename the part to spacer.

Go into the Assembly 1 and play with inserting this part and changing the size. That is all that is involved with making configurations! (You can only have one set of configurations per part studio but they can affect multiple parts in that part studio.)

Index

A
Applications, 55
Arc
    Center Point, 113

B
Belt generator, 121
Bill of Materials, *see* BOM
BOM, 137
    Flattened, 138
Brass Inserts, 59

C
Circle
    Center Point, 30
Classroom, 7
Configurations, 147
Constraint, 17
    Coincident, 78
    Horizontal, 17, 63
    Midpoint, 21, 35
    Parallel, 134
Construction Line, 21, 35, 43

D
Derived parts, 56
Dimension, 18, 21, 36

E
Export, 32
Extrude, 19, 31

F
Fastened Mates, *see* Mates, Fastened
Fillet, 58
Fillet, Sketch, 52
Fix, 89

G
Gear Relation, 96, 105

H
Hudson Bay, 8

I
Inscribed Polygon, 16
Insert, 118
Insert Tool, 55, 87

K
Keyboard Shortcuts, 143

L
Line, 43
Linear Pattern
    Assembly, 102
    Sketch, 64
Loft, 70
Loony Belt Gen, 121

M
Mate Connector, 89
Mates
    Fastened, 12, 24
    Revolute, 93

Midpoint, *see* Constraint, Midpoint
Mirror, 45

O
Offset, 31, 113

P
Part Studio in Context, 131
Plane, 69
Polygon, 16
Preferences, 5

R
Rectangle
    Center Point, 36
    Corner, 83
Revolute Mate, *see* Mates, Revolute
Revolute Offset, 128
Revolve, 61
Rib, 66

S
Shell, 71
Sketch, 15

T
Team Tiles, 9
Text, 20, 35
Timing Belt, 121

U
Use, 49

V
Variables, 32
Version, 85, 129

W
Workspace Units, 11, 145

Z
Zoom to Fit, 50